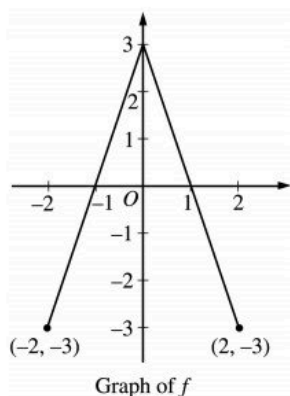


5-8

1.



The graph of the function f shown above consists of two line segments. Let g be the function given by $g(x) = \int_0^x f(t) dt$.

- Find $g(-1)$, $g'(-1)$, and $g''(-1)$.
- For what values of x in the open interval $(-2, 2)$ is g increasing? Explain your reasoning.
- For what values of x in the open interval $(-2, 2)$ is the graph of g concave down? Explain your reasoning.
- On the axes provided, sketch the graph of g on the closed interval $[-2, 2]$.

Part A

1 point is earned for $g(-1)$

1 point is earned for $g'(-1)$

1 point is earned for $g''(-1)$

$$g(-1) = \int_0^{-1} f(t) dt = - \int_{-1}^0 f(t) dt = -\frac{3}{2}$$

$$g'(-1) = f(-1) = 0$$

$$g''(-1) = f'(-1) = 3$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for $g(-1)$

1 point is earned for $g'(-1)$

1 point is earned for $g''(-1)$

$$g(-1) = \int_0^{-1} f(t) dt = - \int_{-1}^0 f(t) dt = -\frac{3}{2}$$

$$g'(-1) = f(-1) = 0$$

$$g''(-1) = f'(-1) = 3$$

5-8

Part B

1 point is earned for the interval

1 point is earned for the answer

 g is increasing on $-1 < x < 1$ because $g'(x) = f(x) > 0$ on this interval.

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the interval

1 point is earned for the answer

 g is increasing on $-1 < x < 1$ because $g'(x) = f(x) > 0$ on this interval.**Part C**

1 point is earned for the interval

1 point is earned for the reason

The graph of g is concave down on $0 < x < 2$ because $g''(x) = f'(x) < 0$ on this interval.

or

because $g'(x) = f(x)$ is decreasing on this interval.

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the interval

1 point is earned for the reason

The graph of g is concave down on $0 < x < 2$ because $g''(x) = f'(x) < 0$ on this interval.

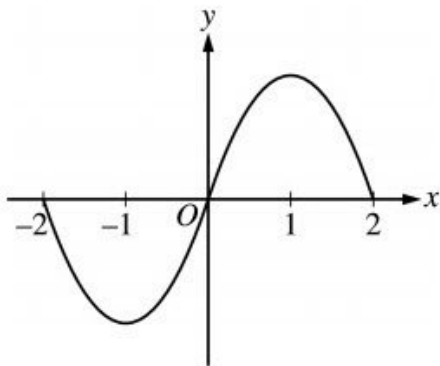
or

because $g'(x) = f(x)$ is decreasing on this interval.**Part D**1 point is earned for $g(-2) = g(0) = g(2) = 0$

1 point is earned for appropriate increasing/decreasing and concavity behavior

<-1> vertical asymptote

5-8



✓

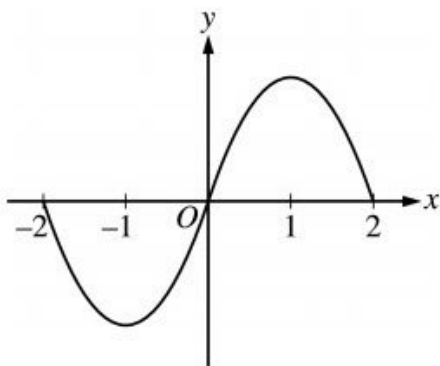
0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $g(-2) = g(0) = g(2) = 0$

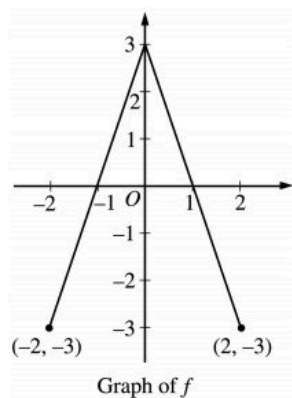
1 point is earned for appropriate increasing/decreasing and concavity behavior

<-1> vertical asymptote



5-8

2.



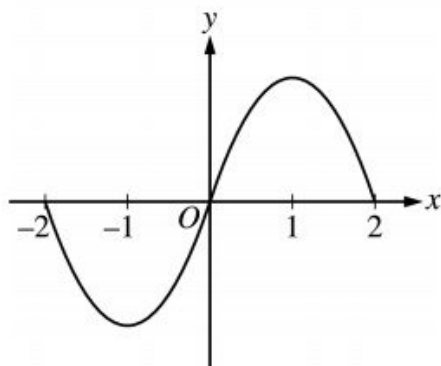
The graph of the function f shown above consists of two line segments. Let g be the function given by $g(x) = \int_0^x f(t) dt$.

On the axes provided, sketch the graph of g on the closed interval $[-2, 2]$.

Part D

1 point is earned for $g(-2) = g(0) = g(2) = 0$

1 point is earned for appropriate increasing/decreasing and concavity behavior <-1> vertical asymptote



✓

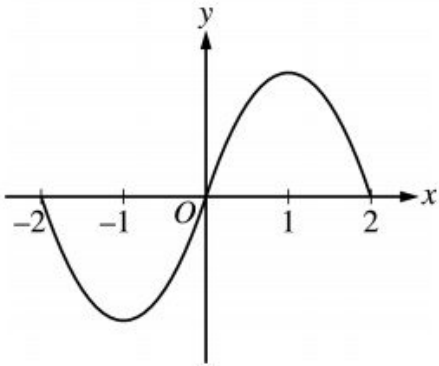
0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $g(-2) = g(0) = g(2) = 0$

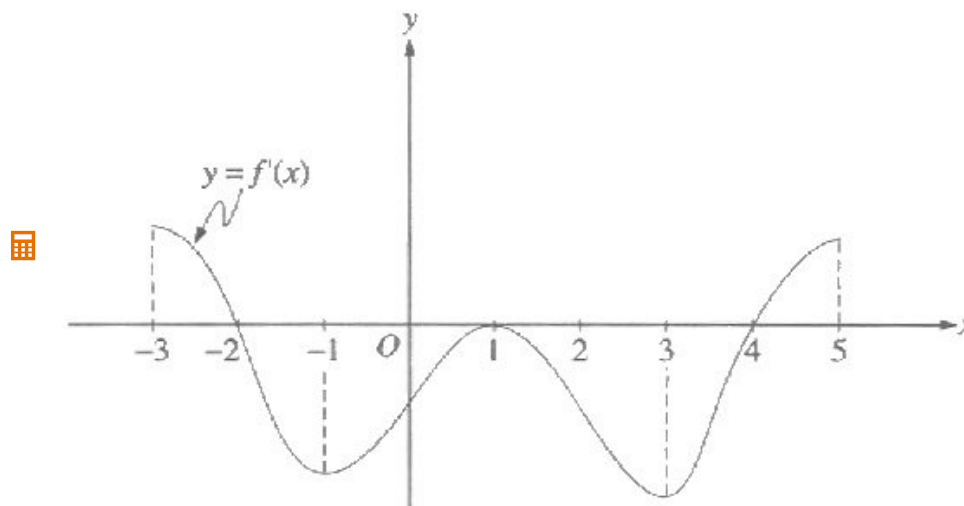
1 point is earned for appropriate increasing/decreasing and concavity behavior <-1> vertical asymptote

5-8



5-8

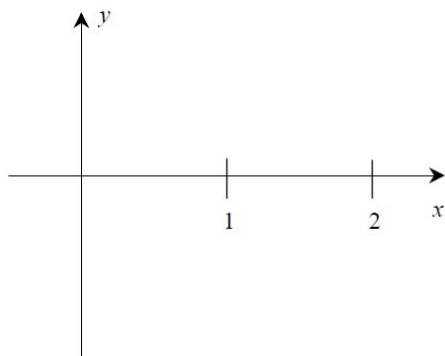
3.



Note: This is the graph of the derivative of f , not the graph of f .

The figure above shows the graph of f' , the derivative of a function f . The domain of f is the set of all real numbers x such that $-3 < x < 5$.

- For what values of x does f have a relative maximum? Why?
- For what values of x does f have a relative minimum? Why?
- On what intervals is the graph of f concave upward? Use f' to justify your answer.
- Suppose that $f(1) = 0$. In the xy -plane provided, draw a sketch that shows the general shape of the graph of the function f on the open interval $0 < x < 2$.



Part A

1 point is earned for the correct answer $x = -2$

$x = -2$

1 point is earned for the correct reason

$f'(x)$ changes from positive to negative at $x = -2$ OR f is increasing to the left of $x = -2$ and decreasing to the right of $x = -2$. OR

$f'(-2) = 0$ and $f''(-2) < 0$

0/2 for point other than -2 or 5

3/4 points between Part A and Part B if maximum and minimum are reversed

5-8



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the correct answer $x = -2$

1 point is earned for the correct reason

$f'(x)$ changes from positive to negative at $x = -2$ OR f is increasing to the left of $x = -2$ and decreasing to the right of $x = -2$.

OR $f'(-2) = 0$ and $f''(-2) < 0$

0/2 for point other than -2 or 4

3/4 points between Part A and Part B if reverse maximum and minimum

Part B

1 point is earned for the correct answer $x = 4$

$x = 4$

1 point is earned for the correct reason

$f'(x)$ changes from negative to positive at $x = 4$ OR f is decreasing to the left of $x = 4$ and increasing to the right of $x = 4$.

3/4 points between Part A and Part B if reverse maximum and minimum

<-1> for mismatching if clear



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the correct answer $x = 4$

$x = 4$

1 point is earned for the correct reason

$f'(x)$ changes from negative to positive at $x = 4$ OR f is decreasing to the left of $x = 4$ and increasing to the right of $x = 4$.

3/4 points between Part A and Part B if reverse maximum and minimum

<-1> for mismatching if clear

Part C

1 point is earned for the correct interval $(-1, 1)$

1 point is earned for the correct interval $(3, 5)$

5-8 $(-1, 1)$ and $(3, 5)$ 1 point is earned for the correct reason using f' f' is increasing on these intervals

Max 1/3 for one incorrect interval

0/3 for two incorrect intervals



0	1	2	3
---	---	---	---

The student response earns all of the following points:

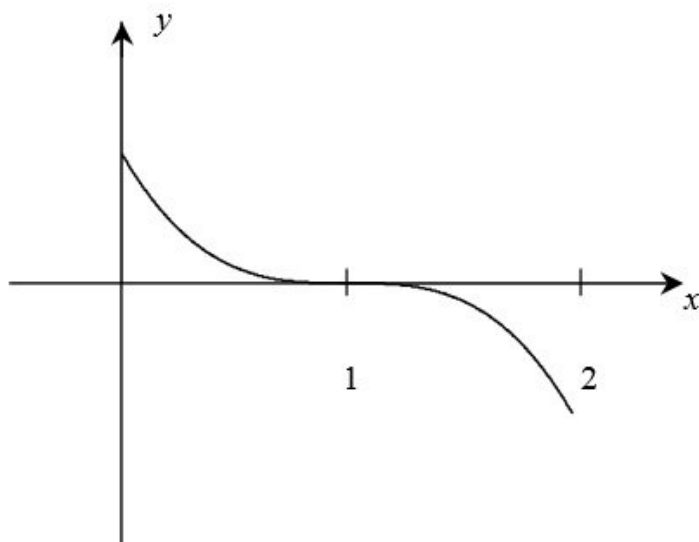
1 point is earned for the correct interval $(-1, 1)$ 1 point is earned for the correct interval $(3, 5)$ $(-1, 1)$ and $(3, 5)$ 1 point is earned for the correct reason using f' f' is increasing on these intervals

Max 1/3 for one incorrect interval

0/3 for two incorrect intervals

Part D

2 points are earned for the correct graph

<-1> $f(1) \neq 0$

<-1> not decreasing

<-1> incorrect concavity

5-8

<-1> $f'(1) \neq 0$

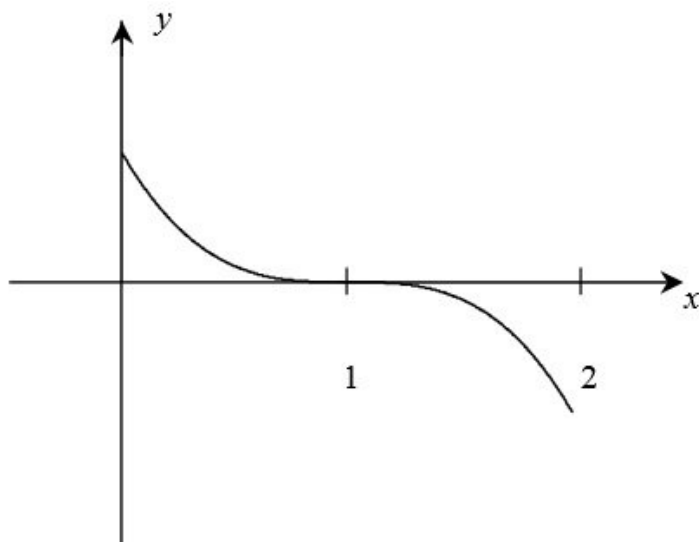
Incorrect use of standard notation deduct point only once



0	1	2
---	---	---

The student response earns all of the following points:

2 points are earned for the correct graph

<-1> $f(1) \neq 0$

<-1> not decreasing

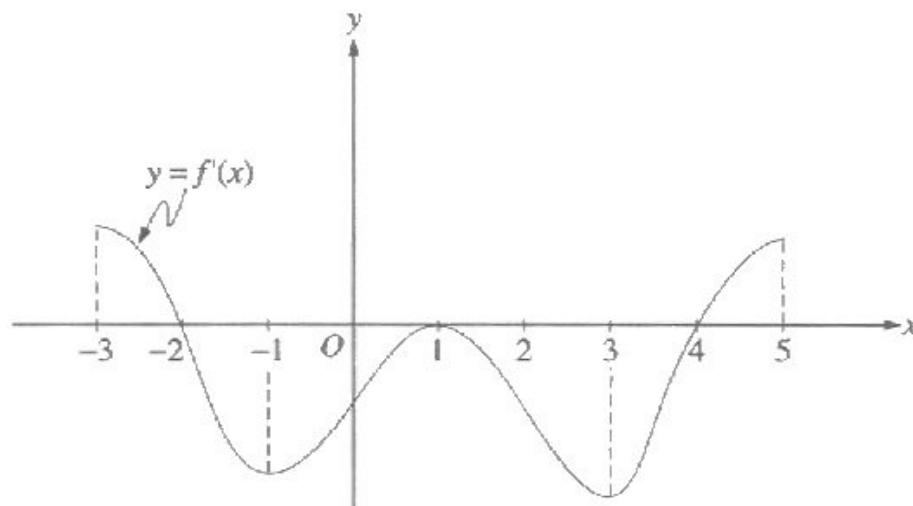
<-1> incorrect concavity

<-1> $f'(1) \neq 0$

Incorrect use of standard notation deduct point only once

5-8

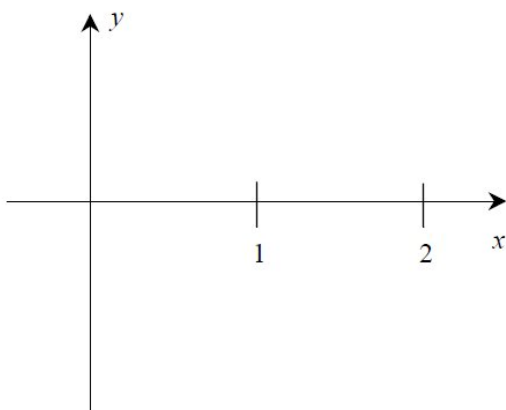
4.



Note: This is the graph of the derivative of f , not the graph of f .

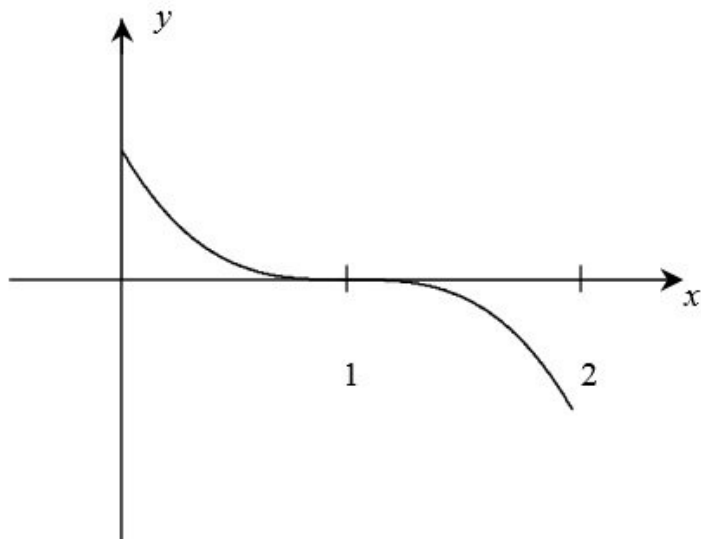
The figure above shows the graph of f' , the derivative of a function f . The domain of f is the set of all real numbers x such that $-3 < x < 5$.

Suppose that $f(1) = 0$. In the xy -plane provided, draw a sketch that shows the general shape of the graph of the function f on the open interval $0 < x < 2$.

**Part D**

2 points are earned for the correct graph

5-8

< -1 > $f(1) \neq 0$

< -1 > not decreasing

< -1 > incorrect concavity

< -1 > $f'(1) \neq 0$

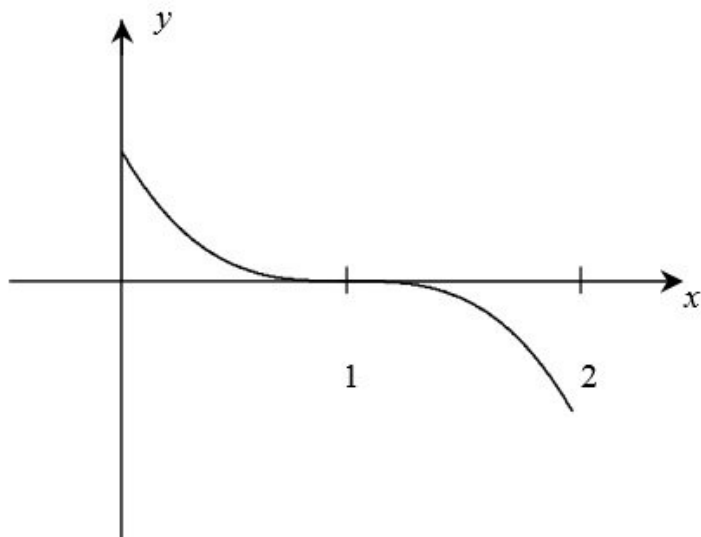
Incorrect use of standard notation deduct point only once

✓

0	1	2
---	---	---

The student response earns two of the following points:

2 points are earned for the correct graph

< -1 > $f(1) \neq 0$

5-8

< -1 > not decreasing

< -1 > incorrect concavity

< -1 > $f'(1) \neq 0$

Incorrect use of standard notation deduct point only once

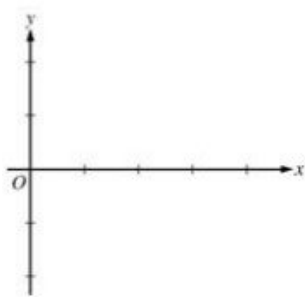
5-8

5.

x	0	$0 < x < 1$	1	$1 < x < 2$	2	$2 < x < 3$	3	$3 < x < 4$
$f(x)$	-1	Negative	0	Positive	2	Positive	0	Negative
$f'(x)$	4	Positive	0	Positive	DNE	Negative	-3	Negative
$f''(x)$	-2	Negative	0	Positive	DNE	Negative	0	Positive

Let f be a function that is continuous on the interval $[0, 4)$. The function f is twice differentiable except at $x = 2$. The function f and its derivatives have the properties indicated in the table above, where DNE indicates that the derivatives of f do not exist at $x = 2$.

- (a) For $0 < x < 4$, find all values of x at which f has a relative extremum. Determine whether f has a relative maximum or a relative minimum at each of these values. Justify your answer.
- (b) On the axes provided, sketch the graph of a function that has all the characteristics of f .



- (c) Let g be the function defined by $g(x) = \int_1^x f(t)dt$ on the open interval $(0, 4)$. For $0 < x < 4$, find all values of x at which g has a relative extremum. Determine whether g has a relative maximum or a relative minimum at each of these values. Justify your answer.

- (d) For the function g defined in part (c), find all values of x , for $0 < x < 4$, at which the graph of g has a point of inflection. Justify your answer.

Part A

1 point is earned for the relative extremum at $x = 2$

1 point is earned for the relative maximum with justification

f has a relative maximum at $x = 2$ because f' changes from positive to negative at $x = 2$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the relative extremum at $x = 2$

1 point is earned for the relative maximum with justification

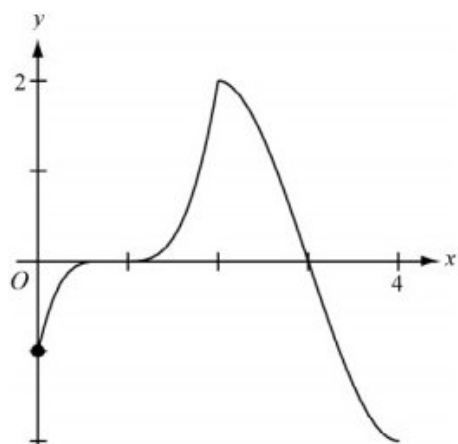
f has a relative maximum at $x = 2$ because f' changes from positive to negative at $x = 2$.

Part B

1 point is earned for points at $x = 0, 1, 2, 3$ and behavior at $(2, 2)$

5-8

1 point is earned for appropriate increasing/decreasing and concavity behavior



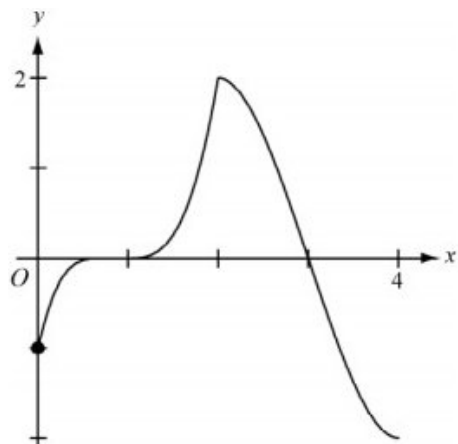
✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for points at $x = 0, 1, 2, 3$ and behavior at $(2, 2)$

1 point is earned for appropriate increasing/decreasing and concavity behavior



Part C

1 point is earned for: $g'(x) = f(x)$

1 point is earned for the critical points

1 point is earned for the answer with justification

$g'(x) = f(x) = 0$ at $x = 1, 3$.

g' changes from negative to positive at $x = 1$ so g has a relative minimum at $x = 1$. g' changes from positive to negative at $x = 3$ so g has a relative maximum at $x = 3$.

5-8



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for: $g'(x) = f(x)$

1 point is earned for the critical points

1 point is earned for the answer with justification

$g'(x) = f(x) = 0$ at $x = 1, 3$.

g' changes from negative to positive at $x = 1$ so g has a relative minimum at $x = 1$. g' changes from positive to negative at $x = 3$ so g has a relative maximum at $x = 3$.

Part D

1 point is earned for: $x = 2$

1 point is earned for the answer with justification

The graph of g has a point of inflection at $x = 2$ because $g'' = f'$ changes sign at $x = 2$.



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for: $x = 2$

1 point is earned for the answer with justification

The graph of g has a point of inflection at $x = 2$ because $g'' = f'$ changes sign at $x = 2$.

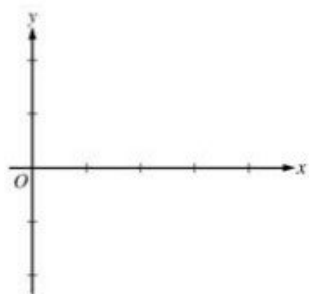
5-8

6.

x	0	$0 < x < 1$	1	$1 < x < 2$	2	$2 < x < 3$	3	$3 < x < 4$
$f(x)$	-1	Negative	0	Positive	2	Positive	0	Negative
$f'(x)$	4	Positive	0	Positive	DNE	Negative	-3	Negative
$f''(x)$	-2	Negative	0	Positive	DNE	Negative	0	Positive

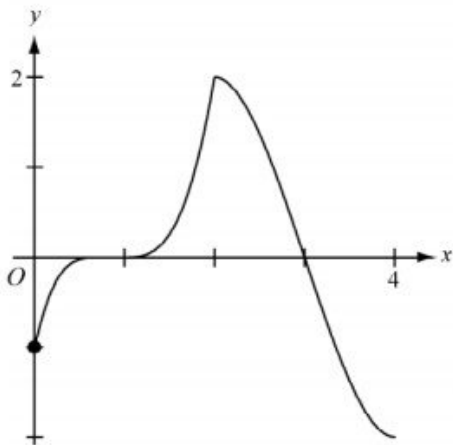
Let f be a function that is continuous on the interval $[0, 4)$. The function f is twice differentiable except at $x = 2$. The function f and its derivatives have the properties indicated in the table above, where DNE indicates that the derivatives of f do not exist at $x = 2$.

On the axes provided, sketch the graph of a function that has all the characteristics of f .

**Part B**

1 point is earned for points at $x = 0, 1, 2, 3$ and behavior at $(2, 2)$

1 point is earned for appropriate increasing/decreasing and concavity behavior



✓

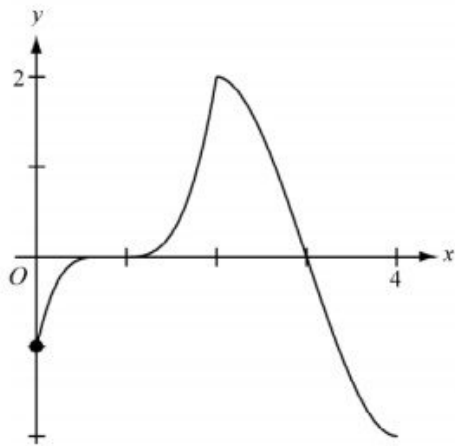
0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for points at $x = 0, 1, 2, 3$ and behavior at $(2, 2)$

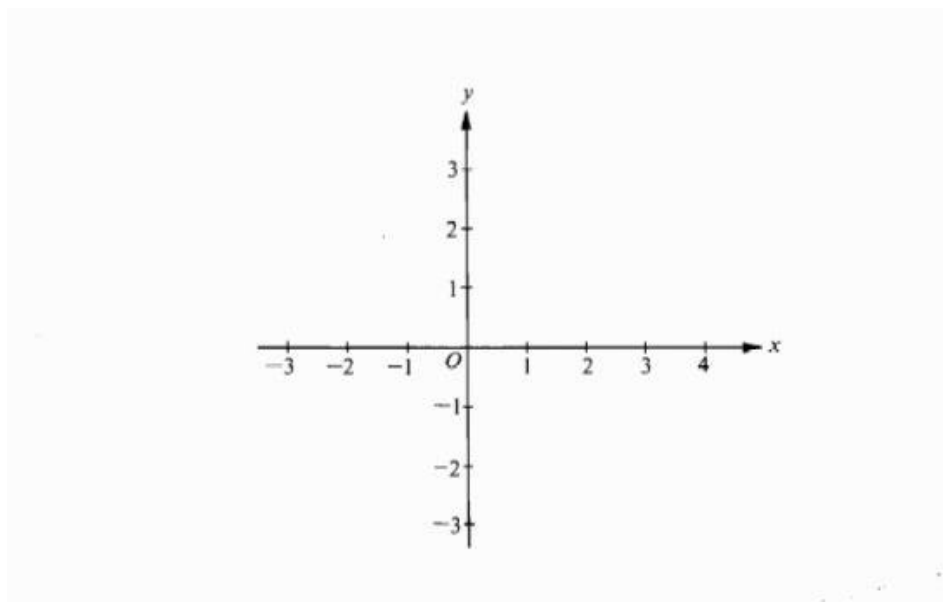
1 point is earned for appropriate increasing/decreasing and concavity behavior

5-8



5-8

7. Let f be the function defined by $f(x) = 2xe^{-x}$ for all real numbers x .
- Write an equation of the horizontal asymptote for the graph of f .
 - Find the x -coordinate of each critical point of f . For each such x , determine whether $f(x)$ is a relative maximum, a relative minimum, or neither.
 - For what values of x is the graph of f concave down?
 - Using the results found in parts (a), (b), and (c), sketch the graph of $y = f(x)$ in the xy -plane provided below.

**Part A**

1 point is earned for the correct equation

$$y = 0$$



0	1
---	---

The student response earns all of the following points:

1 point is earned for the correct equation

$$y = 0$$

Part B

1 point is earned for correct derivative

1 point is earned for critical value for f'

1 point is earned for identifying critical point as relative maximum

$$\begin{aligned} f'(x) &= 2(-xe^{-x} + e^{-x}) \\ &= 2e^{-x}(1 - x) \end{aligned}$$

5-8critical point at $x = 1$ relative maximum at $x = 1$ 

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for correct derivative

1 point is earned for critical value for f'

1 point is earned for identifying critical point as relative maximum

$$\begin{aligned} f'(x) &= 2(-xe^{-x} + e^{-x}) \\ &= 2e^{-x}(1 - x) \end{aligned}$$

critical point at $x = 1$ relative maximum at $x = 1$ **Part C**1 point is earned for correct $f''(x)$ for $f'(x)$ found in (b)

1 point is earned for correct interval

$$\begin{aligned} f''(x) &= 2e^{-x}(-1) + (-2e^{-x})(1 - x) \\ &= 2e^{-x}(x - 2) \end{aligned}$$

Concave down when

$$\begin{aligned} 2e^{-x}(x - 2) &< 0 \\ (x - 2) &< 0 \\ x &< 2 \end{aligned}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for correct $f''(x)$ for $f'(x)$ found in (b)

1 point is earned for correct interval

5-8

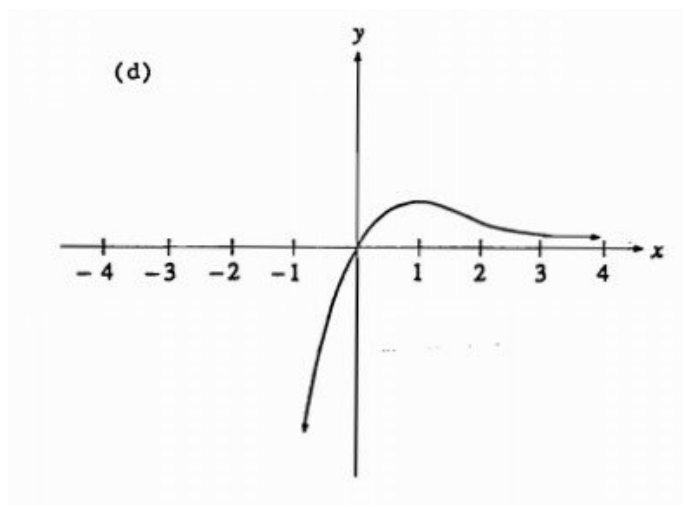
$$\begin{aligned}
 f''(x) &= 2e^{-x}(-1) + (-2e^{-x})(1-x) \\
 &= 2e^{-x}(x-2)
 \end{aligned}$$

Concave down when

$$\begin{aligned}
 2e^{-x}(x-2) &< 0 \\
 (x-2) &< 0 \\
 x &< 2
 \end{aligned}$$

Part D

3 points are earned for the graph consistent with information found in (a), (b), and (c)



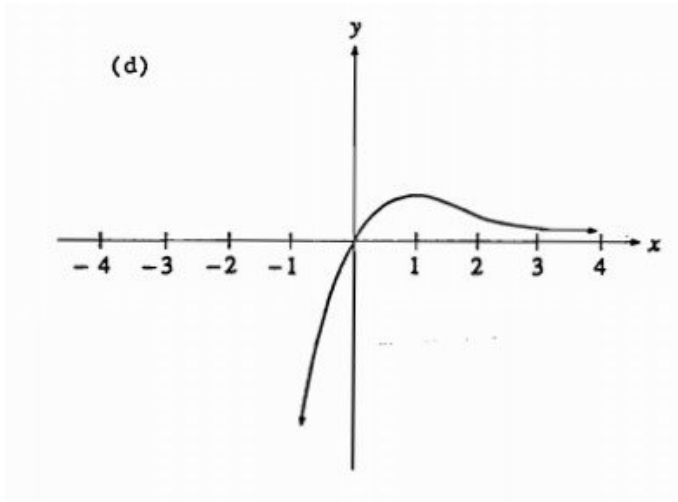
✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

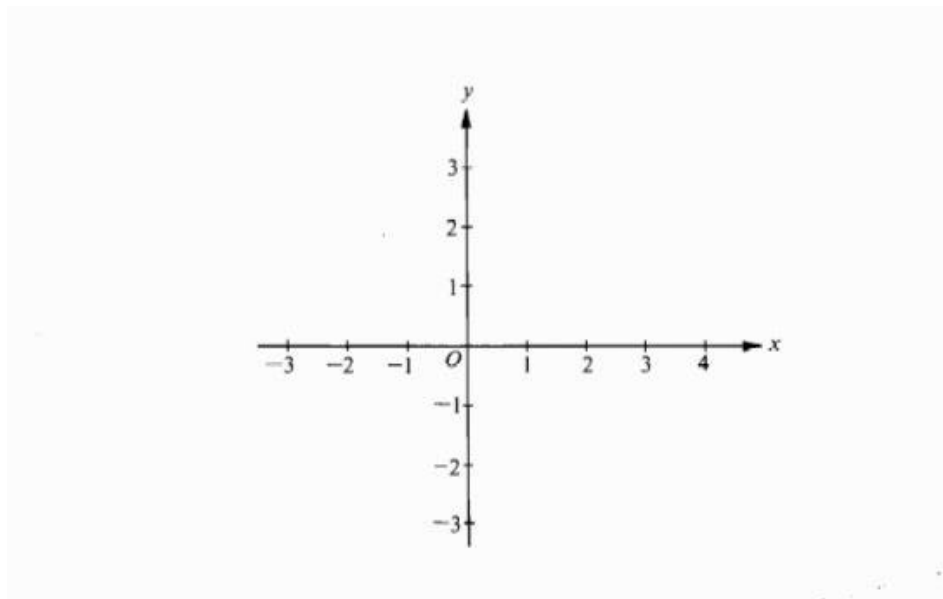
3 points are earned for the graph consistent with information found in (a), (b), and (c)

5-8

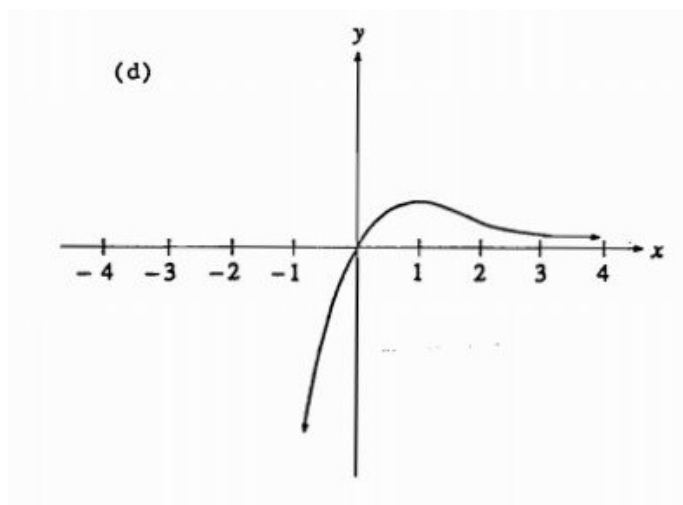


5-8

8. Let F be the function defined by $f(x) = 2xe^{-x}$ for all real numbers x .
Using the results found in parts (a), (b), and (c), sketch the graph of $y = f(x)$ in the xy -plane provided below.

**Part D**

3 points are earned for the graph consistent with information found in (a), (b), and (c)



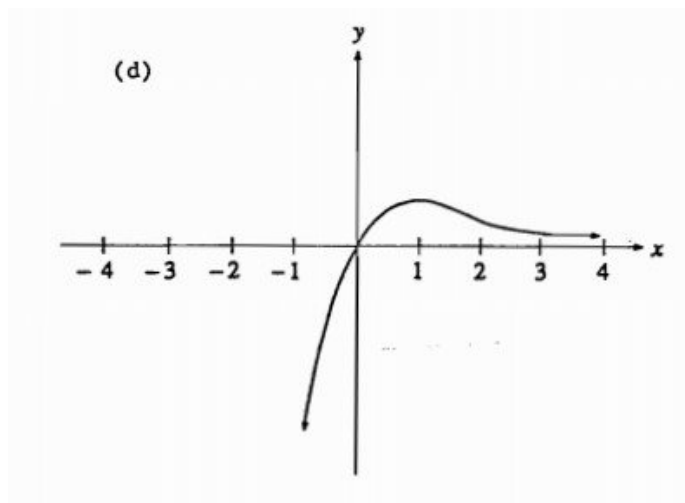
✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

3 points are earned for the graph consistent with information found in (a), (b), and (c)

5-8

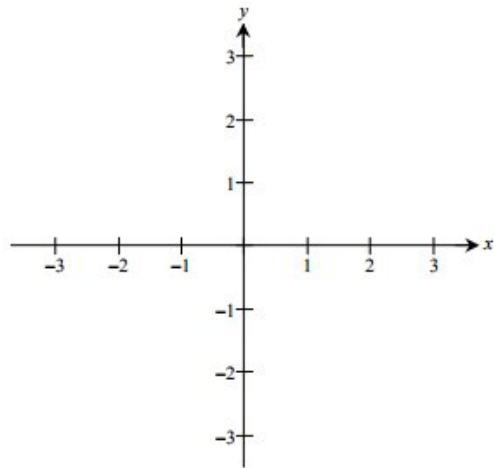


5-8

9. Let f be a function that is even and continuous on the closed interval $[-3,3]$. The function f and its derivatives have the properties indicated in the table below.

x	0	$0 < x < 1$	1	$1 < x < 2$	2	$2 < x < 3$
$f(x)$	1	Positive	0	Negative	-1	Negative
$f'(x)$	Undefined	Negative	0	Negative	Undefined	Positive
$f''(x)$	Undefined	Positive	0	Negative	Undefined	Negative

- (a) Find the x -coordinate of each point at which f attains an absolute maximum value or an absolute minimum value. For each x -coordinate you give, state whether f attains an absolute maximum or an absolute minimum.
- (b) Find the x -coordinate of each point of inflection on the graph of f . Justify your answer.
- (c) In the xy -plane provided below, sketch the graph of a function with all the given characteristics of f .



Part A

- 1 point is earned for first point
- 1 point is earned for second point
- 1 point is earned for third point
- 1 point is **deducted** for each additional point or incorrect point
- Absolute maximum at $x = 0$
- Absolute minimum at $x = \pm 2$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

- 1 point is earned for first point
- 1 point is earned for second point
- 1 point is earned for third point
- 1 point is **deducted** for each additional point or incorrect point

5-8

Absolute maximum at $x = 0$

Absolute minimum at $x = \pm 2$

Part B

1 point is earned for first point identified

1 point is earned for second point identified

1 point is earned for justification (required on $[0,3)$ only)

1 point is **deducted** for each additional point or incorrect point

The points of inflection are at $x = \pm 1$ because the sign of $f''(x)$ changes at $x = 1$ and f is even.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for first point identified

1 point is earned for second point identified

1 point is earned for justification (required on $[0,3)$ only)

1 point is **deducted** for each additional point or incorrect point

The points of inflection are at $x = \pm 1$ because the sign of $f''(x)$ changes at $x = 1$ and f is even.

Part C

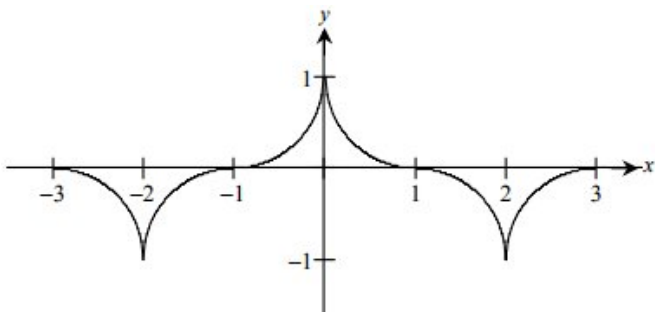
3 points are earned for sketch without error

1 point is **deducted** for each error:

- symmetry
- slopes on $[0,3)$
- concavity on $[0,3)$
- discontinuity on $[0,3)$
- $f(0), f(1), f(2)$ incorrect or extra intercept on $[0,3)$

Note: No more than 1 point can be lost in any one category.

5-8



✓

0	1	2	3
---	---	---	---

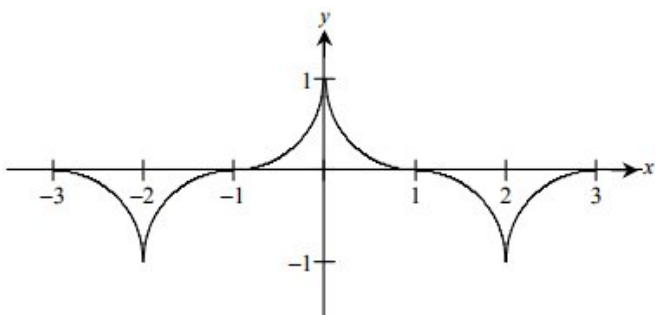
The student response earns all of the following points:

3 points are earned for sketch without error

1 point is **deducted** for each error:

- symmetry
- slopes on $[0,3)$
- concavity on $[0,3)$
- discontinuity on $[0,3)$
- $f(0), f(1), f(2)$ incorrect or extra intercept on $[0,3)$

Note: No more than 1 point can be lost in any one category.

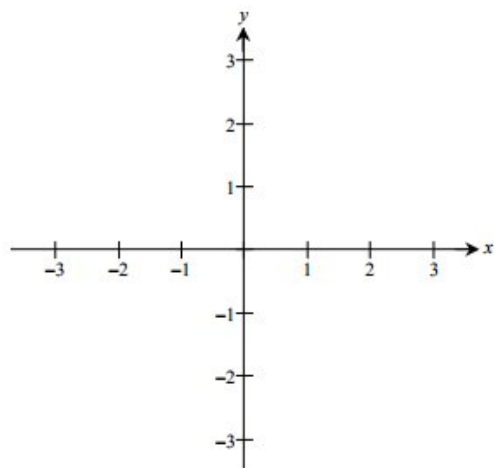


5-8

10. Let f be a function that is even and continuous on the closed interval $[-3, 3]$. The function f and its derivatives have the properties indicated in the table below.

x	0	$0 < x < 1$	1	$1 < x < 2$	2	$2 < x < 3$
$f(x)$	1	Positive	0	Negative	-1	Negative
$f'(x)$	Undefined	Negative	0	Negative	Undefined	Positive
$f''(x)$	Undefined	Positive	0	Negative	Undefined	Negative

In the xy -plane provided below, sketch the graph of a function with all the given characteristics of f .



Part C

3 point is earned for sketch without error

1 point is deducted for each error:

symmetry

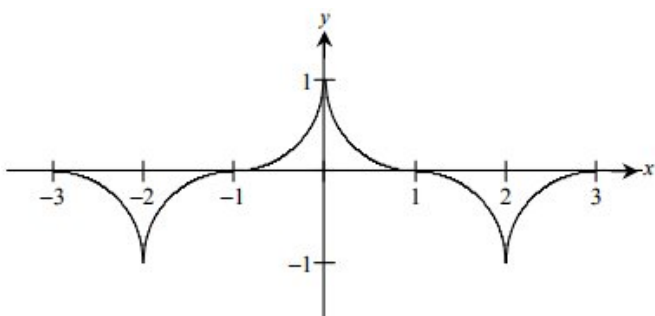
slopes on $[0, 3)$

concavity on $[0, 3)$

discontinuity on $[0, 3)$

$f(0), f(1), f(2)$ incorrect or extra intercept on $[0, 3)$

Note: No more than 1 point can be lost in any one category.



5-8



0	1	2	3
---	---	---	---

The student response earns three of the following points:

3 point is earned for sketch without error

1 point is deducted for each error:

symmetry

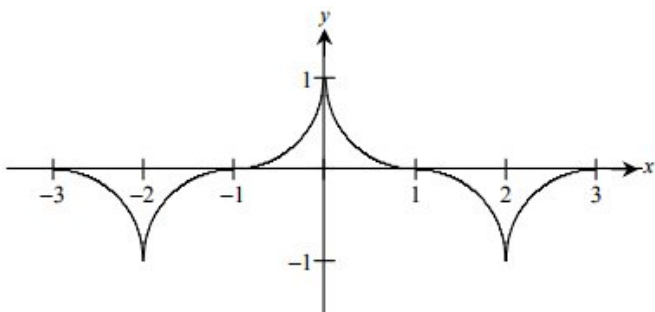
slopes on $[0,3)$

concavity on $[0,3)$

discontinuity on $[0,3)$

$f(0), f(1), f(2)$ incorrect or extra intercept on $[0,3)$

Note: No more than 1 point can be lost in any one category.



5-8

11.  Let f be the function given by $f(x) = 2xe^{2x}$.

(a) Find $\lim_{x \rightarrow -\infty} f(x)$ and $\lim_{x \rightarrow \infty} f(x)$.

(b) Find the absolute minimum value of f . Justify that your answer is an absolute minimum.

(c) What is the range of f ?

(d) Consider the family of functions defined by $y = bxe^{bx}$, where b is a nonzero constant. Show that the absolute minimum value of bxe^{bx} is the same for all nonzero values of b .

Part A

1 point is earned for 0 as $x \rightarrow -\infty$

1 point is earned for ∞ or DNE as $x \rightarrow \infty$

$$\lim_{x \rightarrow -\infty} 2xe^{2x} = 0$$

$$\lim_{x \rightarrow \infty} 2xe^{2x} = \infty \text{ or DNE}$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for 0 as $x \rightarrow -\infty$

1 point is earned for ∞ or DNE as $x \rightarrow \infty$

$$\lim_{x \rightarrow -\infty} 2xe^{2x} = 0$$

$$\lim_{x \rightarrow \infty} 2xe^{2x} = \infty \text{ or DNE}$$

Part B

1 point is earned if solves $f'(x) = 0$

1 point is earned if evaluates f at student's critical point

0/1 if not local minimum from student's derivative

1 point is earned if justifies absolute minimum value

0/1 for a local argument

0/1 without explicit symbolic derivative

Note: 0/3 if no absolute minimum based on student's derivative

$$f'(x) = 2e^{2x} + 2x \cdot 2 \cdot e^{2x} = 2e^{2x}(1 + 2x) = 0$$

$$\text{if } x = -1/2$$

$$f(-1/2) = -1/e \text{ or } -0.368 \text{ or } -0.367$$

5-8

$-1/e$ is an absolute minimum value because:

(i) $f'(x) < 0$ for all $x < -1/2$ and $f'(x) > 0$ for all $x > -1/2$

-or-

(ii) $f'(x)$ $\begin{array}{c} - \quad + \\ \hline -1/2 \end{array}$

and $x = -1/2$ is the only critical number

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned if solves $f'(x) = 0$

1 point is earned if evaluates f at student's critical point

0/1 if not local minimum from student's derivative

1 point is earned if justifies absolute minimum value

0/1 for a local argument

0/1 without explicit symbolic derivative

Note: 0/3 if no absolute minimum based on student's derivative

$$f'(x) = 2e^{2x} + 2x \cdot 2 \cdot e^{2x} = 2e^{2x}(1 + 2x) = 0$$

if $x = -1/2$

$$f(-1/2) = -1/e \text{ or } -0.368 \text{ or } -0.367$$

$-1/e$ is an absolute minimum value because:

(i) $f'(x) < 0$ for all $x < -1/2$ and $f'(x) > 0$ for all $x > -1/2$

-or-

(ii) $f'(x)$ $\begin{array}{c} - \quad + \\ \hline -1/2 \end{array}$

and $x = -1/2$ is the only critical number

Part C

1 point is earned for the answer

Note: must include the left-hand endpoint; exclude the right-hand "endpoint"

Range of $f = [-1/e, \infty)$ or $[-0.367, \infty)$ or $[-0.368, \infty)$

5-8



0	1
---	---

The student response earns all of the following points:

1 point is earned for the answer

Note: must include the left-hand endpoint; exclude the right-hand “endpoint”

Range of $f = [-1/e, \infty)$ or $[-0.367, \infty)$ or $[-0.368, \infty)$

Part D

1 point is earned if sets $y' = be^{bx}(1 + bx) = 0$

1 point is earned for solving student's $y' = 0$

1 point is earned if evaluates y at a critical number and gets a value independent of b

Note: 0/3 if only considering specific values of b

$$y' = be^{bx} + b^2xe^{bx} = be^{bx}(1 + bx) = 0$$

if $x = -1/b$

At $x = -1/b$, $y = -1/e$

y has an absolute minimum value of $-1/e$ for all nonzero b



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned if sets $y' = be^{bx}(1 + bx) = 0$

1 point is earned for solving student's $y' = 0$

1 point is earned if evaluates y at a critical number and gets a value independent of b

Note: 0/3 if only considering specific values of b

$$y' = be^{bx} + b^2xe^{bx} = be^{bx}(1 + bx) = 0$$

if $x = -1/b$

At $x = -1/b$, $y = -1/e$

y has an absolute minimum value of $-1/e$ for all nonzero b

5-8

12. Let f be the function given by $f(x) = 2xe^{2x}$.
What is the range of f ?

Part C

The student response earns none of the following points:

1 point is earned for the answer

Note: must include the left-hand endpoint; exclude the right-hand “endpoint”

Range of $f = [-1/e, \infty)$
 or $[-0.367, \infty)$
 or $[-0.368, \infty)$



0	1
---	---

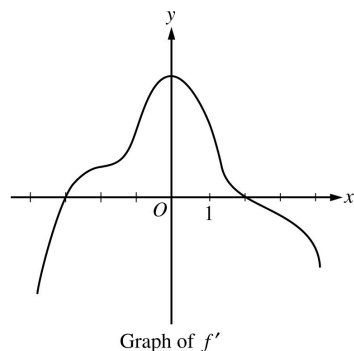
The student response earns none of the following points:

1 point is earned for the answer

Note: must include the left-hand endpoint; exclude the right-hand “endpoint”

Range of $f = [-1/e, \infty)$
 or $[-0.367, \infty)$
 or $[-0.368, \infty)$

13.



The graph of f' , the derivative of the function f , is shown above. Which of the following statements must be true?

- I. f has a relative minimum at $x = -3$.
- II. The graph of f has a point of inflection at $x = -2$.
- III. The graph of f is concave down for $0 < x < 4$.

- (A) I only
- (B) II only
- (C) III only
- (D) I and II only

(E) I and III only

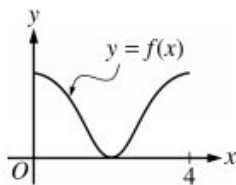


5-8

14. For $x > 0$, f is a function such that $f'(x) = \frac{\ln x}{x}$ and $f''(x) = \frac{1 - \ln x}{x^2}$. Which of the following is true?

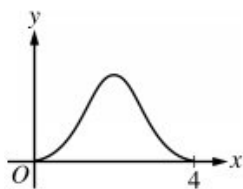
- (A) f is decreasing for $x > 1$, and the graph of f is concave down for $x > e$.
 (B) f is decreasing for $x > 1$, and the graph of f is concave up for $x > e$.
 (C) f is increasing for $x > 1$, and the graph of f is concave down for $x > e$. ✓
 (D) f is increasing for $x > 1$, and the graph of f is concave up for $x > e$.
 (E) f is increasing for $0 < x < e$, and the graph of f is concave down for $0 < x < e^{3/2}$.

15.

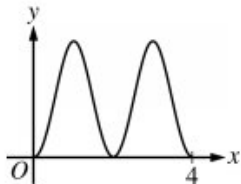


The graph of $y = f(x)$ on the closed interval $[0, 4]$ is shown above. Which of the following could be the graph $y = f'(x)$?

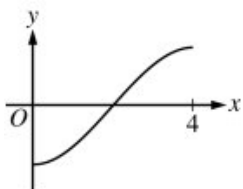
(A)



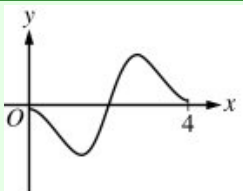
(B)



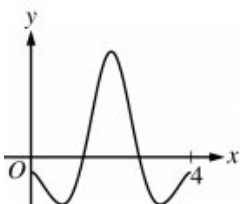
(C)



(D)

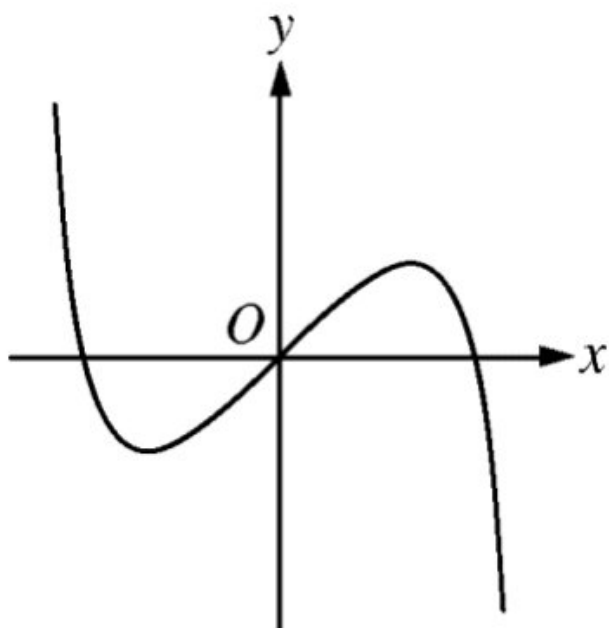


(E)



5-8

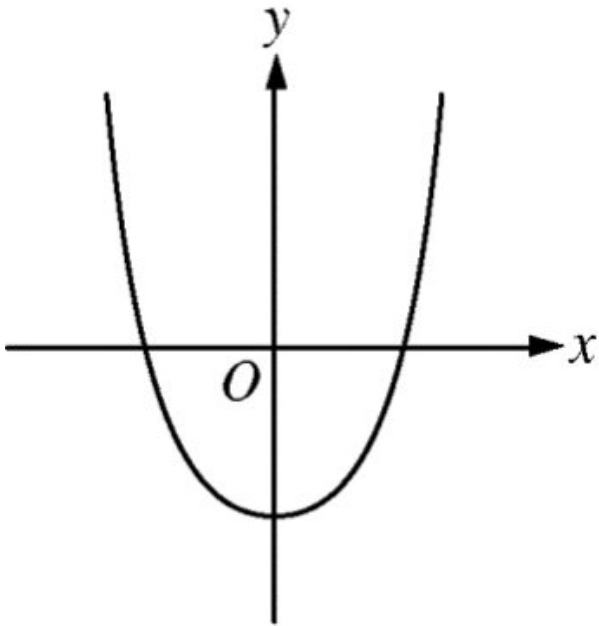
16.

Graph of f

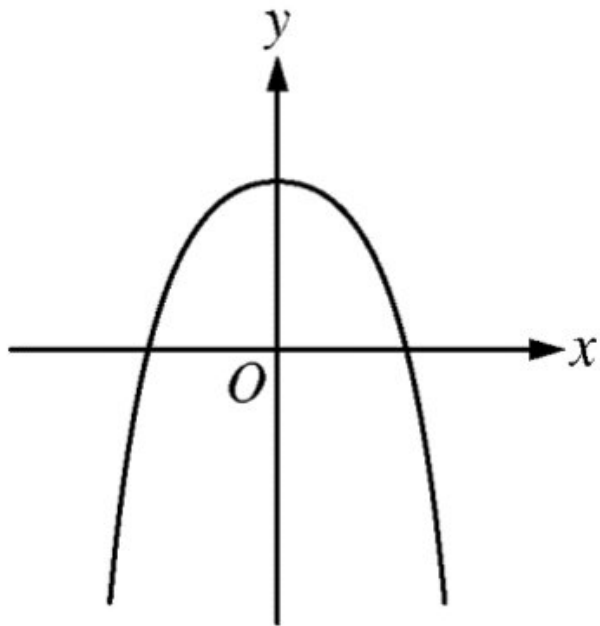
The graph of the function f is shown in the figure above. Which of the following could be the graph of f' , the derivative of f ?

5-8

(A)

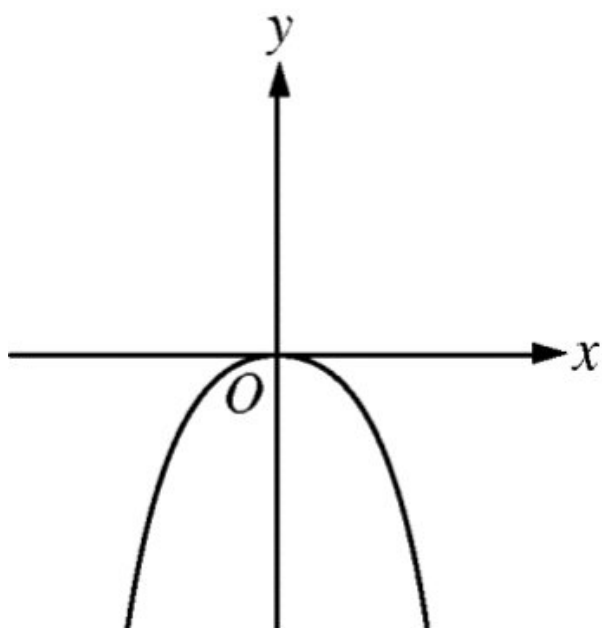


(B)

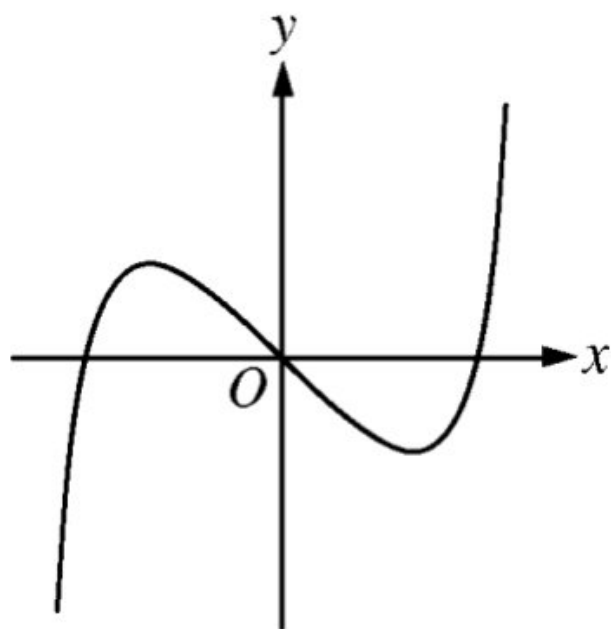


5-8

(C)

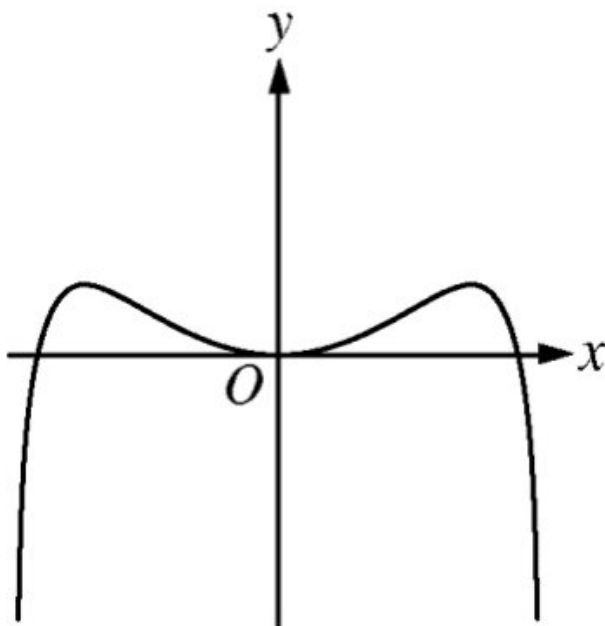


(D)

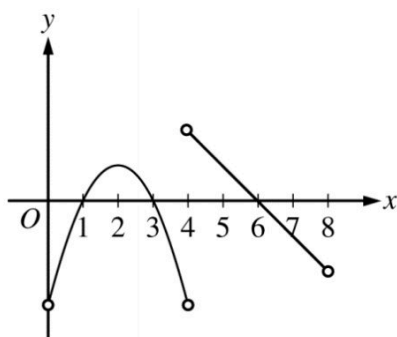


5-8

(E)

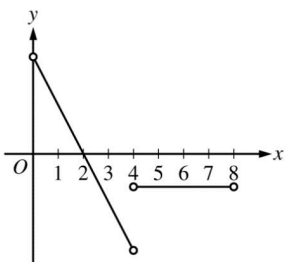


17.

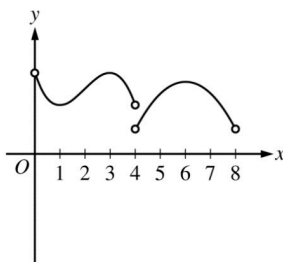
Graph of f'

The graph of f' , the derivative of a function f , is shown above. Which of the following could be the graph of f ?

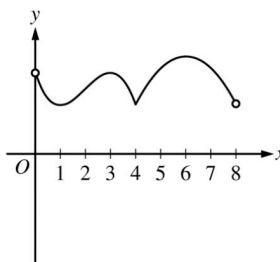
I.



II.



III.



- (A) I only
 (B) II only
 (C) III only
 (D) I and II only
 (E) II and III only



5-8

18. For all x in the closed interval $[2,5]$, the function f has a positive first derivative and a negative second derivative. Which of the following could be a table of values for f ?

(A)

x	$f(x)$
2	7
3	9
4	12
5	16

(B)

x	$f(x)$
2	7
3	11
4	14
5	16



(C)

x	$f(x)$
2	16
3	12
4	9
5	7

(D)

x	$f(x)$
2	16
3	14
4	11
5	7

(E)

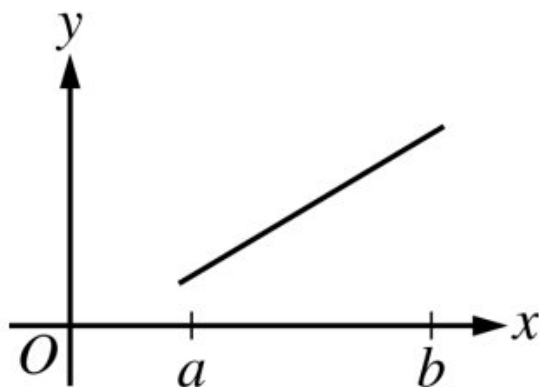
x	$f(x)$
2	16
3	13
4	10
5	7

5-8

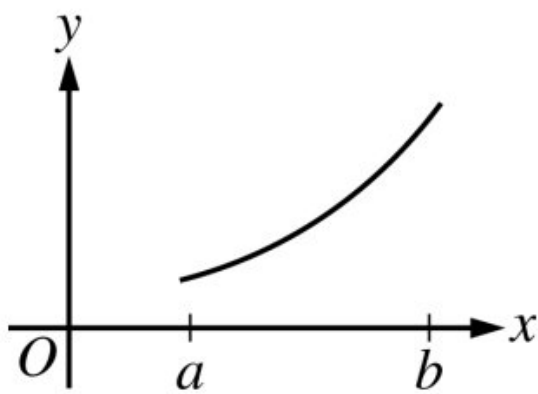
19. Let f be a twice-differentiable function on the open interval (a,b) . If $f'(x) > 0$ on (a,b) and $f''(x) < 0$ on (a,b) , which of the following could be the graph of f ?

5-8

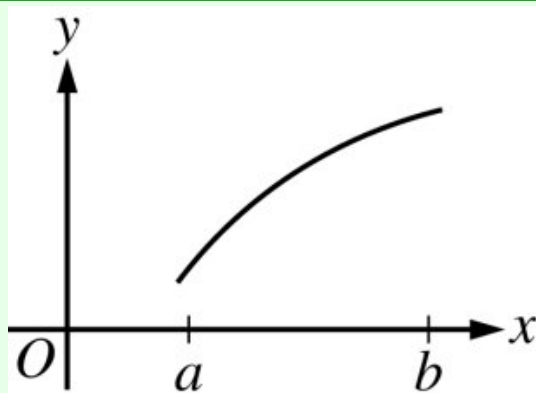
(A)



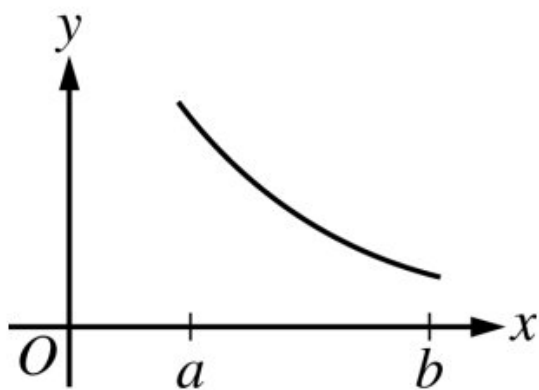
(B)



(C)

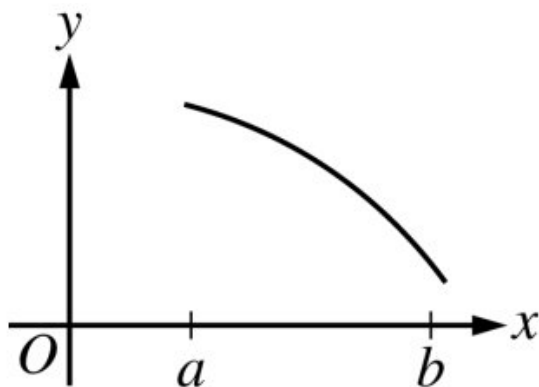


(D)



5-8

(E)



20.

x	1	2	3	4	5
$f(x)$	9	4	0	-3	-5

The table above gives values of a function f at selected values of x .

If f is twice-differentiable on the interval $1 \leq x \leq 5$, which of the following statements could be true?

(A) f' is negative and decreasing for $1 \leq x \leq 5$.

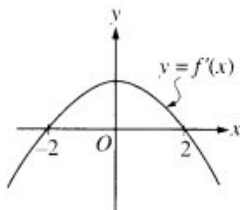
(B) f' is negative and increasing for $1 \leq x \leq 5$. ✓

(C) f' is positive and decreasing for $1 \leq x \leq 5$.

(D) f' is positive and increasing for $1 \leq x \leq 5$.

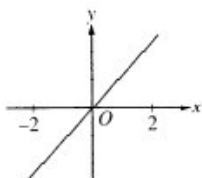
5-8

21.

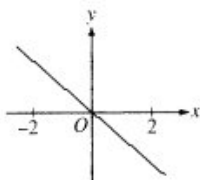


The graph of the derivative of f is shown in the figure above. Which of the following could be the graph of f ?

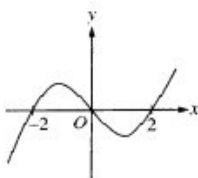
(A)



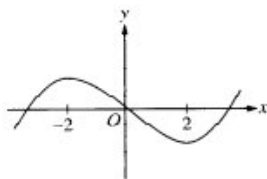
(B)



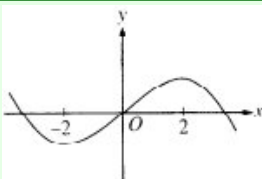
(C)



(D)

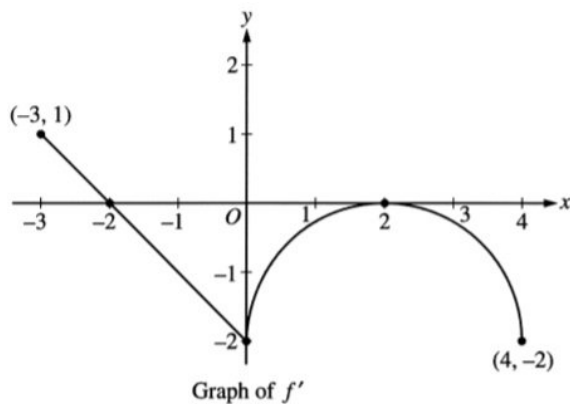


(E)



5-8

22.



Let f be a function defined on the closed interval $-3 \leq x \leq 4$ with $f(0) = 3$. The graph of f' , the derivative of f , consists of one line segment and a semicircle, as shown above.

Find the x -coordinate of each point of inflection of the graph of f on the open interval $-3 < x < 4$. Justify your answer.

Part B

1 point is earned for the correct answers $x = 0$ and $x = 2$ only

1 point is earned for the justification

$x = 0$ and $x = 2$

f' changes from decreasing to increasing at $x = 0$ and from increasing to decreasing at $x = 2$



0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for the correct answers $x = 0$ and $x = 2$ only

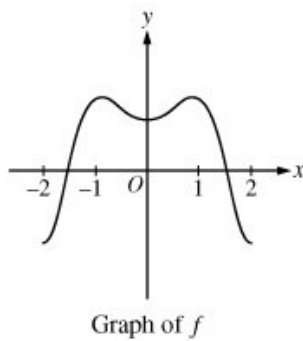
1 point is earned for the justification

$x = 0$ and $x = 2$

f' changes from decreasing to increasing at $x = 0$ and from increasing to decreasing at $x = 2$

5-8

23.

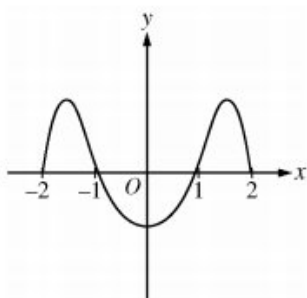


The graph of the function f is shown above for $-2 \leq x \leq 2$.

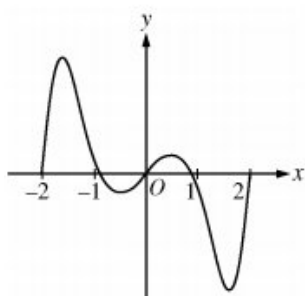
Which of the following could be the graph of an antiderivative of f ?

5-8

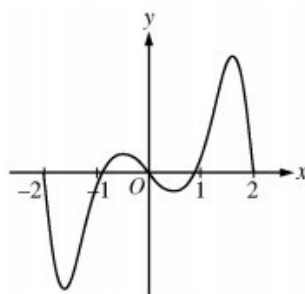
(A)



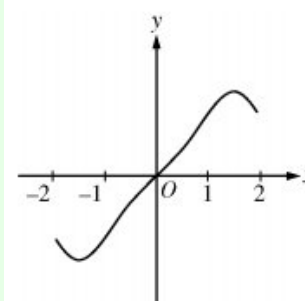
(B)



(C)

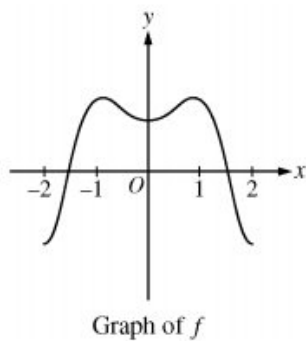


(D)



5-8

24.

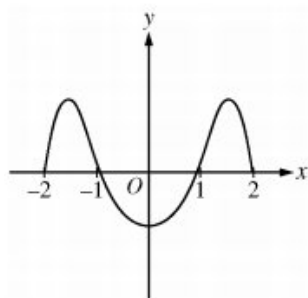


The graph of the function f is shown above for $-2 \leq x \leq 2$.

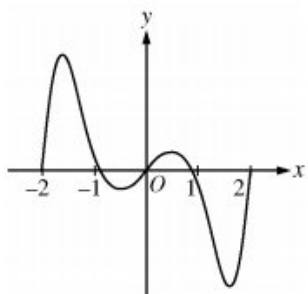
Which of the following could be the graph of an antiderivative of f ?

5-8

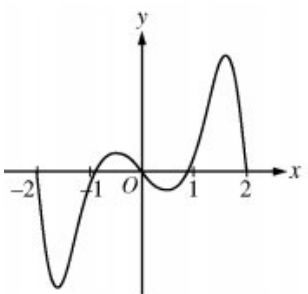
(A)



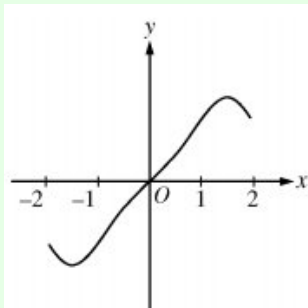
(B)



(C)



(D)



5-8

25. The function f is continuous on the closed interval and twice differentiable on the open interval $(2,4)$. If $f'(3) = 2$ and $f''(x) < 0$ on the open interval $(2,4)$, which of the following could be a table of values for f ?

(A)

x	$f(x)$
2	2.5
3	5
4	6.5



(B)

x	$f(x)$
2	2.5
3	5
4	7

(C)

x	$f(x)$
2	3
3	5
4	6.5

(D)

x	$f(x)$
2	3
3	5
4	7

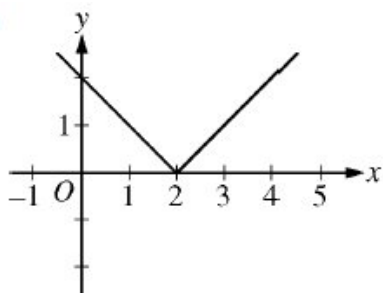
(E)

x	$f(x)$
2	3.5
3	5
4	7.5

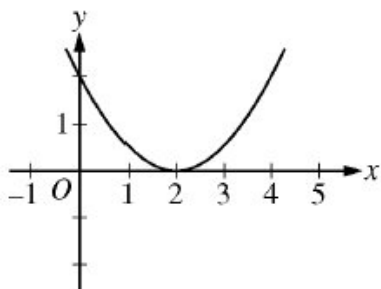
5-8

26. If $f'(x) = |x - 2|$ which of the following could be the graph of $y = f(x)$?

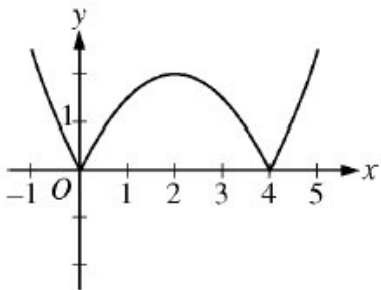
(A)



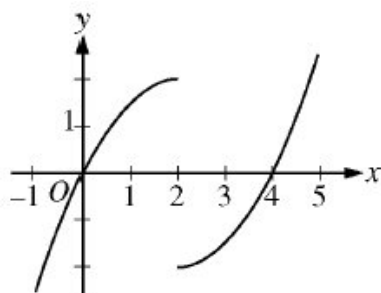
(B)



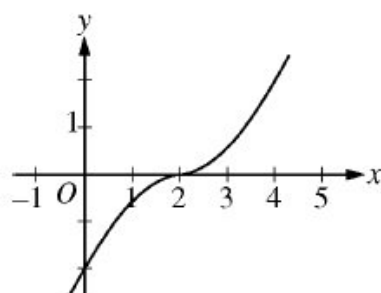
(C)



(D)



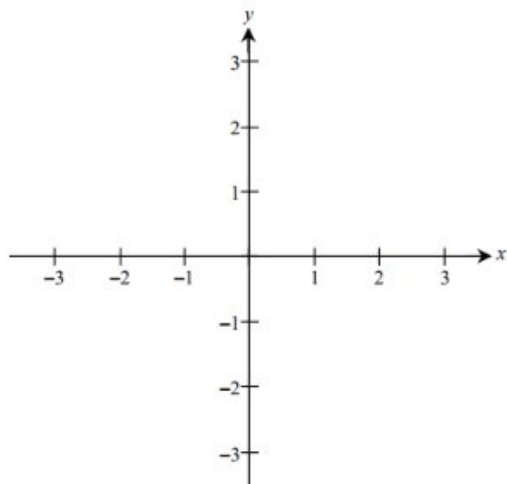
(E)



5-8

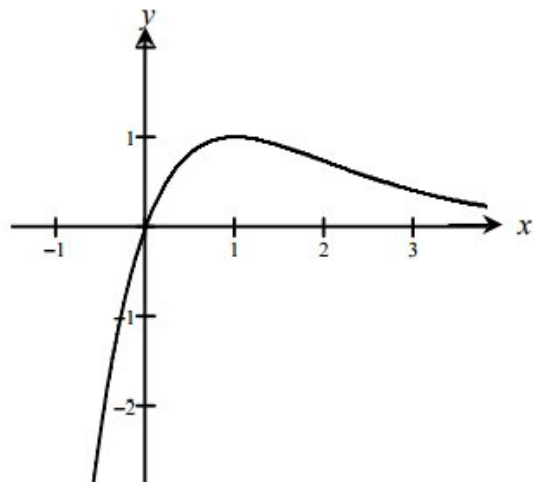
27. Let f be the function defined by $f(x) = xe^{1-x}$ for all real numbers x .

Using the results found in parts (a), (b), and (c), sketch the graph of f in the xy -plane provided below. (Indicate all intercepts.)

**Part D**

3 points are earned for answer without mistake

1 point is **deducted** for each error: intercept, domain, increasing, decreasing, range, inflection points.



✓

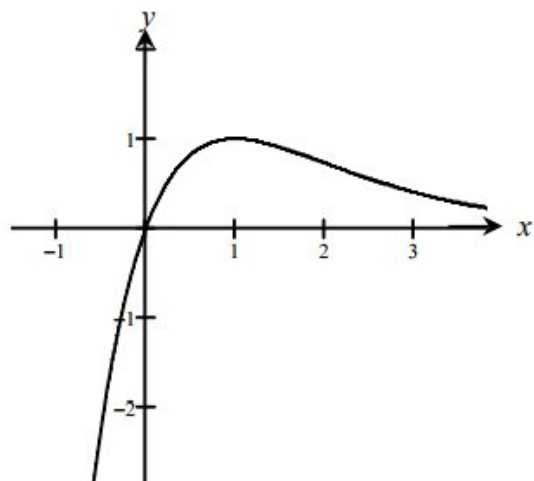
0	1	2	3
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The student response earns three of the following points:

3 points are earned for answer without mistake

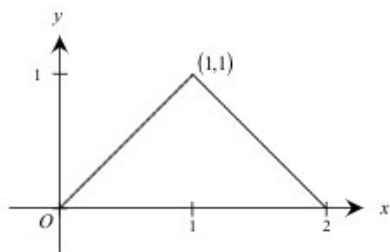
1 point is **deducted** for each error: intercept, domain, increasing, decreasing, range, inflection points.

5-8



5-8

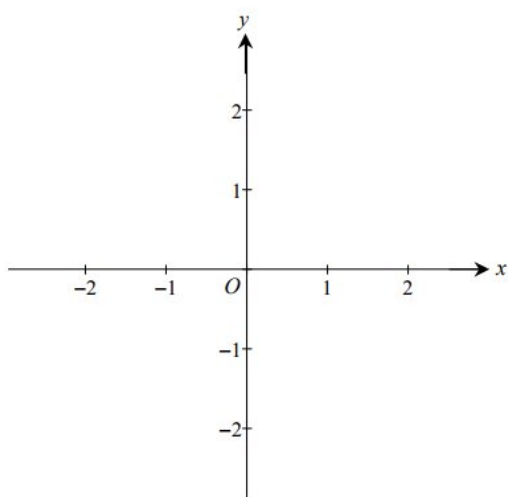
28.



Note: This is the graph of the derivative of f ,
not the graph of f .

f' , the derivative of f . The domain of f is the set of all x such that $0 < x < 2$.

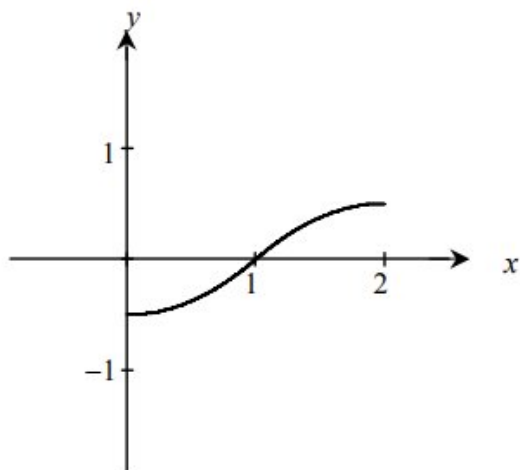
In the xy -plane provided below, sketch the graph of $y=f(x)$

**Part C**

1 point is earned for increasing and continuous on $(0, 2)$

1 point is earned for concave up on $(0, 1)$ and concave down on $(1, 2)$

1 point is earned for $f(1) = 0$



5-8



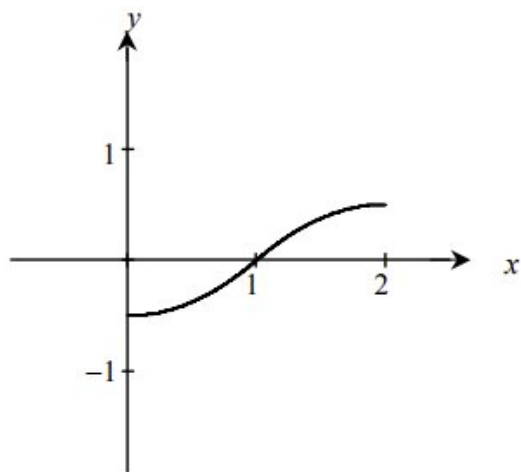
0	1	2	3
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The student response earns three of the following points:

1 point is earned for increasing and continuous on $(0,2)$

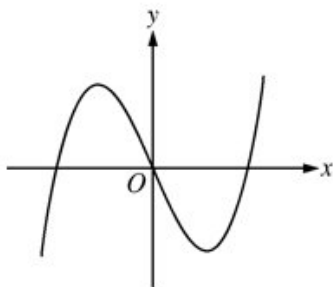
1 point is earned for concave up on $(0,1)$ and concave down on $(1,2)$

1 point is earned for $f(1) = 0$



5-8

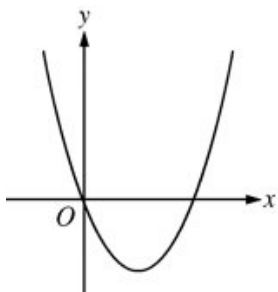
29.

Graph of f

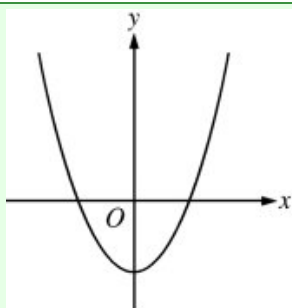
The graph of a function f is shown above. Which of the following could be the graph of f' , the derivative of f ?

5-8

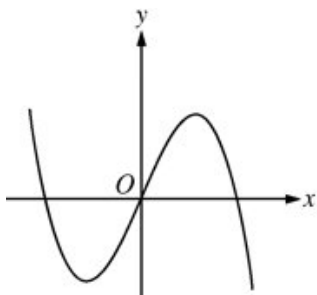
(A)



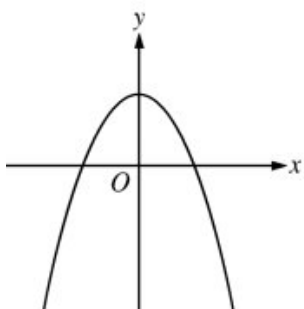
(B)



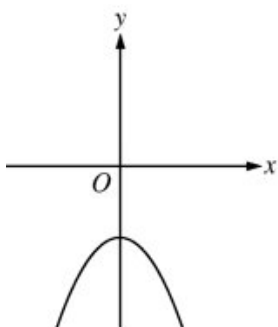
(C)



(D)

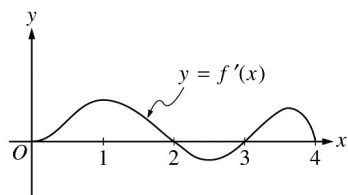


(E)



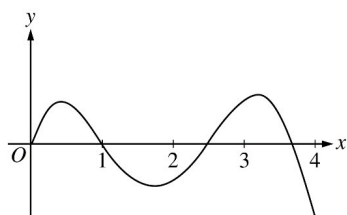
5-8

30.

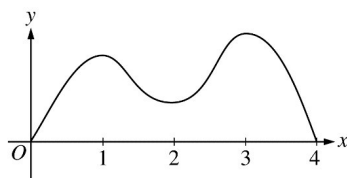


The figure above shows the graph of f' the derivative of the function f . If $f(0) = 0$, which of the following could be the graph of f ?

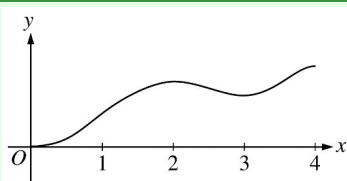
(A)



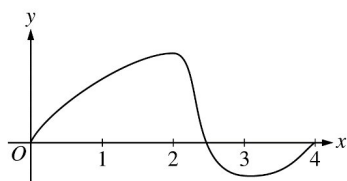
(B)



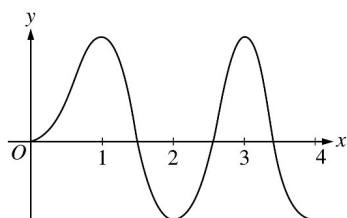
(C)



(D)

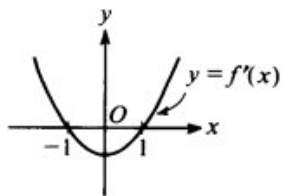


(E)



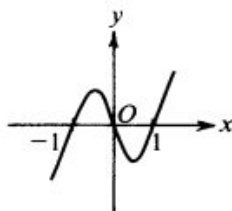
5-8

31.

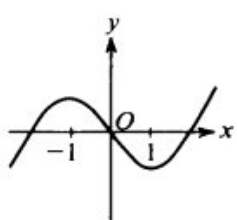


The graph of the derivative of f is shown in the figure above. Which of the following could be the graph of f ?

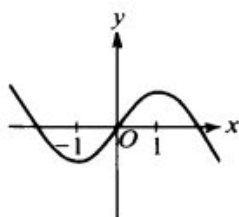
(A)



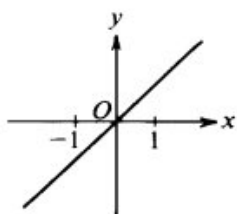
(B)



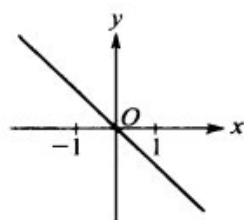
(C)



(D)

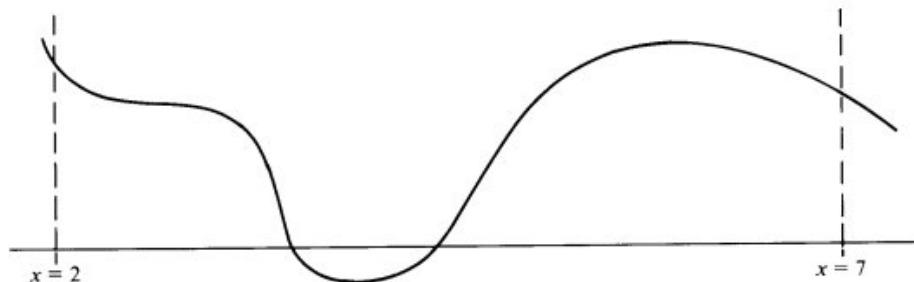


(E)



5-8

32.



The graph of $y=f(x)$ on the closed interval $[2,7]$ is shown above. How many points of inflection does this graph have on this interval?

(A) One

(B) Two

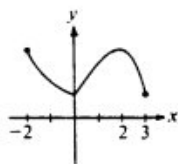
(C) Three

(D) Four

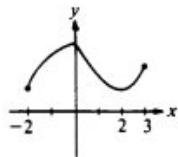
(E) Five

33. Let f be a function that is continuous on the closed interval $[-2,3]$ such that $f'(0)$ does not exist, $f'(2) = 0$, and $f''(x) < 0$ for all x except $x = 0$. Which of the following could be the graph of f ?

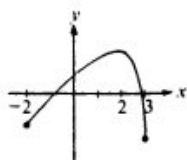
(A)



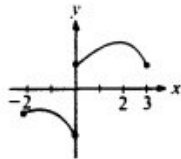
(B)



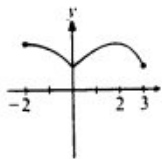
(C)



(D)

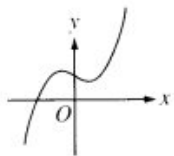


(E)



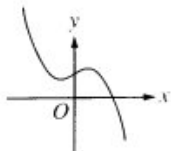
5-8

34.

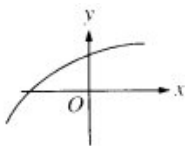


The graph of $y = h(x)$ is shown above. Which of the following could be the graph of $y = h'(x)$?

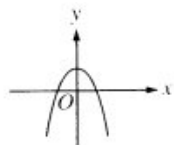
(A)



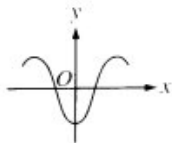
(B)



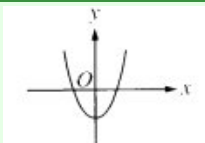
(C)



(D)



(E)



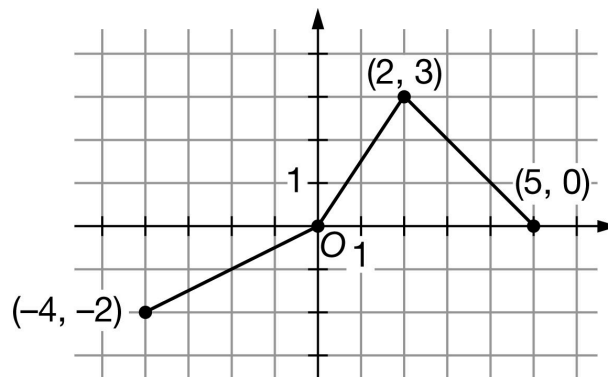
5-8

35. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g'

The graph of the function g' , the derivative of a function g , on the interval $-4 \leq x \leq 5$ consists of three line segments, as shown in the figure above. It is known that $g(-4) = 7$.

- Find the x -coordinate of each critical point of g on the open interval $-4 < x < 5$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for g . Justify your answers.
- Find the absolute maximum value of g on the interval $-4 \leq x \leq 5$. Justify your answer.
- Is $x = 2$ the location of a point of inflection for the graph of g ? Give a reason for your answer.
- Is $x = 0$ the location of a point of inflection for the graph of g ? Give a reason for your answer.
- Find $\lim_{x \rightarrow -2} \frac{g'(x)+1}{x^2-4}$. Give a reason for your answer.
- Let h be the function defined by $h(x) = x \cdot g(x)$. It is known that $g(2) = 6$. Find the value of $h'(2)$, or explain why it does not exist. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point:

- $x = 0$ only
- $x = 0$ and any additional x -coordinates not in $-4 < x < 5$

Does not earn the point:

- $x = 0$ and any additional x -coordinates in $-4 < x < 5$
- $x = 0$ with any incorrect supporting work

5-8

Do not read work related to additional values not in $-4 < x < 5$; this work does not affect whether the point is earned.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g'(x) = 0 \Rightarrow x = 0$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: Relative minimum because

- $g'(x)$ changes from negative to positive at $x = 0$
- the slope of g changes from negative to positive at $x = 0$
- the graph changes from negative to positive at $x = 0$

Does not earn the point: Relative minimum because

- g is decreasing to the left of $x = 0$ and increasing to the right of $x = 0$ (formulaic)
- Sign chart without written explanation
- $g'(x)$ changes sign at $x = 0$

Note: If additional x -coordinates in $-4 < x < 5$ are identified in part (a), the point is eligible ONLY if the additional value is declared as neither. If declared a relative minimum or maximum, the response is NOT eligible to earn the point.

Note: If additional x -coordinates not in $-4 < x < 5$ are identified in part (a), do not read these responses. Only read for values in the interval.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 0$ is the location of a relative minimum for g because $g'(x)$ changes from negative to positive there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

5-8

Note: The work for this part may be found in part (b). In that case, read part (b) for part (c).

First point: Considers both endpoints

Earns the first point by identifying $x = -4$ and $x = 5$ as candidates in a table or in a written statement

Second point: Uses initial condition

Uses $g(-4) = 7$ in a calculation for $g(0)$ or $g(5)$

Third point: Answer with justification

The response must declare as an answer the maximum value of 10.5.

The ordered pair $(5, 10.5)$ is not a declaration of the maximum value of g and does not earn the point.

If a Candidates Test is used for justification, all function values must be correct.

x	$g(x)$
-4	7
0	3
6	10.5



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Considers both endpoints
- ☐ Uses initial condition
- ☐ Answer with justification

Solution:

The only critical point on $-4 < x < 5$ is a relative minimum, so the absolute maximum value of g on $-4 \leq x \leq 5$ must occur at an endpoint.

$$g(-4) = 7$$

$$\begin{aligned} g(5) &= g(-4) + \int_{-4}^5 g'(t) \, dt \\ &= 7 + \int_{-4}^0 g'(t) \, dt + \int_0^5 g'(t) \, dt \\ &= 7 - \frac{1}{2} \cdot 4 \cdot 2 + \frac{1}{2} \cdot 3 \cdot 5 = 10.5 \end{aligned}$$

Therefore, the absolute maximum value of g on $-4 \leq x \leq 5$ is $g(5) = 10.5$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = 2$ is the location of a point of inflection because

5-8

- g' changes from increasing to decreasing
- g'' changes from positive to negative at $x = 2$
- the slope of g' changes from positive to negative at $x = 2$
- g'' changes sign at $x = 2$
- a relative extrema exists on the graph of g' at $x = 2$
- a relative maximum exists on the graph of g' at $x = 2$

Does not earn the point:

- the concavity at $x = 2$ changes [Must appeal to the change in sign of g'']
- g' changes from positive to negative
- the graph changes from positive to negative
- Only states $g''(x) = 0$ or $g''(x)$ does not exist (DNE)
- a relative minimum exists on the graph of g' at $x = 2$
- The acceleration changes from positive to negative



0	1
---	---

The student response accurately includes the criteria below.

☐

Answer with reason

Solution:

$x = 2$ is the location of a point of inflection for the graph of g because g' changes from increasing to decreasing there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = 0$ is not the location of a point of inflection because

- g' is increasing for $x < 0$ and $x > 0$
- g' is increasing through $x = 0$
- g'' is always positive before and after $x = 0$
- the slope of g' is always positive before and after $x = 0$
- g'' does not change sign at $x = 0$
- a relative extrema does not exist on the graph of g' at $x = 0$

Does not earn the point:

5-8

- the concavity does not change [Must appeal to the lack of change in sign of g'']
- g' is always positive
- g' does not change sign



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 0$ is not the location of a point of inflection for the graph of g because g' is increasing on both sides of $x = 0$.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Confirm $\lim_{x \rightarrow -2} (g'(x) + 1) = 0$ and $\lim_{x \rightarrow -2} (x^2 - 4) = 0$

Does not need to reference continuity or confirm g' and g'' are continuous at $x = -2$

Earns the point:

· $\lim_{x \rightarrow -2} (g'(x) + 1) = 0$ and $\lim_{x \rightarrow -2} (x^2 - 4) = 0$ [with or without limit notation]

· $\lim_{x \rightarrow -2} \frac{g'(x)+1}{x^2-4} \rightarrow \frac{0}{0}$

· $\lim_{x \rightarrow -2} \frac{g'(x)+1}{x^2-4} = \text{indeterminate form } \frac{0}{0}$

Does not earn the point in the presence of limit $= \frac{0}{0}$, i.e., $\lim_{x \rightarrow -2} \frac{g'(x)+1}{x^2-4} = \frac{0}{0}$

Second point: Applies L'Hospital's Rule

Earns the point for a ratio of the derivatives:

· $\lim_{x \rightarrow -2} \frac{g'(x)}{2x}$

· $\frac{g''(-2)}{2(-2)}$

· $\frac{g'(x)}{2x}$

Note: If there is an error in the derivative, the response will not earn the point.

Third point: Correct answer for limit

Eligibility: Earns the second point

5-8

Must be our answer $-\frac{1}{8}$ or equivalent



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

g' and g'' are continuous at $x = -2$. $\Rightarrow \lim_{x \rightarrow -2} g'(x) = g'(-2)$ and $\lim_{x \rightarrow -2} g''(x) = g''(-2)$.

$$\lim_{x \rightarrow -2} (g'(x) + 1) = g'(-2) + 1 = -1 + 1 = 0$$

$$\lim_{x \rightarrow -2} (x^2 - 4) = (-2)^2 - 4 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow -2} \frac{g'(x) + 1}{x^2 - 4} &= \lim_{x \rightarrow -2} \frac{g''(x)}{2x} \\ &= \frac{g''(-2)}{2 \cdot (-2)} = \frac{\frac{1}{2}}{-4} = -\frac{1}{8} \end{aligned}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

If the Product rule is not used, the response does not earn any points.

First point: Product rule

Applies product rule: $h'(x) = x \cdot g'(x) + g(x)$

Product rule may appear as $h'(2) = 2 \cdot g'(2) + g(2)$

If there is an error in the derivative, the response earns no points.

Second point: Correct Answer

Eligibility: Earns the first point

Must be our answer, $h'(2) = 12$ or equivalent



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

5-8☐ Product rule☐ Answer**Solution:**

$$h'(x) = x \cdot g'(x) + g(x)$$

$$h'(2) = 2 \cdot g'(2) + g(2) = 2 \cdot 3 + 6 = 12$$

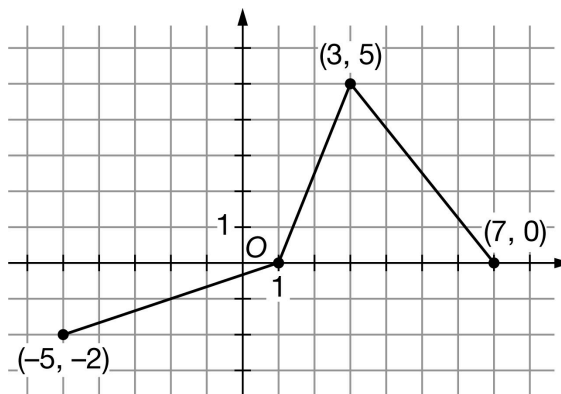
5-8

36. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of h'

The graph of the function h' , the derivative of a function h , on the interval $-5 \leq x \leq 7$ consists of three line segments, as shown in the figure above. It is known that $h(-5) = 6$.

- Find the x -coordinate of each critical point of h on the open interval $-5 < x < 7$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for h . Justify your answers.
- Find the absolute maximum value of h on the interval $-5 \leq x \leq 7$. Justify your answer.
- Is $x = 3$ the location of a point of inflection for the graph of h ? Give a reason for your answer.
- Is $x = 1$ the location of a point of inflection for the graph of h ? Give a reason for your answer.
- Find $\lim_{x \rightarrow -2} \frac{h'(x)+1}{x^2-4}$. Give a reason for your answer.
- Let g be the function defined by $g(x) = x \cdot h(x)$. It is known that $h(3) = 5$. Find the value of $g'(3)$, or explain why it does not exist. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point:

- $x = 1$ only
- $x = 1$ and any additional x -coordinates not in $-5 < x < 7$

Does not earn the point:

- $x = 1$ and any additional x -coordinates in $-5 < x < 7$
- $x = 1$ with any incorrect supporting work

5-8

Do not read work related to additional values not in $-5 < x < 7$; this work does not affect whether the point is earned.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h'(x) = 0 \Rightarrow x = 1$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: Relative minimum because

- $h'(x)$ changes from negative to positive at $x = 1$
- the slope of h changes from negative to positive at $x = 1$
- the graph changes from negative to positive at $x = 1$

Does not earn the point: Relative minimum because

- h is decreasing to the left of $x = 1$ and increasing to the right of $x = 1$ (formulaic)
- Sign chart without written explanation
- $h'(x)$ changes sign at $x = 1$

Note: If additional x -coordinates in $-5 < x < 7$ are identified in part (a), the point is eligible ONLY if the additional value is declared as neither. If declared a relative minimum or maximum, the response is not eligible to earn the point.

Note: If additional x -coordinates not in $-5 < x < 7$ are identified in part (a), do not read these responses. Only read for values in the interval.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 1$ is the location of a relative minimum for h because $h'(x)$ changes from negative to positive there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

5-8

Note: The work for this part may be found in part (b). In that case, read part (b) for part (c).

First point: Considers both endpoints

Earns the first point by identifying $x = -5$ and $x = 7$ as candidates in a table or in a written statement

Second point: Uses initial condition

Uses $h(-5) = 6$ in calculation for $h(1)$ or $h(7)$

Third point: Answer with justification

The response must declare as an answer the absolute maximum value of 15.

The ordered pair $(7, 15)$ is not a declaration of the maximum value of h and does not earn the point.

If a Candidates Test is used for justification, all function values must be correct.

x	$h(x)$
-5	6
1	0
7	15



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Considers both endpoints
- ☐ Uses initial condition
- ☐ Answer with justification

Solution:

The only critical point on $-5 < x < 7$ is a relative minimum, so the absolute maximum value of h on $-5 \leq x \leq 7$ must occur at an endpoint.

$$h(-5) = 6$$

$$\begin{aligned} h(7) &= h(-5) + \int_{-5}^7 h'(t) \, dt \\ &= 6 + \int_{-5}^1 h'(t) \, dt + \int_1^7 h'(t) \, dt \\ &= 6 - \frac{1}{2} \cdot 6 \cdot 2 + \frac{1}{2} \cdot 6 \cdot 5 = 15 \end{aligned}$$

Therefore, the absolute maximum value of h on $-5 \leq x \leq 7$ is $h(7) = 15$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = 3$ is the location of a point of inflection because

5-8

- h' changes from increasing to decreasing
- h'' changes from positive to negative at $x = 3$
- the slope of h' changes from positive to negative at $x = 3$
- h'' changes sign at $x = 3$
- a relative extrema exists on the graph of h' at $x = 3$
- a relative maximum exists on the graph of h' at $x = 2$

Does not earn the point:

- the concavity changes at $x = 3$ [Must appeal to the change in sign of h'']
- h' changes from positive to negative
- the graph changes from positive to negative
- Only states $h''(x) = 0$ OR $h''(x)$ DOES NOT EXIST (DNE)
- a relative minimum exists on the graph of h' at $x = 2$
- the acceleration changes from positive to negative



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 3$ is the location of a point of inflection for the graph of h because h' changes from increasing to decreasing there .

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = 1$ is not the location of a point of inflection because

- h' is increasing for $x < 1$ and $x > 1$
- h' is increasing through $x = 1$
- h'' is always positive before and after $x = 1$
- the slope of h' is always positive before and after $x = 1$
- h'' does not change sign at $x = 1$
- a relative extrema does not exist on the graph of h' at $x = 1$

Does not earn the point:

5-8

- the concavity does not change [Must appeal to the change in sign of h'']
- h' is always positive
- h' does not change sign



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 1$ is not the location of a point of inflection for the graph of h because h' is increasing on both sides of this point.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Confirm $\lim_{x \rightarrow -2} (h'(x) + 1) = 0$ and $\lim_{x \rightarrow -2} (x^2 - 4) = 0$

Does not need to reference continuity or confirm h' and h'' are continuous at $x = -2$

Earns the point:

· $\lim_{x \rightarrow -2} (h'(x) + 1) = 0$ and $\lim_{x \rightarrow -2} (x^2 - 4) = 0$ [with or without limit notation]

· $\lim_{x \rightarrow -2} \frac{h'(x)+1}{x^2-4} \rightarrow \frac{0}{0}$

· $\lim_{x \rightarrow -2} \frac{h'(x)+1}{x^2-4} = \text{indeterminate form } \frac{0}{0}$

Does not earn the point in the presence of limit $= \frac{0}{0}$, i.e., $\lim_{x \rightarrow -2} \frac{h'(x)+1}{x^2-4} = \frac{0}{0}$

Second point: Applies L'Hospital's Rule

Earns the point for a ratio of the derivatives:

· $\lim_{x \rightarrow -2} \frac{h'(x)}{2x}$

· $\frac{h''(-2)}{2(-2)}$

· $\frac{h'(x)}{2x}$

Note: If there is an error in the derivative, the response will not earn the point.

Third point: Correct answer for limit

Eligibility: Earns the second point

5-8

Must be our answer $-\frac{1}{12}$ or equivalent



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

h' and h'' are continuous at $x = -2$. $\Rightarrow \lim_{x \rightarrow -2} h'(x) = h'(-2)$ and $\lim_{x \rightarrow -2} h''(x) = h''(-2)$.

$$\lim_{x \rightarrow -2} h'(x) + 1 = -1 + 1 = 0$$

$$\lim_{x \rightarrow -2} (x^2 - 4) = (-2)^2 - 4 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow -2} \frac{h'(x) + 1}{x^2 - 4} &= \lim_{x \rightarrow -2} \frac{h''(x)}{2x} \\ &= \frac{h''(-2)}{2 \cdot (-2)} = \frac{\frac{1}{3}}{-4} = -\frac{1}{12} \end{aligned}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

If the Product rule is not used, the response does not earn any points.

1st point: Product Rule

Applies product rule: $g'(x) = x \cdot h'(x) + h(x)$

Product rule may appear as $g'(3) = 3 \cdot h'(3) + h(3)$

If there is an error in the derivative, the response earns no points in this part.

Second point: Correct Answer

Eligibility: Earns the 1st point

Must be our answer, $g'(3) = 20$ or equivalent



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

5-8☐ Product rule☐ Answer**Solution:**

$$g'(x) = x \cdot h'(x) + h(x)$$

$$g'(3) = 3 \cdot h'(3) + h(3) = 3 \cdot 5 + 5 = 20$$

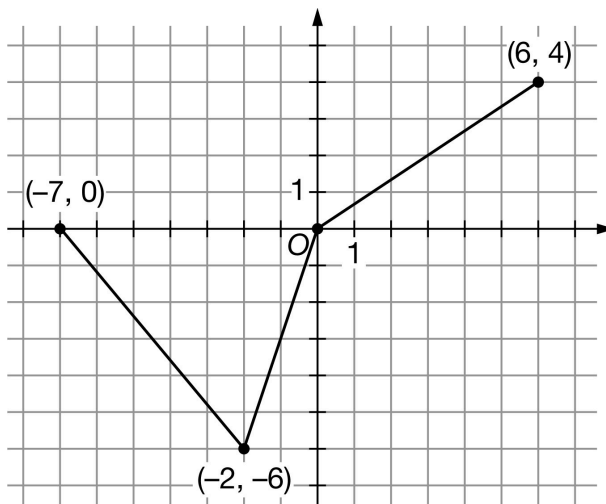
5-8

37. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of h'

The graph of the function h' , the derivative of a function h , on the interval $-7 \leq x \leq 6$ consists of three line segments, as shown in the figure above. It is known that $h(-7) = 2$.

- Find the x -coordinate of each critical point of h on the open interval $-7 < x < 6$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for h . Justify your answers.
- Find the absolute maximum value of h on the interval $-7 \leq x \leq 6$. Justify your answer.
- Is $x = -2$ the location of a point of inflection for the graph of h ? Give a reason for your answer.
- Is $x = 0$ the location of a point of inflection for the graph of h ? Give a reason for your answer.
- Find $\lim_{x \rightarrow 3} \frac{2h'(x) - 4}{x^2 - 9}$. Give a reason for your answer.
- Let g be the function defined by $g(x) = x \cdot h(x)$. It is known that $h(-2) = -13$. Find the value of $g'(-2)$, or explain why it does not exist. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point:

- $x = 0$ only
- $x = 0$ and any additional x -coordinates not in $-7 < x < 6$

Does not earn the point:

5-8

- $x = 0$ and any additional x -coordinates in $-7 < x < 6$
- $x = 0$ with any incorrect supporting work

Do not read work related to additional values not in $-7 < x < 6$; this work does not affect whether the point is earned.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h'(x) = 0 \Rightarrow x = 0$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: Relative minimum because

- $h'(x)$ changes from negative to positive at $x = 0$
- the slope of h changes from negative to positive at $x = 0$
- the graph changes from negative to positive at $x = 0$

Does not earn the point: Relative minimum because

- h is decreasing to the left of $x = 0$ and increasing to the right of $x = 0$ (formulaic)
- Sign chart without written explanation
- $h'(x)$ changes sign at $x = 0$
- Any incorrect argument

Note: If additional x -coordinates in $-7 < x < 6$ are identified in part (a), the point is eligible ONLY if the additional value is declared as neither. If declared a relative minimum or maximum, the response is not eligible to earn the point.

Note: If additional x -coordinates not in $-7 < x < 6$ are identified in part (a), do not read reasons for these responses. Only read for values in the interval.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

5-8

$x = 0$ is the location of a relative minimum for h because $h'(x)$ changes from negative to positive there

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Note: The work for this part may be found in part (b). In that case, read part (b) for part (c).

First point: Considers both endpoints

Earns the first point by identifying $x = -7$ and $x = 6$ as candidates in a table or in a written statement

Second point: Uses initial condition

Uses $h(-7) = 2$ in a calculation for $h(0)$ and/or $h(6)$

Third point: Answer with justification

The response must declare as an answer the maximum value of 2.

The ordered pair $(-7, 2)$ is not a declaration of the maximum value of h and does not earn the point.

If a Candidates Test is used for justification, all function values must be correct.

x	$h(x)$
-7	2
0	-19
6	-7



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Considers both endpoints
- ☐ Uses initial condition
- ☐ Answer with justification

Solution:

The only critical point on $-7 < x < 6$ is a relative minimum, so the absolute maximum value of h on $-7 \leq x \leq 6$ must occur at an endpoint.

$$h(-7) = 2$$

$$h(6) = h(-7) + \int_{-7}^6 h'(t) dt$$

$$= 2 + \int_{-7}^0 h'(t) dt + \int_0^6 h'(t) dt$$

$$= 2 - \frac{1}{2} \cdot 7 \cdot 6 + \frac{1}{2} \cdot 6 \cdot 4 = -7$$

Therefore, the absolute maximum value of h on $-7 \leq x \leq 6$ is $h(-7) = 2$.

5-8

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = -2$ is the location of a point of inflection because

- h' changes from decreasing to increasing
- h'' changes from negative to positive at $x = -2$
- the slope of h' changes from negative to positive at $x = -2$
- h'' changes sign at $x = -2$
- a relative extrema exists on the graph of h' at $x = -2$
- a relative minimum exists on the graph of h' at $x = -2$

Does not earn the point:

- the concavity at $x = -2$ changes [Must appeal to the change in sign of h'']
- h' changes from negative to positive
- the graph changes from negative to positive
- Only states $h''(x) = 0$ OR $h''(x)$ DOES NOT EXIST (DNE)
- a relative maximum exists on the graph of h' at $x = -2$
- The acceleration changes from negative to positive



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = -2$ is the location of a point of inflection for the graph of h because h' changes from decreasing to increasing there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = 0$ is not the location of a point of inflection because

- h' is increasing for $x < 0$ and $x > 0$
- h' is increasing through $x = 0$
- h'' is always positive before and after $x = 0$

5-8

- the slope of h' is always positive before and after $x = 0$
- h'' does not change sign at $x = 0$
- a relative extrema does not exist on the graph of h' at $x = 0$

Does not earn the point:

- the concavity does not change [Must appeal to the lack of change in sign of h'']
- h' is always positive
- h' does not change sign



0	1
---	---

The student response accurately includes the criteria below.

☐

Answer with reason

Solution:

$x = 0$ is not the location a point of inflection for the graph of h because h' is increasing on both sides of this point.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Confirm $\lim_{x \rightarrow 3} (2h'(x) - 4) = 0$ and $\lim_{x \rightarrow 3} (x^2 - 9) = 0$

Does not need to reference continuity or confirm h' and h'' are continuous at $x = -2$

Earns the point:

$$\cdot \lim_{x \rightarrow 3} (2h'(x) - 4) = 0 \text{ and } \lim_{x \rightarrow 3} (x^2 - 9) = 0 \text{ [with or without limit notation]}$$

$$\cdot \lim_{x \rightarrow 3} \frac{2h'(x) - 4}{x^2 - 9} \rightarrow \frac{0}{0}$$

$$\cdot \lim_{x \rightarrow 3} \frac{2h'(x) - 4}{x^2 - 9} = \text{indeterminate form } \frac{0}{0}$$

Does not earn the point in the presence of limit $= \frac{0}{0}$, i.e., $\lim_{x \rightarrow 3} \frac{2h'(x) - 4}{x^2 - 9} = \frac{0}{0}$

Second point: Applies L'Hospital's Rule

Earns the point for a ratio of the derivatives:

$$\cdot \lim_{x \rightarrow 3} \frac{2h'(x)}{2x}$$

$$\cdot \frac{2h''(3)}{2(3)}$$

5-8

$$\frac{2h'(x)}{2x}$$

Note: If there is an error in the derivative, the response will not earn the point.

Third point: Correct answer for limit

Eligibility: Earns the second point

Must be our answer $\frac{2}{9}$ or equivalent



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

h' and h'' are continuous at $x = 3$. $\Rightarrow \lim_{x \rightarrow 3} h'(x) = h'(3)$ and $\lim_{x \rightarrow 3} h''(x) = h''(3)$.

$$\lim_{x \rightarrow 3} 2h'(x) - 4 = 2 \cdot 2 - 4 = 0$$

$$\lim_{x \rightarrow 3} (x^2 - 9) = 3^2 - 9 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow 3} \frac{2h'(x) - 4}{x^2 - 9} &= \lim_{x \rightarrow 3} \frac{2h''(x)}{2x} \\ &= \frac{h''(3)}{3} = \frac{\frac{2}{3}}{3} = \frac{2}{9} \end{aligned}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

If the Product rule is not used, the response does not earn any points.

First point: Product Rule

$$\text{Applies product rule: } g'(x) = x \cdot h'(x) + h(x)$$

$$\text{Product rule may appear as } g'(-2) = -2 \cdot h'(-2) + h(-2)$$

If there is an error in the derivative, the response earns no points.

Second point: Correct Answer

Eligibility: Earns the first point

5-8

Must be our answer, $g'(-2) = -1$ or equivalent



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Product rule
- ☐ Answer

Solution:

$$g'(x) = x \cdot h'(x) + h(x)$$

$$g'(-2) = -2 \cdot h'(-2) + h(-2) = -2 \cdot (-6) + (-13) = -1$$

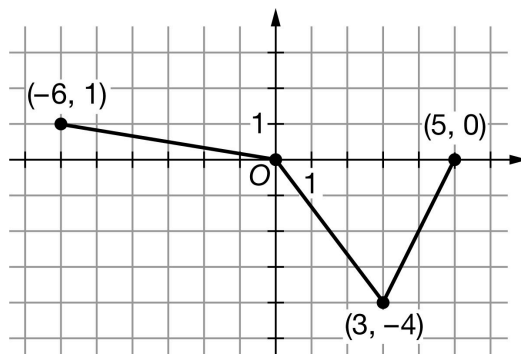
5-8

38. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of h'

The graph of the function h' , the derivative of a function h , on the interval $-6 \leq x \leq 5$ consists of three line segments, as shown in the figure above. It is known that $h(-6) = 5$.

- Find the x -coordinate of each critical point of h on the open interval $-6 < x < 5$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for h . Justify your answers.
- Find the absolute minimum value of h on the interval $-6 \leq x \leq 5$. Justify your answer.
- Is $x = 3$ the location of a point of inflection for the graph of h ? Give a reason for your answer.
- Is $x = 0$ the location of a point of inflection for the graph of h ? Give a reason for your answer.
- Find $\lim_{x \rightarrow -3} \frac{2h'(x)-1}{x^2-9}$. Give a reason for your answer.
- Let g be the function defined by $g(x) = 2x \cdot h(x)$. It is known that $h(3) = 2$. Find the value of $g'(3)$, or explain why it does not exist. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point:

- $x = 0$ only
- $x = 0$ and any additional x -coordinates not in $-6 < x < 5$

Does not earn the point:

- $x = 0$ and any additional x -coordinates in $-6 < x < 5$
- $x = 0$ with any incorrect supporting work

Do not read work related to additional values not in $-6 < x < 5$; this work does not affect whether the point is earned.

5-8



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h'(x) = 0 \Rightarrow x = 0$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: Relative maximum because

- $h'(x)$ changes from positive to negative at $x = 0$
- the slope of h changes from positive to negative at $x = 0$
- the graph changes from positive to negative at $x = 0$

Does not earn the point: Relative maximum because

- h is increasing to the left of $x = 0$
- and decreasing to the right of $x = 0$
- (formulaic)
- Sign chart without written explanation
- $h'(x)$ changes sign at $x = 0$

Note: If additional x -coordinates in $-6 < x < 5$ are identified in part (a), the point is eligible ONLY if the additional value is declared as neither. If declared a relative minimum or maximum, the response is not eligible to earn the point.

Note: If additional x -coordinates not in $-6 < x < 5$ are identified in part (a), do not read these responses. Only read for values in the interval.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 0$ is the location of a relative maximum for h because $h'(x)$ changes from positive to negative there.

5-8

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Note: The work for this part may be found in part (b). In that case, read part (b) for part (c).

First point: Considers both endpoints

Earns the 1st point by identifying $x = -6$ and $x = 5$ as candidates in a table or in a written statement

Second point: Uses initial condition

Uses $h(-6) = 5$ in a calculation for $h(0)$ or $h(5)$

Third point: Answer with justification

The response must declare as an answer the minimum value of -2 .

The ordered pair $(5, -2)$ is not a declaration of the minimum value of g and does not earn the point.

If a Candidates Test is used for justification, all function values must be correct.

x	$h(x)$
-6	5
0	8
5	-2



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Considers both endpoints
- ☐ Uses initial condition
- ☐ Answer with justification

Solution:

The only critical point on $-6 < x < 5$ is a relative maximum, so the absolute minimum value of h on $-6 \leq x \leq 5$ must occur at an endpoint.

$$h(-6) = 5$$

$$\begin{aligned} h(5) &= h(-6) + \int_{-6}^5 h'(t) dt \\ &= 5 + \int_{-6}^0 h'(t) dt + \int_0^5 h'(t) dt \\ &= 5 + \frac{1}{2} \cdot 6 \cdot 1 - \frac{1}{2} \cdot 5 \cdot 4 = -2 \end{aligned}$$

Therefore, the absolute minimum value of h on $-6 \leq x \leq 5$ is $h(5) = -2$.

5-8

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = 3$ is the location of a point of inflection because

- h' changes from decreasing to increasing
- h'' shifts from negative to positive at $x = 3$
- the slope of h' shifts from negative to positive at $x = 3$
- h'' changes sign at $x = 3$
- a relative extrema exists on the graph of h' at $x = 3$
- a relative minimum exists on the graph of h' at $x = 3$

Does not earn the point:

- the concavity at $x = 3$ changes. [Must appeal to the change in sign of h'']
- h' changes from negative to positive
- the graph changes from negative to positive
- Only states $h''(x) = 0$ OR $h''(x)$ DOES NOT EXIST (DNE)
- a relative maximum exists on the graph of h' at $x = 3$
- the acceleration changes from negative to positive



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 3$ is the location of a point of inflection for the graph of h because h' changes from decreasing to increasing there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = 0$ is not the location of a point of inflection because

- h' is decreasing for $x < 0$ and $x > 0$
- h' is decreasing through $x = 0$
- h'' is always negative before and after $x = 0$

5-8

- the slope of h' is always negative before and after $x = 0$
- h'' does not change sign at $x = 0$
- a relative extrema does not exist on the graph of h' at $x = 0$

Does not earn the point:

- the concavity does not change. [Must appeal to the lack of change in sign of h'']
- h' is always negative
- h' does not change sign



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 0$ is not the location of a point of inflection for the graph of h because h' is decreasing on both sides of this point.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Confirm $\lim_{x \rightarrow -3} (2h'(x) - 1) = 0$ and $\lim_{x \rightarrow -3} (x^2 - 9) = 0$

Does not need to reference continuity or confirm h' and h'' are continuous at $x = -3$

Earns the point:

- $\lim_{x \rightarrow -3} (2h'(x) - 1) = 0$ and $\lim_{x \rightarrow -3} (x^2 - 9) = 0$ [with or without limit notation]
- $\lim_{x \rightarrow -3} \frac{2h'(x) - 1}{x^2 - 9} \rightarrow 0$
- $\lim_{x \rightarrow -3} \frac{2h'(x) - 1}{x^2 - 9} = \text{indeterminate form } \frac{0}{0}$

Does not earn the point in the presence of limit $= \frac{0}{0}$, i.e., $\lim_{x \rightarrow -3} \frac{2h'(x) - 1}{x^2 - 9} = \frac{0}{0}$

Second point: Applies L'Hospital's Rule

Earns the point for a ratio of the derivatives:

- $\lim_{x \rightarrow -3} \frac{2h'(x)}{2x}$
- $\frac{2h'(-3)}{2(-3)}$

5-8

$$\frac{2h'(x)}{2x}$$

Note: If there is an error in the derivative, the response will not earn the point.

Third point: Correct answer for limit

Eligibility: Earns the second point

Must be our answer $\frac{1}{18}$ or equivalent



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:

h' and h'' are continuous at $x = -3$. $\Rightarrow \lim_{x \rightarrow -3} h'(x) = h'(-3)$ and $\lim_{x \rightarrow -3} h''(x) = h''(-3)$.

$$\lim_{x \rightarrow -3} 2h'(x) - 1 = 2 \cdot \frac{1}{2} - 1 = 0$$

$$\lim_{x \rightarrow -3} (x^2 - 9) = 3^2 - 9 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\lim_{x \rightarrow -3} \frac{2h'(x) - 1}{x^2 - 9} = \lim_{x \rightarrow -3} \frac{2h''(x)}{2x}$$

$$= \frac{h''(-3)}{-3} = \frac{-\frac{1}{6}}{-3} = \frac{1}{18}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

If the Product rule is not used, the response does not earn any points.

First point: Product Rule

$$\text{Applies product rule: } g'(x) = 2x \cdot h'(x) + 2 \cdot h(x)$$

$$\text{Product rule may appear as } g'(3) = 6 \cdot h'(3) + 2 \cdot h(3)$$

If there is an error in the derivative, the response earns no points in this part.

Second point: Correct Answer

Eligibility: Earns the first point

5-8

Must be our answer, $g'(3) = -20$ or equivalent



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Product rule
- ☐ Answer

Solution:

$$g'(x) = 2x \cdot h'(x) + 2h(x)$$

$$g'(3) = 2 \cdot 3 \cdot h'(3) + 2h(3) = 6 \cdot (-4) + 2 \cdot 2 = -20$$

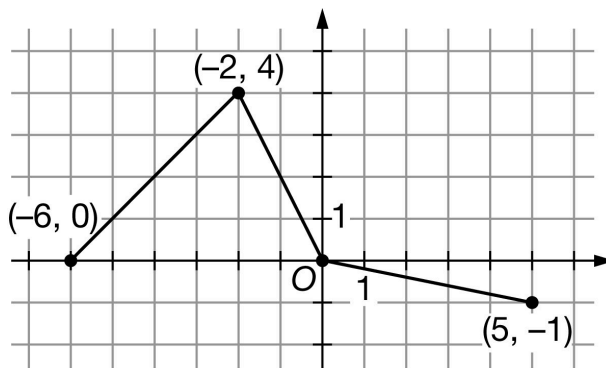
5-8

39. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g'

The graph of the function g' , the derivative of a function g , on the interval $-6 \leq x \leq 5$ consists of three line segments, as shown in the figure above. It is known that $g(-6) = -1$.

- Find the x -coordinate of each critical point of g on the open interval $-6 < x < 5$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for g . Justify your answers.
- Find the absolute minimum value of g on the interval $-6 \leq x \leq 5$. Justify your answer.
- Is $x = -2$ the location of a point of inflection for the graph of g ? Give a reason for your answer.
- Is $x = 0$ the location of a point of inflection for the graph of g ? Give a reason for your answer.
- Find $\lim_{x \rightarrow -1} \frac{g'(x)-2}{x^2-1}$. Give a reason for your answer.
- Let h be the function defined by $h(x) = 3x \cdot g(x)$. It is known that $g(-2) = 7$. Find the value of $h'(-2)$, or explain why it does not exist. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point:

- $x = 0$ only
- $x = 0$ and any additional x -coordinates not in $-6 < x < 5$

Does not earn the point:

- $x = 0$ and any additional x -coordinates in $-6 < x < 5$
- $x = 0$ with any incorrect supporting work

5-8

Do not read work related to additional values not in $-6 < x < 5$; this work does not affect whether the point is earned.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g'(x) = 0 \Rightarrow x = 0$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: Relative maximum because

- $g'(x)$ changes from positive to negative at $x = 0$
- the slope of g changes from positive to negative at $x = 0$
- the graph changes from positive to negative at $x = 0$

Does not earn the point: Relative maximum because

- g is increasing to the left of $x = 0$ and decreasing to the right of $x = 0$ (formulaic)
- Sign chart without written explanation
- $g'(x)$ changes sign at $x = 0$

Note: If additional x -coordinates in $-6 < x < 5$ are identified in part (a), the point is eligible ONLY if the additional value is declared as neither. If declared a relative minimum or maximum, the response is not eligible to earn the point.

Note: If additional x -coordinates not in $-6 < x < 5$ are identified in part (a), do not read these responses. Only read for values in the interval



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 0$ is the location of a relative maximum for g because $g'(x)$ changes from positive to negative there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

5-8

Note: The work for this part may be found in part (b). In that case, read part (b) for part (c).

First point: Considers both endpoints

Earns the first point by identifying $x = -6$ and $x = 5$ as candidates in a table or in a written statement

Second point: Uses initial condition

Uses $g(-6) = -1$ in calculation for $g(0)$ or $g(5)$

Third point: Answer with justification

The response must declare as an answer the absolute minimum value of -1 .

The ordered pair $(-6, -1)$ is not a declaration of the minimum value of g and does not earn the point.

If a Candidates Test is used for justification, all function values must be correct.

x	$g(x)$
-6	-1
0	11
5	8.5



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Considers both endpoints
- ☐ Uses initial condition
- ☐ Answer with justification

Solution:

The only critical point on $-6 < x < 5$ is a relative maximum, so the absolute minimum value of g on $-6 \leq x \leq 5$ must occur at an endpoint.

$$g(-6) = -1$$

$$\begin{aligned} g(5) &= g(-6) + \int_{-6}^5 g'(t) \, dt \\ &= -1 + \int_{-6}^0 g'(t) \, dt + \int_0^5 g'(t) \, dt \\ &= -1 + \frac{1}{2} \cdot 6 \cdot 4 - \frac{1}{2} \cdot 5 \cdot 1 = \frac{17}{2} \end{aligned}$$

Therefore, the absolute minimum value of g on $-6 \leq x \leq 5$ is $g(-6) = -1$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = -2$ is the location of a point of inflection because

5-8

- g' changes from increasing to decreasing
- g'' changes from positive to negative at $x = -2$
- the slope of g' changes from positive to negative at $x = -2$
- g'' changes sign at $x = -2$
- a relative extrema exists on the graph of g' at $x = -2$
- a relative maximum exists on the graph of g' at $x = -2$

Does not earn the point:

- the concavity at $x = -2$ changes [Must appeal to the change in sign of g'']
- g' changes from positive to negative
- the graph changes from positive to negative
- Only states $g''(x) = 0$ OR $g''(x)$ DOES NOT EXIST (DNE)
- a relative minimum exists on the graph of g' at $x = -2$
- The acceleration changes from positive to negative



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason
Solution:

$x = -2$ is the location of a point of inflection for the graph of g because g' changes from increasing to decreasing there.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Earns the point: $x = 0$ is not the location of a point of inflection because

- g' is decreasing for $x < 0$ and $x > 0$
- g' is decreasing through $x = 0$
- g'' is always negative before and after $x = 0$
- the slope of g' is always negative before and after $x = 0$
- g'' does not change sign at $x = 0$.
- a relative extrema does not exist on the graph of g' at $x = 0$

Does not earn the point:

5-8

- the concavity does not change [Must appeal to the lack of change in sign of g'']
- g' is always negative
- g' does not change sign



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 0$ is not the location of a point of inflection for the graph of g because g' is decreasing on both sides of this point.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

First point: Confirm $\lim_{x \rightarrow -1} (g'(x) - 2) = 0$ and $\lim_{x \rightarrow -1} (x^2 - 1) = 0$

Does not need to reference continuity or confirm g' and g'' are continuous at $x = -1$

Earns the point:

· $\lim_{x \rightarrow -1} (g'(x) - 2) = 0$ and $\lim_{x \rightarrow -1} (x^2 - 1) = 0$ [with or without limit notation]

· $\lim_{x \rightarrow -1} \frac{g'(x) - 2}{x^2 - 1} \rightarrow 0$

· $\lim_{x \rightarrow -1} \frac{g'(x) - 2}{x^2 - 1} = \text{indeterminate form } \frac{0}{0}$

Does not earn the point in the presence of limit $= \frac{0}{0}$, i.e., $\lim_{x \rightarrow -1} \frac{g'(x) - 2}{x^2 - 1} = \frac{0}{0}$

Second point: Applies L'Hospital's Rule

Earns the point for a ratio of the derivatives:

· $\lim_{x \rightarrow -1} \frac{g'(x)}{2x}$

· $\frac{g''(-1)}{2(-1)}$

· $\frac{g'(x)}{2x}$

Note: If there is an error in the derivative, the response will not earn the point.

Third point: Correct answer for limit

Eligibility: Earns the second point

5-8

Must be our answer 1 or equivalent



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Limits of numerator and denominator equal 0
- ☐ Application of L'Hospital's Rule
- ☐ Answer

Solution:
 g' and g'' are continuous at $x = -1$. $\Rightarrow \lim_{x \rightarrow -1} g'(x) = g'(-1)$ and $\lim_{x \rightarrow -1} g''(x) = g''(-1)$.

$$\lim_{x \rightarrow -1} g'(x) - 2 = 2 - 2 = 0$$

$$\lim_{x \rightarrow -1} (x^2 - 1) = 1^2 - 1 = 0$$

Therefore, L'Hospital's Rule can be applied.

$$\begin{aligned} \lim_{x \rightarrow -1} \frac{g'(x) - 2}{x^2 - 1} &= \lim_{x \rightarrow -1} \frac{g''(x)}{2x} \\ &= \frac{g''(-1)}{2 \cdot (-1)} = \frac{-2}{-2} = 1 \end{aligned}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

If the Product rule is not used, the response does not earn any points.

First point: Product Rule

$$\text{Applies product rule: } h'(x) = 3x \cdot g'(x) + 3 \cdot g(x)$$

$$\text{Product rule may appear as } h'(-2) = -6 \cdot g'(-2) + 3 \cdot g(-2)$$

If there is an error in the derivative, the response earns no points in this part.

Second point: Correct Answer

Eligibility: Earns the first point

Must be our answer, $h'(-2) = -3$ or equivalent

0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

5-8☐ Product rule☐ Answer**Solution:**

$$h'(x) = 3x \cdot g'(x) + 3g(x)$$

$$h'(-2) = 3 \cdot (-2) \cdot g'(-2) + 3g(-2) = -6 \cdot 4 + 3 \cdot 7 = -3$$

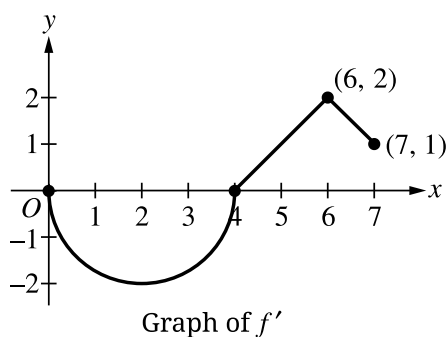
5-8

40. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Let f be a differentiable function with $f(4) = 3$. On the interval $0 \leq x \leq 7$, the graph of f' , the derivative of f , consists of a semicircle and two line segments, as shown in the figure.

- Find $f(0)$ and $f(5)$.
- Find the x -coordinates of all points of inflection of the graph of f for $0 < x < 7$. Justify your answer.
- Let g be the function defined by $g(x) = f(x) - x$. On what intervals, if any, is g decreasing for $0 \leq x \leq 7$? Show the analysis that leads to your answer.
- For the function g defined in part (c), find the absolute minimum value on the interval $0 \leq x \leq 7$. Justify your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response with answers of only $f(0) = \pm 2\pi$, or only $f(5) = \frac{1}{2}$, or both earns 1 of the 3 points.

A response displaying $f(5) = \frac{7}{2}$ and a missing or incorrect value for $f(0)$ earns 2 of the 3 points.

The second and third points can be earned in either order.

Read unlabeled values from left to right and from top to bottom as $f(0)$ and $f(5)$. A single value must be labeled as either $f(0)$ or $f(5)$ in order to earn any points.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

5-8

- ☐ Area of either region – OR – $\int_0^4 f'(x) \, dx$ – OR – $\int_4^5 f'(x) \, dx$
- ☐ $f(0)$
- ☐ $f(5)$

Solution:

$$f(0) = f(4) + \int_4^0 f'(x) \, dx = 3 - \int_0^4 f'(x) \, dx = 3 + 2\pi$$

$$f(5) = f(4) + \int_4^5 f'(x) \, dx = 3 + \frac{1}{2} = \frac{7}{2}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response that gives only one of $x = 2$ or $x = 6$, along with a correct justification, earns 1 of the 2 points.

A response that claims that there is a point of inflection at any value other than $x = 2$ or $x = 6$ earns neither point.

To earn the second point a response must use correct reasoning based on the graph of f' . Examples of correct reasoning include:

- Correctly discussing the signs of the slopes of the graph of f'
- Citing $x = 2$ and $x = 6$ as the locations of local extrema on the graph of f'

Examples of reasoning not (sufficiently) connected to the graph of f' include:

- Reasoning based on sign changes in f'' unless the connection is made between the sign of f'' and the slopes of the graph of f'
- Reasoning based only on the concavity of the graph of f

The second point cannot be earned by use of vague or undefined terms such as “it” or “the function” or “the derivative.”

Responses that report inflection points as ordered pairs must report the points $(2, 3 + \pi)$ and $(6, 5)$ in order to earn the first point. If the y-coordinates are reported incorrectly, the response remains eligible for the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Justification

Solution:

The graph of f has a point of inflection at each of $x = 2$ and $x = 6$, because $f'(x)$ changes from decreasing to increasing at $x = 2$ and from increasing to decreasing at $x = 6$.

Part C

5-8

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point can be earned for $f'(x) \leq 1$ or the equivalent, in words or symbols.

Endpoints do not need to be included in the interval to be eligible for the second point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ $g'(x) = f'(x) - 1$
- ☐ Interval with reason

Solution:

$$g'(x) = f'(x) - 1$$

$$f'(x) - 1 \leq 0 \Rightarrow f'(x) \leq 1$$

The graph of g is decreasing on the interval $0 \leq x \leq 5$ because $g'(x) \leq 0$ on this interval.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A justification that uses a local argument, such as “ g' changes from negative to positive (or g changes from decreasing to increasing) at $x = 5$ ” must also state that $x = 5$ is the only critical point.

If $g'(x) = 0$ (or equivalent) is not declared explicitly, a response that isolates $x = 5$ as the only critical number belonging to $(0, 7)$ earns the first point.

A response that imports $g'(x) = f'(x)$ from part (c) is eligible for the first point but not the second.

In this case, consideration of $x = 4$ as the only critical number on $(0, 7)$ earns the first point.

Solution using Candidates Test:

$$g'(x) = f'(x) - 1 = 0 \Rightarrow x = 5, x = 7$$

x	$g(x)$
0	$3 + 2\pi$
5	$-\frac{3}{2}$
7	$-\frac{1}{2}$

The absolute minimum value of g on the interval $0 \leq x \leq 7$ is $-\frac{3}{2}$.

When using a Candidates Test, a response may import an incorrect value of $f(0) = g(0) > -\frac{3}{2}$ from part (a). The second point can only be earned for an answer of $-\frac{3}{2}$.

5-8



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Considers $g'(x) = 0$
- ☐ Answer with justification

Solution:

g is continuous, $g'(x) < 0$ for $0 < x < 5$, and $g'(x) > 0$ for $5 < x < 7$.

Therefore, the absolute minimum occurs at $x = 5$, and $g(5) = f(5) - 5 = \frac{7}{2} - 5 = -\frac{3}{2}$ is the absolute minimum value of g .

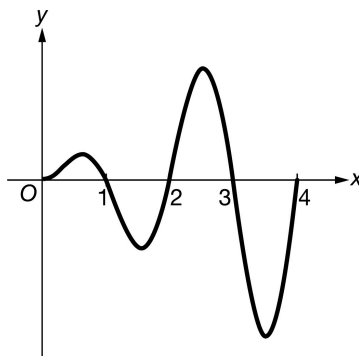
5-8

41. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f'

The figure above shows the graph of f' , the derivative of a twice-differentiable function f , on the closed interval $[0, 4]$. The areas of the regions bounded by the graph of f' and the x -axis on the intervals $[0, 1]$, $[1, 2]$, $[2, 3]$, and $[3, 4]$ are 2, 6, 10, and 14, respectively. The graph of f' has horizontal tangents at $x = 0.6$, $x = 1.6$, $x = 2.5$, and $x = 3.5$. It is known that $f(2) = 5$.

On what open intervals contained in $(0, 4)$ is the graph of f both decreasing and concave down? Give a reason for your answer.

Find the absolute minimum value of f on the interval $[0, 4]$. Justify your answer.

Evaluate $\int_0^4 f(x)f'(x) \, dx$.

The function g is defined by $g(x) = x^3 f(x)$. Find $g'(2)$. Show the work that leads to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ answer
- ☐ reason

Solution:

The graph of f is decreasing and concave down on the intervals $(1, 1.6)$ and $(3, 3.5)$ because f' is negative and decreasing on these intervals.

5-8

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The second point can be earned without earning the first point. The integrals that lead to the answer must be shown.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ considers $x = 2$ as a candidate
- ☐ answer with justification

Solution:

The graph of f' changes from negative to positive only at $x = 2$.

$$f(2) = 5$$

$$f(0) = f(2) + \int_2^0 f'(x) dx = f(2) - \int_0^2 f'(x) dx = 5 - (2 - 6) = 9$$

$$f(4) = f(2) + \int_2^4 f'(x) dx = 5 + (10 - 14) = 1$$

On the interval $[0, 4]$, the absolute minimum value of f is $f(4) = 1$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The third point, for the answer, cannot be earned if no points are earned for the antiderivative.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ antiderivative of the form $a(f(x))^2$
- ☐ earned the first point, and $a = \frac{1}{2}$
- ☐ answer

Solution:

$$\begin{aligned} \int_0^4 f(x)f'(x) dx &= \frac{1}{2}(f(x))^2 \Big|_{x=0}^{x=4} = \frac{1}{2}((f(4))^2 - (f(0))^2) \\ &= \frac{1}{2}(1^2 - 9^2) = -40 \end{aligned}$$

5-8

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The second point requires a correct expression for the product rule; a correct numerical answer without supporting work will not earn the point.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ product rule
- ☐ answer, with supporting work

Solution:

$$g'(x) = 3x^2 f(x) + x^3 f'(x)$$

$$g'(2) = 3 \cdot 2^2 f(2) + 2^3 f'(2) = 12 \cdot 5 + 8 \cdot 0 = 60$$

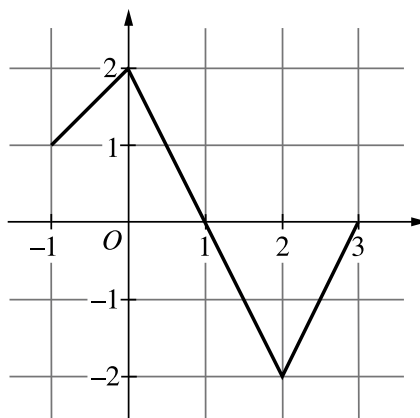
5-8

42. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f

Let f be the continuous function defined on the closed interval $[-1, 3]$ whose graph, consisting of three line segments, is shown. Let g be the function given by $g(x) = \int_0^x f(t) \, dt$.

- On what intervals, if any, is g decreasing? Give a reason for your answer.
- Find $g''(-\frac{1}{2})$, or explain why it does not exist.
- What is the absolute minimum value of g on the closed interval $[-1, 3]$? Justify your answer.
- Let h be the function defined by $h(x) = \ln(g(x))$ for $0 < x < 2$. Find $h'(\frac{3}{2})$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for $g'(x) = f(x)$ in any part of the problem.

A response of $g''(x) = f'(x)$ or $g''(-\frac{1}{2}) = f'(-\frac{1}{2})$ in part (b) is sufficient to earn the first point.

The interval $[1, 3]$ can be reported $(1, 3)$, $(1, 3]$, or $(1, 3]$.

The second point can be earned without the first by presenting a correct interval and a statement that $f(x) < 0$ on this interval.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

5-8

- ☐ $g'(x) = f(x)$ in any part of response
- ☐ Answer with reason

Solution:

$g'(x) = f(x) < 0$ on the open interval $(1, 3)$. Therefore, g is decreasing on the closed interval $[1, 3]$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

An answer of 1 with no supporting work earns the point.

Any work shown must be correct in order to earn the point.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$$g''(x) = f'(x)$$

$$g''\left(-\frac{1}{2}\right) = f'\left(-\frac{1}{2}\right) = 1$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for any presented consideration of $x = -1$ and $x = 3$.

A response that finds only that $g(-1) = -1.5$ and declares this as the minimum earns only the second point.

A response that evaluates $g(x)$ at any values of x in addition to $x = -1$, $x = 1$, and $x = 3$ does not earn the third point.

To earn the third point, a response must:

- have earned the first point,
- have no errors in the evaluation of $g(-1)$ and $g(3)$,
- conclude that $g(3) > g(-1)$, and
- either evaluate $g(1)$ as part of the Candidates Test or include a global argument that specifically addresses the behavior of $g(x)$ at $x = 1$ (e.g., “ $g'(x) = f(x)$ changes signs from positive to negative at $x = 1$ ” or “ $g'(1) = f(1) = 0$ and there is a maximum at $x = 1$ ”).

5-8



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Considers $x = -1$ and $x = 3$
- ☐ Answer
- ☐ Justification

Solution:

Because $g'(x) = f(x) > 0$ for $-1 < x < 1$ and $g'(x) = f(x) < 0$ for $1 < x < 3$, the minimum value of g must occur at either $x = -1$ or $x = 3$.

$$g(-1) = \int_0^{-1} f(t) dt = - \int_{-1}^0 f(t) dt = -1.5$$

$$g(3) = \int_0^3 f(t) dt = -1$$

Therefore, the minimum value of g is $g(-1) = -1.5$.

Alternate solution using Candidates Test:

$$g'(x) = f(x) = 0 \Rightarrow x = 1$$

$$g(-1) = \int_0^{-1} f(t) dt = - \int_{-1}^0 f(t) dt = -1.5$$

$$g(1) = \int_0^1 f(t) dt = 1$$

$$g(3) = \int_0^3 f(t) dt = -1$$

Therefore, the minimum value of g is $g(-1) = -1.5$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for either $h'(x) = \frac{1}{g(x)} \cdot g'(x)$, $h'\left(\frac{3}{2}\right) = \frac{1}{g\left(\frac{3}{2}\right)} \cdot g'\left(\frac{3}{2}\right)$, or $\frac{1}{\frac{3}{4}} \cdot (-1)$.

To earn the second point a response must use the correct value of either $g\left(\frac{3}{2}\right)$ or $g'\left(\frac{3}{2}\right)$ (or both) in an expression for $h'\left(\frac{3}{2}\right)$.

The minimal answer that earns all 3 points is $\frac{1}{\frac{3}{4}} \cdot (-1)$.

An answer of $-\frac{4}{3}$ with incorrect or without supporting work does not earn any points.

The third point is earned only for $-\frac{4}{3}$ or equivalent.

5-8



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ $h'(x)$
- ☐ Correct value of $g\left(\frac{3}{2}\right)$ or $g'\left(\frac{3}{2}\right)$
- ☐ Answer

Solution:

$$h'(x) = \frac{1}{g(x)} \cdot g'(x)$$

$$h'\left(\frac{3}{2}\right) = \frac{1}{g\left(\frac{3}{2}\right)} \cdot g'\left(\frac{3}{2}\right) = \frac{1}{\frac{3}{4}} \cdot (-1) = -\frac{4}{3}$$

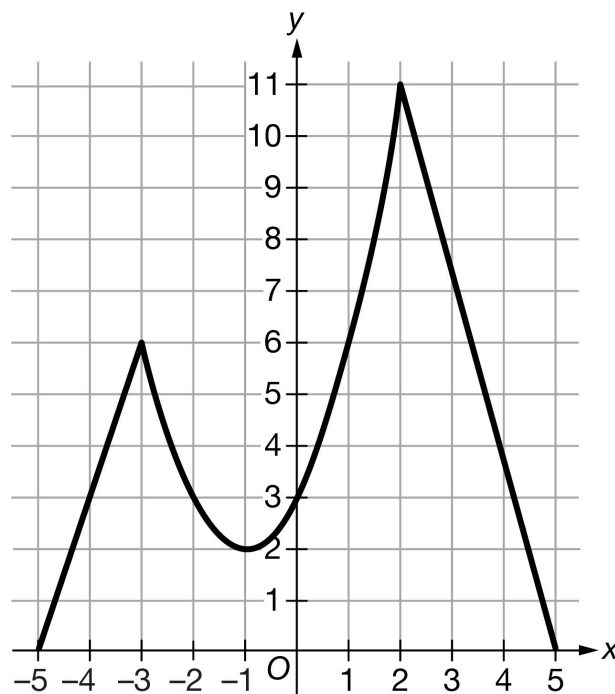
5-8

43. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f

The continuous function f is defined on the closed interval $[-5, 5]$. The graph of f consists of a parabola and two line segments, as shown in the figure above. Let g be a function such that $g'(x) = f(x)$.

- (a) Fill in the missing entries in the table below to describe the behavior of f' and f'' . Indicate Positive, Negative, or 0. Give reasons for your answers.

x	$-5 < x < -3$	$-3 < x < -1$	$-1 < x < 2$	$2 < x < 5$
$f(x)$	Positive	Positive	Positive	Positive
$f'(x)$				
$f''(x)$				

- (b) There is no value of x in the open interval $(-1, 5)$ at which $f'(x) = \frac{f(5) - f(-1)}{5 - (-1)}$. Explain why this does not violate the Mean Value Theorem.
- (c) Find all values of x in the open interval $(-5, 5)$ at which the graph of g has a point of inflection. Explain your reasoning.
- (d) At what value of x does g attain its absolute maximum on the closed interval $[-5, 5]$? Give a reason for your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

5-8



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ behavior for $f'(x)$
- ☐ behavior for $f''(x)$
- ☐ reasons

Solution:

x	$-5 < x < -3$	$-3 < x < -1$	$-1 < x < 2$	$2 < x < 5$
$f(x)$	Positive	Positive	Positive	Positive
$f'(x)$	Positive	Negative	Positive	Negative
$f''(x)$	0	Positive	Positive	0

$f'(x)$ is positive for $-5 < x < -3$ and $-1 < x < 2$ because f is increasing there.

$f'(x)$ is negative for $-3 < x < -1$ and $2 < x < 5$ because f is decreasing there.

$f''(x)$ is positive for $-3 < x < -1$ and $-1 < x < 2$ because the graph of f is concave up there.

$f''(x) = 0$ for $-5 < x < -3$ and $2 < x < 5$ because the graph of f is linear there.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct explanation.

Solution:

This does not violate the Mean Value Theorem because f is not differentiable at $x = 2$ due to a sharp corner. The Mean Value Theorem cannot be applied on the interval $-1 < x < 5$.

Part C

The response is not eligible for any points if incorrect values of x are included.

A response may earn the first point for two correct values of x without explanations or with incomplete, incorrect explanations.

A maximum of 1 point may be earned for three correct values of x without explanations.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

5-8



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ One correct value of x with explanation OR Two correct values of x without explanations
- ☐ A second correct value of x with explanation
- ☐ A third correct value of x with explanation

Solution:

The graph of g has a point of inflection at $x = -3$, $x = -1$, and $x = 2$.

$g'(x) = f(x)$ changes from increasing to decreasing at $x = -3$ and $x = 2$, and $g'(x) = f(x)$ changes from decreasing to increasing at $x = -1$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ answer
- ☐ reason

Solution:

Because $g'(x) = f(x) > 0$ on the interval $(-5, 5)$, g is increasing on the interval $(-5, 5)$. Therefore, the absolute maximum value of g on the closed interval $[-5, 5]$ occurs at the right endpoint $x = 5$.

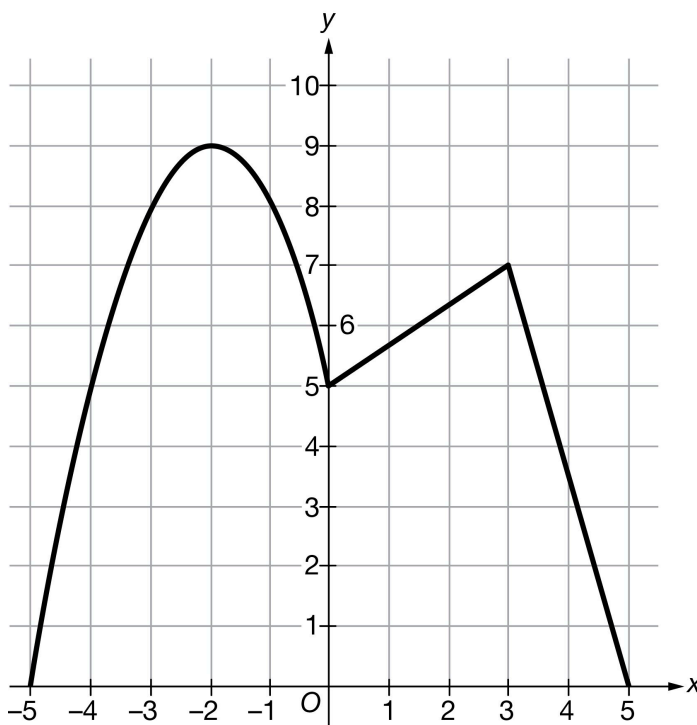
5-8

44. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f

The continuous function f is defined on the closed interval $[-5, 5]$. The graph of f consists of a parabola and two line segments, as shown in the figure above. Let g be a function such that $g'(x) = f(x)$.

- (a) Fill in the missing entries in the table below to describe the behavior of f' and f'' . Indicate Positive, Negative, or 0. Give reasons for your answers.

x	$-5 < x < -2$	$-2 < x < 0$	$0 < x < 3$	$3 < x < 5$
$f(x)$	Positive	Positive	Positive	Positive
$f'(x)$				
$f''(x)$				

- (b) There is no value of x in the open interval $(-1, 3)$ at which $f'(x) = \frac{f(3) - f(-1)}{3 - (-1)}$. Explain why this does not violate the Mean Value Theorem.
- (c) Find all values of x in the open interval $(-5, 5)$ at which the graph of g has a point of inflection. Explain your reasoning.
- (d) At what value of x does g attain its absolute maximum on the closed interval $[-5, 5]$? Give a reason for your answer.

Part A

5-8

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ behavior for $f'(x)$
- ☐ behavior for $f''(x)$
- ☐ reasons

Solution:

x	$-5 < x < -2$	$-2 < x < 0$	$0 < x < 3$	$3 < x < 5$
$f(x)$	Positive	Positive	Positive	Positive
$f'(x)$	Positive	Negative	Positive	Negative
$f''(x)$	Negative	Negative	0	0

$f'(x)$ is positive for $-5 < x < -2$ and $0 < x < 3$ because f is increasing there.

$f'(x)$ is negative for $-2 < x < 0$ and $3 < x < 5$ because f is decreasing there.

$f''(x)$ is negative for $-5 < x < -2$ and $-2 < x < 0$ because the graph of f is concave down there.

$f''(x) = 0$ for $0 < x < 3$ and $3 < x < 5$ because the graph of f is linear there.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The student response accurately includes a correct explanation.

Solution:

This does not violate the Mean Value Theorem because f is not differentiable at $x = 0$ due to a sharp corner. The Mean Value Theorem cannot be applied on the interval $-1 < x < 3$.

Part C

Two values of x must be correct without explanation to earn one point, and no incorrect values included.

A maximum of 1 point may be earned for one correct value with explanation.

A maximum of 2 points may be earned for two correct values with explanation.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

5-8



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ values of x (2 points full credit)
- ☐ explanation

Solution:

The graph of g has a point of inflection at $x = -2$, $x = 0$, and $x = 3$.

$g'(x) = f(x)$ changes from increasing to decreasing at $x = -2$ and $x = 3$, and $g'(x) = f(x)$ changes from decreasing to increasing at $x = 0$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ answer
- ☐ reason

Solution:

Because $g'(x) = f(x) > 0$ on the interval $(-5, 5)$, g is increasing on the interval $(-5, 5)$. Therefore, the absolute maximum value of g on the closed interval $[-5, 5]$ occurs at the right endpoint $x = 5$.

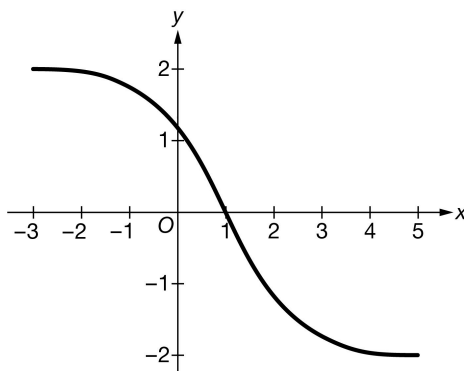
5-8

45. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g

The graph of the continuous function g is shown above for $-3 \leq x \leq 5$. The function g is twice differentiable except at $x = 1$.

Let f be the function with $f(1) = 3$ and derivative given by $f'(x) = (6x^2 - 5x)e^x$.

- (a) Find the x -coordinate of each critical point of f . Classify each critical point as the location of a relative minimum, a relative maximum, or neither. Justify your answers.
- (b) Find all values of x at which the graph of f has a point of inflection. Give reasons for your answers.

Fill in the missing entries in the table below to describe the behavior of g' and g'' on the interval $-3 \leq x \leq 5$. Indicate Positive or Negative. Give reasons for your answers.

x	$x = -3$	$-3 < x < 1$	$x = 1$	$1 < x < 5$	$x = 5$
$g(x)$	2	Positive	0	Negative	-2
$g'(x)$	0		$-\frac{3}{2}$		0
$g''(x)$	0		Undefined		0

- (d) Let h be the function defined by $h(x) = f(x)g(x)$. Is h increasing or decreasing on the interval $1 < x < 5$? Give a reason for your answer.

Part A

Note: Sign charts are a useful tool to investigate and summarize the behavior of a function. By itself a sign chart for $f'(x)$ or $f''(x)$ is not a sufficient response for a justification.

The first point requires reference to $f'(x) = 0$.

5-8

A maximum of 1 out of 3 points is earned for only one correct critical point with correct identification and justification, and no incorrect critical points are included.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ critical points
- ☐ relative maximum at $x = 0$ with justification.
- ☐ relative minimum at $x = \frac{5}{6}$ with justification.

Solution:

$$f'(x) = (6x^2 - 5x)e^x = 0 \Rightarrow 6x^2 - 5x = 0 \Rightarrow x(6x - 5) = 0 \Rightarrow x = 0, x = \frac{5}{6}$$

f has a relative maximum at $x = 0$ because $f'(x)$ changes from positive to negative there.

f has a relative minimum at $x = \frac{5}{6}$ because $f'(x)$ changes from negative to positive there.

Part B

Note: Sign charts are a useful tool to investigate and summarize the behavior of a function. By itself a sign chart for $f'(x)$ or $f''(x)$ is not a sufficient response for a justification.

A maximum of 1 out of 2 points is earned if only one point of inflection with reason and no incorrect points of inflection.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $f''(x)$
- ☐ answers with reasons

Solution:

$$\begin{aligned} f''(x) &= (12x - 5)e^x + (6x^2 - 5x)e^x = e^x(6x^2 + 7x - 5) = 0 \\ \Rightarrow 6x^2 + 7x - 5 &= (3x + 5)(2x - 1) = 0 \\ \Rightarrow x &= -\frac{5}{3} \text{ and } x = \frac{1}{2} \end{aligned}$$

Because $f''(x)$ changes sign at $x = -\frac{5}{3}$ and $x = \frac{1}{2}$, the graph of f has points of inflection there.

Part C

5-8

A maximum of 1 out of 2 points is earned if only behavior for $g'(x)$ with reason OR behavior for $g''(x)$ with reason.

Select a point value to view scoring criteria, solutions, and/or examples to score the response



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ behavior for $g'(x)$ and $g''(x)$
- ☐ reasons

Solution:

x	$x = -3$	$-3 < x < 1$	$x = 1$	$1 < x < 5$	$x = 5$
$g(x)$	2	Positive	0	Negative	-2
$g'(x)$	0	Negative	$-\frac{3}{2}$	Negative	0
$g''(x)$	0	Negative	Undefined	Positive	0

$g'(x)$ is negative for $-3 < x < 1$ and $1 < x < 5$ because g is decreasing there.

$g''(x)$ is negative for $-3 < x < 1$ because the graph of g is concave down there, and $g''(x)$ is positive for $1 < x < 5$ because the graph of g is concave up there.

Part D

The response is not eligible for the second point with an incorrect product rule.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ product rule
- ☐ answer with reason

Solution:

$$h'(x) = f'(x)g(x) + f(x)g'(x)$$

Because $f(1) = 3$ and $f'(x) > 0$ on $1 < x < 5$, it follows that $f(x) > 0$ on $1 < x < 5$. Because $g(x) < 0$ and $g'(x) < 0$ on $1 < x < 5$, it follows that $h'(x) < 0$. Thus h is decreasing on the interval $1 < x < 5$.

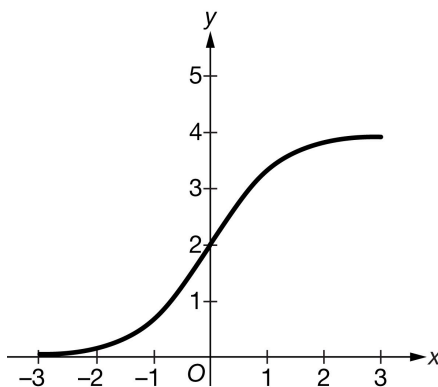
5-8

46. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g

The graph of the continuous function g is shown above for $-3 \leq x \leq 3$. The function g is twice differentiable, except at $x = 0$.

Let f be the function with $f(0) = -1$ and derivative given by $f'(x) = (x^2 - \frac{5}{4})e^x$.

- Find the x -coordinate of each critical point of f . Classify each critical point as the location of a relative minimum, a relative maximum, or neither. Justify your answers.
- Find all values of x at which the graph of f has a point of inflection. Give reasons for your answers.
- Fill in the missing entries in the table below to describe the behavior of g' and g'' on the interval $-3 \leq x \leq 3$. Indicate Positive or Negative. Give reasons for your answers.

x	$x = -3$	$-3 < x < 0$	$x = 0$	$0 < x < 3$	$x = 3$
$g(x)$	0	Positive	2	Positive	4
$g'(x)$	0		2		0
$g''(x)$	0		Undefined		0

- Let h be the function defined by $h(x) = f(x)g(x)$. Is h increasing or decreasing at $x = 0$? Give a reason for your answer.

Part A

Note: Sign charts are a useful tool to investigate and summarize the behavior of a function. By itself a sign chart for $f'(x)$ or $f''(x)$ is not a sufficient response for a justification.

The first point requires reference to $f'(x) = 0$.

A maximum of 1 out of 3 points is earned for only one correct critical point with correct identification and justification, and no incorrect critical points are included.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

5-8



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ critical points
- ☐ relative maximum at $x = -\frac{\sqrt{5}}{2}$ with justification
- ☐ relative minimum at $x = \frac{\sqrt{5}}{2}$ with justification

Solution:

$$f'(x) = (x^2 - \frac{5}{4})e^x = 0 \Rightarrow x^2 = \frac{5}{4} \Rightarrow x = -\frac{\sqrt{5}}{2}, x = \frac{\sqrt{5}}{2}$$

f has a relative maximum at $x = -\frac{\sqrt{5}}{2}$ because $f'(x)$ changes from positive to negative there.

f has a relative minimum at $x = \frac{\sqrt{5}}{2}$ because $f'(x)$ changes from negative to positive there.

Part B

Note: Sign charts are a useful tool to investigate and summarize the behavior of a function. By itself a sign chart for $f'(x)$ or $f''(x)$ is not a sufficient response for a justification.

A maximum of 1 out of 2 points is earned if only one point of inflection with reason and no incorrect points of inflection.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ $f''(x)$
- ☐ answers with reasons

Solution:

$$f''(x) = 2xe^x + (x^2 - \frac{5}{4})e^x = e^x(x^2 + 2x - \frac{5}{4}) = 0$$

$$\Rightarrow x^2 + 2x - \frac{5}{4} = 0$$

$$\Rightarrow x = \frac{-2 \pm \sqrt{4 - 4(1)(-\frac{5}{4})}}{2} = \frac{-2 \pm \sqrt{9}}{2} = \frac{-2 \pm 3}{2}$$

$$\Rightarrow x = -\frac{5}{2} \text{ and } x = \frac{1}{2}$$

Because $f''(x)$ changes sign at $x = -\frac{5}{2}$ and $x = \frac{1}{2}$, the graph of f has points of inflection there.

5-8

Part C

A maximum of 1 out of 2 points is earned if only behavior for $g'(x)$ with reason OR behavior for $g''(x)$ with reason.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ behavior for $g'(x)$ and $g''(x)$
- ☐ reasons

Solution:

x	$x = -3$	$-3 < x < 0$	$x = 0$	$0 < x < 3$	$x = 3$
$g(x)$	0	Positive	2	Positive	4
$g'(x)$	0	Positive	2	Positive	0
$g''(x)$	0	Positive	Undefined	Negative	0

$g'(x)$ is positive for $-3 < x < 0$ and $0 < x < 3$ because g is increasing there.

$g''(x)$ is positive for $-3 < x < 0$ because the graph of g is concave up there, and $g''(x)$ is negative for $0 < x < 3$ because the graph of g is concave down there.

Part D

The response is not eligible for the second point with an incorrect product rule.

Select a point value to view scoring criteria, solutions, and/or examples to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ product rule
- ☐ answer with reason

Solution:

$$h'(x) = f'(x)g(x) + f(x)g'(x)$$

$$h'(0) = f'(0)g(0) + f(0)g'(0)$$

Because $f(0) < 0$, $f'(0) < 0$, $g(0) > 0$, and $g'(0) > 0$, it follows that $h'(0) < 0$ and h is decreasing at $x = 0$.

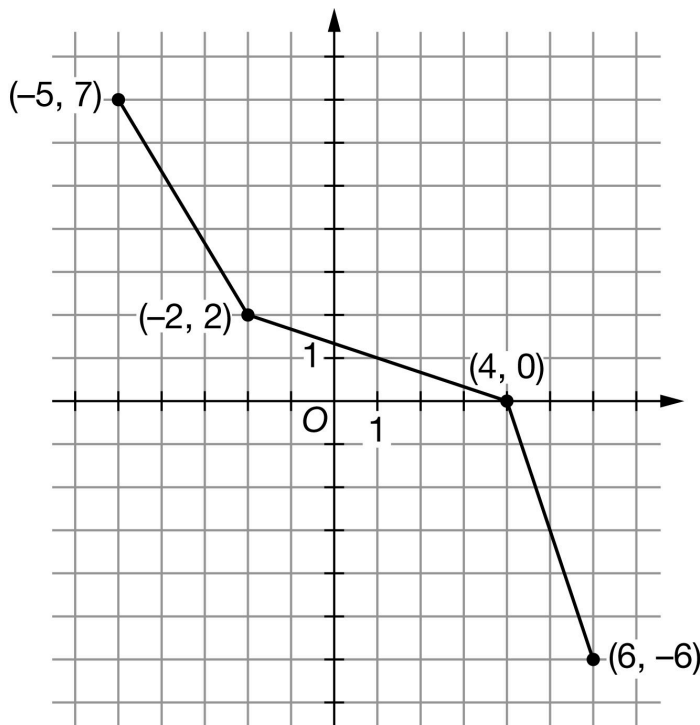
5-8

47. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f

The graph of the function f on the interval $-5 \leq x \leq 6$ consists of three line segments, as shown in the figure above. Let h be the function defined by $h(x) = \int_{-2}^x f(t) dt$ for $-5 \leq x \leq 6$.

- For the function h , is $x = 4$ the location of a relative minimum, a relative maximum, or neither? Give a reason for your answer.
- Find the absolute maximum value of h on the interval $-5 \leq x \leq 6$. Justify your answer.
- On what open intervals contained in $-5 < x < 6$ is the graph of h both increasing and concave down? Give a reason for your answer.
- Find the value of $h''(-2)$, or explain why it does not exist.
- Find $\lim_{x \rightarrow -2} \frac{h(x)}{x-2}$. Show the computations that lead to your answer.
- Let k be the function defined by $k(x) = h(x^2)$. Find the value of $k'(\sqrt{5})$ or explain why it does not exist. Show the computations that lead to your answer. (Note: $\sqrt{5}$ can be keyboarded as sqrt(5))

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

5-8

The response must include the wording local maximum or relative maximum. We also need to see a connection to h . The obvious is $h' = f$. A response of local maximum with the explanation: " f (or the graph) changes from positive to negative" is an implied connection. A response of local maximum with explanation: " h' changes from positive to negative," without the connection of $h' = f$ will not earn the point.

Simply saying " h . changes from increasing to decreasing" is not enough to earn the point.

In this situation, appealing to the Second Derivative Test by claiming that $h''(4) < 0$ is an incorrect statement, so a response with this reason does not earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 4$ is the location of a relative maximum of h because $h'(x) = f(x)$ changes from positive to negative at this point.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The only critical point method:

The first point is for the statement that $x = 4$ is the only critical point. (f is positive on $(-5, 4)$ and negative on $(4, 6)$ is another way to state this.)

Note: Stating that $x = 4$ is the only relative maximum critical point is not enough.

The second point is for arguing that $x = 4$ then produces the absolute maximum. Note that the response does not have to repeat the justification of $x = 4$ being a relative maximum from part (a). If this is not in part (a), then it needs to be here.

The second point cannot be earned without the first point.

The third point is only earned if the first two points are earned and the student presents the correct value of the maximum.

The presentation of a coordinate point $(4, 6)$ as the answer does not earn the third point.

The Candidates Test method:

The first point is for the consideration of $x = 4$. This consideration is more than just the statement $x = 4$. The response must use $x = 4$ and an area calculation in trying to find $h(4)$.

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

Using the y -values of the coordinate points from $f(x)$ is not an attempt to find an area and does not earn this point. These may be seen in a table or in student work as the difference of $f(x)$ values.

The second point is for consideration of the endpoints. The words "must check the endpoints" is not enough.

Both endpoints must be used along with an attempt to find areas for $h(-5)$ and $h(6)$.

5-8

The response cannot simply write the integrals with the correct upper limits. Attempts at both areas must be made.

Using the y -values of the coordinate points from $f(x)$ is not an attempt to find an area and does not earn this point. These may be seen in a table or in student work as the difference of $f(x)$ values.

Common errors that show an attempt to find area:

· $h(6) = 12$ (adding the triangle areas)

· $Ah(-5) = -\frac{15}{2}$ (finding the area of the triangle and forgetting the rectangle). These can be used to earn the second point.

A correct argument that eliminates $x = -5$ by stating $h(-5)$ is negative will earn the second point as long as $x = 6$ is also addressed.

Note: A Candidates Test chart/table that includes $x = -5$, $x = 4$, and $x = 6$, with at least two correct areas and no supporting work will earn the first two points.

The third point is only earned if the first two points are earned and the student presents the correct answer and there are no numerical calculation mistakes in intermediate work.

A coordinate point $(4, 6)$ is not an acceptable answer.

✓

0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Identifies unique critical point
- ☐ Justifies critical point as location of absolute maximum
- ☐ Answer with justification
- ☐ Considers $x = 4$
- ☐ Considers both endpoints
- ☐ Answer with justification

Solution:

Because $x = 4$ is the only point on the interval $-5 \leq x \leq 6$ where $h'(x) = f(x) = 0$, $x = 4$ is the only critical point of h on the interval $-5 \leq x \leq 6$.

Because $x = 4$ is also the location of a relative maximum of h , $h(4)$ is the absolute maximum for h on $-5 \leq x \leq 6$.

$$h(4) = \int_{-2}^4 f(t) dt = 6$$

Therefore, the absolute maximum value of h on $-5 \leq x \leq 6$ is $h(4) = 6$.

Alternate solution:

The absolute maximum value will occur at an endpoint or a critical point that corresponds to a relative maximum.

$x = 4$ is the only critical point in the interval because it is the only point where $h'(x) = f(x) = 0$.

5-8

$$h(4) = \int_{-2}^4 f(t) \, dt = 6$$

$$h(-5) = \int_{-2}^{-5} f(t) \, dt = -\int_{-5}^{-2} f(t) \, dt = -13.5$$

$$h(6) = \int_{-2}^6 f(t) \, dt = 0$$

Therefore, the absolute maximum value of h on $-5 \leq x \leq 6$ is $h(4) = 6$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for providing the interval $(-5, 4)$ or inequality $-5 < x < 4$.

We will allow $[-5, 4]$, $(-5, 4]$, $[-5, 4)$, or an equivalent inequality.

Presenting $(-5, -2) \cup (-2, 4)$ with correct reasoning earns one out of the two points. Any accepted version (open parentheses or brackets) earns the point. The “ $\cup$ ” is not necessary. Be careful, one bracket on either -2 is actually the entire interval and is eligible for both points.

The second point is for the reason.

The student must earn the first point to be eligible for the second point.

The response which reasons “ h' is positive and decreasing” or “ h' is positive and h'' is negative” will earn the second point if the connection $h' = f$ was made here or in a previous part of the problem. If there is no such connection, and the response reasons about the function h , then the second point is not earned.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

h is increasing where $h' = f$ is positive.

h is concave down where $h' = f$ is decreasing.

The graph of h is both increasing and concave down on the open interval $(-5, 4)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the consideration of $h''(x) = f'(x)$.

We are not allowing an import of a previous connection between h'' and f' .

5-8

The explicit connection, $h''(x) = f'(x)$, earns the first point.

It is not assumed that a response which only presents $h'(x) = f(x)$ has made the connection between h'' and f' .

The second point is for the answer with explanation.

This must include "Does not exist" and one of the following:

- limits involving the difference quotient that appear in these scoring guidelines,
- limits involving h' as the argument of the limit,
- or some correct verbal description of the characteristic of the graph of f at $x = -2$.

Note that all of the above descriptions earn the first point for the implicit consideration of $h''(x)$. Disregard notation errors.

Incorrect limit calculations will not earn the second point.

Acceptable verbal descriptions: f' does not exist because f (or "the graph") has a/is a (blank). Fill in the blank with corner, sharp point, sharp turn, kink, cusp, apex of two segments, or no distinct tangent line. We will not accept: a node, doink, peak, bounce, vertex, absolute value, abrupt change in slope, not differentiable, no slope.

Note that the phrase or description must be connected to the graph of f . The phrase $h''(-2)$ does not exist because there is a cusp will not earn this point. Whereas, the phrase $h''(-2)$ does not exist because there is a cusp on f is acceptable and earns both points.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Considers $f'(x)$
- ☐ Answer with explanation

Solution:

$$h''(x) = f'(x)$$

Because $\lim_{x \rightarrow -2^-} \frac{f(x) - f(-2)}{x - (-2)} = -\frac{5}{3}$ and $\lim_{x \rightarrow -2^+} \frac{f(x) - f(-2)}{x - (-2)} = -\frac{1}{3}$, $f'(-2) = h''(-2)$ does not exist.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is for the correct answer with supporting work.

The response must include the answer with some correct supporting work.

The continuity of h does not need to be stated.

An answer of $\frac{h(-2)}{-4}$ will not earn the point.

The answer of 0 with no supporting calculations will not earn the point.

5-8

The minimal answer is $\frac{0}{-4}$.

Copy/calculation errors in the denominator will not earn the point.

Disregard limit notation errors.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

h is continuous. $\Rightarrow \lim_{x \rightarrow -2} h(x) = h(-2)$

$$\lim_{x \rightarrow -2} \frac{h(x)}{x-2} = \frac{h(-2)}{-2-2} = \frac{0}{-4} = 0$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the correct derivative of $k : h'(x^2) \cdot 2x$.

This point is earned for any of $h'(x^2) \cdot 2x$, $f(x^2) \cdot 2x$, $f(5) \cdot 2\sqrt{5}$, or $-3 \cdot 2\sqrt{5}$, provided there are no previous errors shown.

An incorrect chain rule is not eligible for the third point but could still earn the second point.

The second point is for the correct Fundamental Theorem of Calculus (FTC) connection.

$h'(x^2) = f(x^2)$, $h'(5) = f(5)$, or $h'(5) = -3$.

This point would be earned with a response of $f(5) \cdot 2\sqrt{5}$ or $-3 \cdot 2\sqrt{5}$ even with no preceding work.

The third point is for either $-3 \cdot 2\sqrt{5}$ or $-6\sqrt{5}$. Note, the product need not be simplified.

An answer of $-6\sqrt{5}$ with no supporting work earns no points in part (f).



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

5-8

$$k'(x) = h'(x^2) \cdot 2x = f(x^2) \cdot 2x$$

$$k'(\sqrt{5}) = f(5) \cdot 2\sqrt{5} = -3 \cdot 2\sqrt{5} = -6\sqrt{5}$$

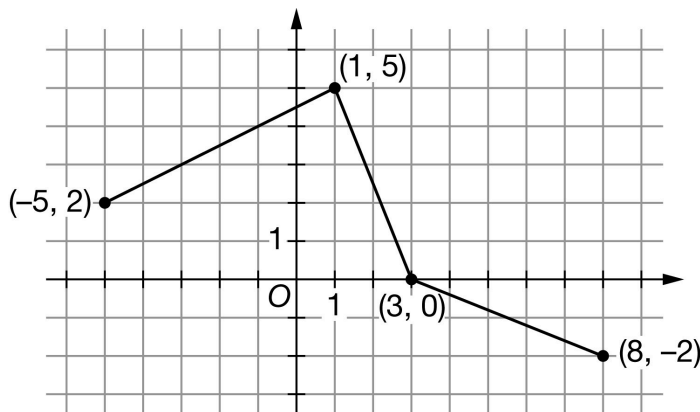
5-8

48. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g

The graph of the function g on the interval $-5 \leq x \leq 8$ consists of three line segments, as shown in the figure above. Let m be the function defined by $m(x) = \int_1^x g(t) \, dt$ for $-5 \leq x \leq 8$.

- For the function m , is $x = 3$ the location of a relative minimum, a relative maximum, or neither? Give a reason for your answer.
- Find the absolute maximum value of m on the interval $-5 \leq x \leq 8$. Justify your answer.
- On what open intervals contained in $-5 < x < 8$ is the graph of m both increasing and concave down? Give a reason for your answer.
- Find the value of $m''(1)$, or explain why it does not exist.
- Find $\lim_{x \rightarrow 1} \frac{m(x)}{x+1}$. Show the computations that lead to your answer.
- Let k be the function defined by $k(x) = m(x^2)$. Find the value of $k'(1)$ or explain why it does not exist. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response must include the wording local maximum or relative maximum. We also need to see a connection to g . The obvious is $m' = g$. A response of local maximum with the explanation: " g (or the graph) changes from positive to negative" is an implied connection. A response of local maximum with explanation: " m' changes from positive to negative" without the connection of $m' = g$ is not acceptable.

Simply saying " m changes from increasing to decreasing" is not acceptable.

In this situation, the Second Derivative Test $m''(3) < 0$ is an incorrect statement and the response with this reason does not earn the point.

5-8



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = 3$ is the location of a relative maximum of m because $m'(x) = g(x)$ changes from positive to negative there.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The only critical point method:

The first point is for the statement that $x = 3$ is the only critical point. (g is positive on $(-5, 3)$ and negative on $(3, 8)$ is another way to state this.)

Note: Stating that $x = 3$ is the only relative maximum critical point is not enough.

The second point is for arguing that $x = 3$ then produces the absolute maximum. Note that the response does not have to repeat the justification of $x = 3$ being a relative maximum from part (a). If this is not in part (a), then it needs to be here.

The second point cannot be earned without the first point.

The third point is only earned if the first two points are earned and the student presents the correct answer.

The presentation of a coordinate point as the answer does not earn the third point.

The Candidates Test method:

The first point is for the consideration of $x = 3$. This consideration is more than just the statement $x = 3$. The response must use $x = 3$ and an area calculation in trying to find $m(3)$.

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

Using the y -values of the coordinate points from $g(x)$ is not an attempt to find area and does not earn this point. This may be seen in a table or in student work as difference of $g(x)$ values.

The second point is for consideration of the endpoints. The words "must check the endpoints" is not enough.

Both endpoints must be used along with an attempt to find area.

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

Using the y -values of the coordinate points from $g(x)$ is not an attempt to find an area and does not earn this point. These may be seen in a table or in student work as the difference of $g(x)$ values.

Common errors that show an attempt to find area:

· $m(-5) = -9$ (finding the area of the triangle and forgetting the rectangle)

· $m(8) = 10$ (adding the triangle areas). These can be used to earn the second point.

5-8

A correct argument that eliminates $x = -5$ by stating $m(-5)$ is negative will earn the second point as long as $x = 8$ is also addressed.

Note: A Candidates Test chart/table that includes $x = -5$, $x = 3$, and $x = 8$, with at least two correct areas and no supporting work will earn the first two points.

The third point is only earned if the first two points are earned and the student presents the correct answer and there are no calculation mistakes in intermediate work.

NOTE: A coordinate point is not an acceptable answer.

✓

0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Identifies unique critical point
- ☐ Justifies critical point as location of absolute maximum
- ☐ Answer with justification
- ☐ Considers $x = 3$
- ☐ Considers both endpoints
- ☐ Answer with justification

Solution:

Because $x = 3$ is the only point on the interval $-5 \leq x \leq 8$ where $m'(x) = g(x) = 0$, $x = 3$ is the only critical point of m on the interval $-5 \leq x \leq 8$.

Because $x = 3$ is also the location of a relative maximum for m , $m(3)$ is the absolute maximum for m in $-5 \leq x \leq 8$.

$$m(3) = \int_1^3 g(t) dt = 5$$

Therefore, the absolute maximum value of m on $-5 \leq x \leq 8$ is $m(3) = 5$.

Alternate solution:

The absolute maximum value will occur at an endpoint or a critical point that corresponds to a relative maximum.

$x = 3$ is the only critical point in the interval because it is the only point where $m'(x) = g(x) = 0$.

$$m(3) = \int_1^3 g(t) dt = 5$$

$$m(-5) = \int_1^{-5} g(t) dt = -\int_{-5}^1 g(t) dt = -21$$

$$m(8) = \int_1^8 g(t) dt = 0$$

Therefore, the absolute maximum value of m on $-5 \leq x \leq 8$ is $m(3) = 5$.

5-8

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for providing the interval $(1, 3)$ or inequality $1 < x < 3$.

We will allow $[1, 3]$, $(1, 3]$, $[1, 3)$, or an equivalent inequality.

The second point is for the reason.

The student must earn the first point to be eligible for the second point.

The response which reasons m' is positive and decreasing or m' is positive and m'' is negative, will earn the second point if the connection $m' = g$ was made here or in a previous part of the problem. If there is no such connection, and the response reasons about the function m , then the second point is not earned.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

m is increasing where $m' = g$ is positive.

m is concave down where $m' = g$ is decreasing.

The graph of m is increasing and concave down on the open interval $(1, 3)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the consideration of $m''(x) = g'(x)$.

We are not allowing an import of a previous connection between m'' and g' .

The explicit connection, $m''(x) = g'(x)$, earns the first point.

It is not assumed that a response which only presents $m'(x) = g(x)$ has made that connection.

The implicit connection will be made in the explanation with a connection to g' , the differentiability of g , or the characteristic of the graph of g .

The second point is for the answer with explanation.

This must include "Does not exist" and one of the following:

- limits involving the difference quotient that appears in these scoring guidelines,
- limits involving g' as the argument of the limit,

5-8

· or some correct verbal description of the characteristic of the graph of g at $x = 1$.

Note that all of the above descriptions earn the first point for the implicit consideration of $g'(x)$. Disregard notation errors.

Incorrect limit calculations will not earn the second point.

Acceptable verbal descriptions: g' does not exist because g (or "the graph") has/is (blank). Fill in the blank with a corner, sharp point, sharp turn, kink, cusp, apex of two segments, or no distinct tangent line. We will not accept: a node, doink, peak, bounce, vertex, absolute value, abrupt change in slope, not differentiable, no slope.

Note that the phrase or description must be connected to the graph of g . The phrase $m''(1)$ does not exist because there is a cusp will not earn this point. Whereas, the phrase $m''(1)$ does not exist because there is a cusp on g is acceptable and earns both points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Considers $g'(x)$
- ☐ Answer with explanation

Solution:

$$m''(x) = g'(x)$$

Because $\lim_{x \rightarrow 1^-} \frac{g(x) - g(1)}{x - 1} = \frac{1}{2}$ and $\lim_{x \rightarrow 1^+} \frac{g(x) - g(1)}{x - 1} = -\frac{5}{2}$, $g'(1) = m''(1)$ does not exist.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is for the correct answer with supporting work.

The response must include the answer with some correct supporting work.

The continuity of m does not need to be stated.

An answer of $\frac{m(1)}{2}$ will not earn the point,

The answer of 0 with no supporting calculations will not earn the point.

The minimal answer is $\frac{0}{2}$.

Copy/calculation errors in the denominator will not earn the point. Disregard notation errors.



0	1
---	---

The student response accurately includes the criteria below.

5-8

☐ Answer**Solution:**

m is continuous. $\Rightarrow \lim_{x \rightarrow 1} m(x) = m(1)$

$$\lim_{x \rightarrow 1} \frac{m(x)}{x+1} = \frac{m(1)}{1+1} = \frac{0}{2} = 0$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the correct derivative of $k : m'(x^2) \cdot 2x$.

This point is earned for any of $m'(x^2) \cdot 2x$, $g(x^2) \cdot 2x$, $g(1) \cdot 2$, or $5 \cdot 2$, provided there are no previous errors shown.

An incorrect chain rule is not eligible for the third point but can still earn the second point.

The second point is for the correct Fundamental Theorem of Calculus (FTC) connection.

$$m'(x^2) = g(x^2), m'(1) = g(1), \text{ or } m'(1) = 5.$$

This point would be earned with a response of $g(1) \cdot 2$, or $5 \cdot 2$, even with no preceding work.

The third point is for either $5 \cdot 2$ or 10. Note, the product need not be simplified.

An answer of 10 with no supporting work earns no points in part (f).



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$k'(x) = m'(x^2) \cdot 2x = g(x^2) \cdot 2x$$

$$k'(1) = g(1) \cdot 2 = 5 \cdot 2 = 10$$

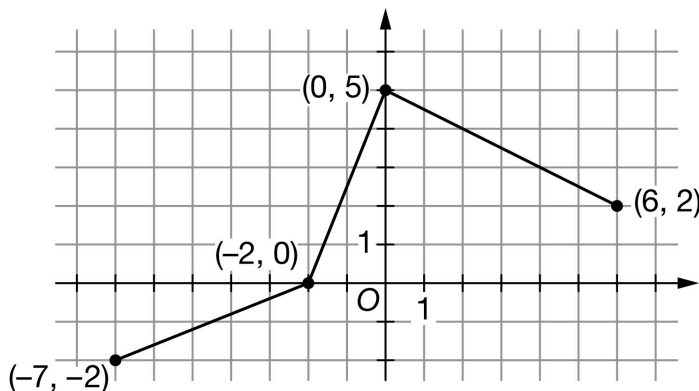
5-8

49. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of h

The graph of the function h on the interval $-7 \leq x \leq 6$ consists of three line segments, as shown in the figure above. Let r be the function defined by $r(x) = \int_0^x h(t) \, dt$ for $-7 \leq x \leq 6$.

- For the function r , is $x = -2$ the location of a relative minimum, a relative maximum, or neither? Give a reason for your answer.
- Find the absolute minimum value of r on the interval $-7 \leq x \leq 6$. Justify your answer.
- On what open intervals contained in $-7 < x < 6$ is the graph of r both increasing and concave down? Give a reason for your answer.
- Find the value of $r''(-2)$, or explain why it does not exist.
- Find $\lim_{x \rightarrow 0} \frac{r(x)}{x-4}$. Show the computations that lead to your answer.
- Let v be the function defined by $v(x) = r(x^2)$. Find the value of $v'(\sqrt{3})$ or explain why it does not exist. Show the computations that lead to your answer. (Note: $\sqrt{3}$ can be keyboarded as sqrt(3))

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response must include the wording local minimum or relative minimum. We also need to see a connection to h . The obvious is $r' = h$. A response of local minimum with the explanation: " h (or the graph) changes from negative to positive" is an implied connection. A response of local minimum with explanation: " r' changes from negative to positive," without the connection of $r' = h$ is not acceptable.

Simply saying " r changes from decreasing to increasing" is not acceptable.

In this situation, the Second Derivative Test $r''(2) > 0$ is an incorrect statement, and a response with this reason does not earn the point.

5-8



0

1

The student response accurately includes the criteria below.

☐

Answer with reason

Solution:

$x = -2$ is the location of a relative minimum of r because $r'(x) = h(x)$ changes from negative to positive there.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The only critical point method:

The first point is for the statement that $x = -2$ is the only critical point. (h is negative on $(-7, -2)$ and positive on $(-2, 6)$ is another way to state this.)

Note: Stating that $x = -2$ is the only relative minimum critical point is not enough.

The second point is for arguing that $x = -2$ then produces the absolute minimum. Note that the response does not have to repeat the justification of $x = -2$ being a relative minimum from part (a). If this is not in part (a), then it needs to be here.

The second point cannot be earned without the first point.

The third point is only earned if the first two points are earned and the student presents the correct answer.

The presentation of a coordinate point as the answer does not earn the third point.

The Candidates Test method:

The first point is for the consideration of $x = -2$. This consideration is more than just the statement $x = -2$. The response must use $x = -2$ and an area calculation in trying to find $r(-2)$.

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

Using the y -values of the coordinate points from $h(x)$ is not an attempt to find area and does not earn this point. This may be seen in a table or in student work as difference of $h(x)$ values.

A common error that shows an attempt to find area is $r(-2) = 5$. This earns the first point.

The second point is for consideration of the endpoints along with an attempt to find area. The words "must check the endpoints" is not enough.

Both endpoints must be used along with an attempt to find area.

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

Using the y -values of the coordinate points from $h(x)$ is not an attempt to find area and does not earn this point. This may be seen in a table or in student work as the difference of $h(x)$ values.

Common errors that show an attempt to find area:

5-8

· $r(-7) = 10$ (adding the triangle areas)

· $r(6) = 9$ (finding the area of the triangle and forgetting the rectangle). These can be used to earn the second point.

A correct argument that eliminates $x = 6$ by stating $r(6)$ is positive will earn the second point as long as $x = -7$ is also addressed.

Note: A Candidates Test chart/table that includes $x = -7$, $x = -2$, and $x = 6$, with at least two correct areas and no supporting work will earn the first two points.

The third point is only earned if the first two points are earned and the student presents the correct answer and there are no numerical calculation mistakes in intermediate work.

Note: A coordinate point is not an acceptable answer.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Identifies unique critical point
- ☐ Justifies critical point as location of absolute minimum
- ☐ Answer with justification
- ☐ Considers $x = -2$
- ☐ Considers both endpoints
- ☐ Answer with justification

Solution:

Because $x = -2$ is the only point on the interval $-7 \leq x \leq 6$ where $r'(x) = h(x) = 0$, $x = -2$ is the only critical point of r on the interval $-7 \leq x \leq 6$.

Because $x = -2$ is also the location of a relative minimum for r , $r(-2)$ is the absolute minimum for r on $-7 \leq x \leq 6$.

$$r(-2) = \int_0^{-2} h(t) dt = - \int_{-2}^0 h(t) dt = -5$$

Therefore, the absolute minimum value of r on $-7 \leq x \leq 6$ is $r(-2) = -5$.

Alternate solution:

The absolute minimum value will occur at an endpoint or a critical point that corresponds to a relative minimum.

$x = -2$ is the only critical point in the interval because it is the only point where $r'(x) = h(x) = 0$.

$$r(-2) = \int_0^{-2} h(t) dt = - \int_{-2}^0 h(t) dt = -5$$

$$r(-7) = \int_0^{-7} h(t) dt = - \int_{-7}^0 h(t) dt = 0$$

$$r(6) = \int_0^6 h(t) dt = 21$$

5-8

Therefore, the absolute minimum value of r on $-7 \leq x \leq 6$ is $r(-2) = -5$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for providing the interval $(0, 6)$ or inequality $0 < x < 6$.

We will allow $[0, 6]$, $(0, 6]$, $[0, 6)$, or an equivalent inequality.

The second point is for the reason.

The student must earn the first point to be eligible for the second point.

The response which reasons r' is positive and decreasing or r' is positive and r'' is negative will earn the second point if the connection $r' = h$ was made here or in a previous part of the problem. If there is no such connection, and the response reasons about the function r , then the second point is not earned.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

r is increasing where $r' = h$ is positive.

r is concave down where $r' = h$ is decreasing.

The graph of r is both increasing and concave down on the open interval $(0, 6)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for a consideration of $r''(x) = h'(x)$.

We are not allowing an import of a previous connection between r'' and h' .

The explicit connection $r''(x) = h'(x)$ earns the first point.

It is not assumed that a response which presents $r'(x) = h(x)$ has made that connection.

The implicit connection will be made in the explanation with a connection to h' , the differentiability of h , or the characteristic of the graph of h .

The second point is for the answer with explanation

This must include "Does not exist" and one of the following:

- limits involving the difference quotient that appears in these scoring guidelines,

5-8

· limits involving h' as the argument of the limit, or some correct verbal description of the characteristic of the graph of h at $x = -2$.

Note that all of the above descriptions earn the first point for the implicit consideration of $h'(x)$. Disregard notation errors.

Incorrect limit calculations will not earn the second point.

Acceptable verbal descriptions: h' does not exist because h (or "the graph") has/is (blank). Fill in the blank with a corner, sharp point, sharp turn, kink, cusp, apex of two segments, or no distinct tangent line. We will not accept: a node, doink, peak, bounce, vertex, absolute value, abrupt change in slope, not differentiable, no slope.

Note that the phrase or description must be connected to the graph of h . The phrase $r''(1)$ does not exist because there is a cusp will not earn this point. Whereas, the phrase $r''(1)$ does not exist because there is a cusp on h is acceptable and earns both points.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Considers $h'(x)$
- ☐ Answer with explanation

Solution:

Because $\lim_{x \rightarrow -2^-} \frac{h(x) - h(-2)}{x - (-2)} = \frac{2}{5}$ and $\lim_{x \rightarrow -2^+} \frac{h(x) - h(-2)}{x - (-2)} = \frac{5}{2}$, $h'(-2) = r''(-2)$ does not exist.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is for the correct answer with supporting work.

The response must include the answer with some correct supporting work.

The continuity of r does not need to be stated.

An answer of $\frac{r(0)}{-4}$ will not earn the point.

The answer of 0 with no supporting calculations will not earn the point.

The minimal answer is $\frac{0}{-4}$.

Copy/calculation errors in the denominator will not earn the point.

Disregard notation errors



0	1
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The student response accurately includes the criteria below.

5-8

☐ Answer**Solution:**

r is continuous. $\Rightarrow \lim_{x \rightarrow 0} r(x) = r(0)$

$$\lim_{x \rightarrow 0} \frac{r(x)}{x-4} = \frac{r(0)}{0-4} = \frac{0}{-4} = 0$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the correct derivative of $v : r'(x^2) \cdot 2x$.

This point is earned for any of $r'(x^2) \cdot 2x$, $h(x^2) \cdot 2x$, $h(3) \cdot 2\sqrt{3}$, or $\frac{7}{2} \cdot 2\sqrt{3}$, provided there are no previous errors shown.

An incorrect chain rule is not eligible for the third point but can still earn the second point

The second point is for the correct Fundamental Theorem of Calculus (FTC) connection.

$$r'(x^2) = h(x^2), r'(3) = h(3), \text{ or } r'(3) = \frac{7}{2}.$$

This point would be earned with a response of $h(3) \cdot 2\sqrt{3}$, or $\frac{7}{2} \cdot 2\sqrt{3}$, even with no preceding work.

The third point is for either $\frac{7}{2} \cdot 2\sqrt{3}$ or $7\sqrt{3}$. Note, the product need not be simplified.

An answer of $7\sqrt{3}$ with no supporting work earns no points in part (f).



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$v'(x) = r'(x^2) \cdot 2x = h(x^2) \cdot 2x$$

$$v'(\sqrt{3}) = h(3) \cdot 2\sqrt{3} = \frac{7}{2} \cdot 2\sqrt{3} = 7\sqrt{3}$$

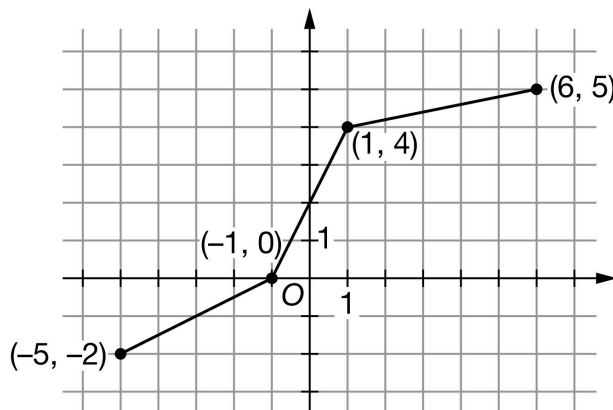
5-8

50. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of h

The graph of the function h on the interval $-5 \leq x \leq 6$ consists of three line segments, as shown in the figure above. Let p be the function defined by $p(x) = \int_1^x h(t) \, dt$ for $-5 \leq x \leq 6$.

- For the function p , is $x = -1$ the location of a relative minimum, a relative maximum, or neither? Give a reason for your answer.
- Find the absolute minimum value of p on the interval $-5 \leq x \leq 6$. Justify your answer.
- On what open intervals contained in $-5 < x < 6$ is the graph of p both decreasing and concave up? Give a reason for your answer.
- Find the value of $p''(1)$, or explain why it does not exist.
- Find $\lim_{x \rightarrow 1} \frac{p(x)}{x+4}$. Show the computations that lead to your answer.
- Let k be the function defined by $k(x) = p(x^2)$. Find the value of $k'(\sqrt{2})$ or explain why it does not exist. Show the computations that lead to your answer. (Note: $\sqrt{2}$ can be keyboarded as sqrt(2))

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response must include the wording local minimum or relative minimum. We also need to see a connection to h . The obvious is $p' = h$. A response of local minimum with the explanation: " h (or the graph) changes from negative to positive" is an implied connection. A response of local minimum with explanation: " p' changes from negative to positive," without the connection of $p' = h$ is not acceptable.

Simply saying " p changes from decreasing to increasing" is not acceptable.

In this situation, the Second Derivative Test $p''(-1) > 0$ is an incorrect statement and a response with this reason would not earn the point.

5-8



0

1

The student response accurately includes the criteria below.

☐

Answer with reason

Solution:

$x = -1$ is the location of a relative minimum of p because $p'(x) = h(x)$ changes from negative to positive there.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The only critical point method:

The first point is for the statement that $x = -1$ is the only critical point. (h is negative on $(-5, -1)$ and positive on $(-1, 6)$ is another way to state this.)

Note: Stating that $x = -1$ is the only relative minimum critical point is not enough.

The second point is for arguing that $x = -1$ then produces the absolute minimum. Note that they do not have to repeat the justification of $x = -1$ being a relative minimum from part (a). If this is not in (a), then it needs to be here.

The second point cannot be earned without the first point.

The third point is only earned if the first two points are earned and the student presents the correct answer.

The presentation of a coordinate point as the answer does not earn the third point.

The Candidates Test method:

The first point is for the consideration of $x = -1$. This consideration is more than just the statement $x = -1$. The response must use $x = -1$ and an area calculation in trying to find $p(-1)$.

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

Using the y -values of the coordinate points from $h(x)$ is not an attempt to find an area and does not earn this point. These may be seen in a table or in student work as the difference of $h(x)$ values.

A common error that shows an attempt to find area is $k(-1) = 4$. This earns the first point.

The second point is for consideration of the endpoints. The words "must check the endpoints" is not enough.

Both endpoints must be used along with an attempt to find area.

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

Using the y -values of the coordinate points from $h(x)$ is not an attempt to find area and does not earn this point. This may be seen in a table or in student work as the difference of $h(x)$ values.

Common errors that show an attempt to find area:

$\cdot p(-5) = 8$ (adding the triangle areas)

5-8

$p(6) = \frac{5}{2}$ (finding the area of the triangle and forgetting the rectangle). These can be used to earn the second point.

A correct argument that eliminates $x = 6$ by stating $p(6)$ is positive will earn the second point as long as $x = -5$ is also addressed.

Note: A Candidates Test chart/table that includes $x = -5$, $x = -1$, and $x = 6$, with at least two correct areas and no supporting work will earn the first two points.

The third point is only earned if the first two points are earned and the student presents the correct answer and there are no calculation mistakes in intermediate work.

A coordinate point is not an acceptable answer.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Identifies unique critical point
- ☐ Justifies critical point as location of absolute minimum
- ☐ Answer with justification
- ☐ Considers $x = -1$
- ☐ Considers both endpoints
- ☐ Answer with justification

Solution:

Because $x = -1$ is the only point on the interval $-5 \leq x \leq 6$ where $p'(x) = h(x) = 0$, $x = -1$ is the only critical point of p on the interval $-5 \leq x \leq 6$.

Because $x = -1$ is also the location of a relative minimum for p , $p(-1)$ is the absolute minimum for p on $-5 \leq x \leq 6$.

$$p(-1) = \int_1^{-1} h(t) dt = -\int_{-1}^1 h(t) dt = -4$$

Therefore, the absolute minimum value of p on $-5 \leq x \leq 6$ is $p(-1) = -4$.

Alternate solution:

The absolute minimum value will occur at an endpoint or a critical point that corresponds to a relative minimum.

$x = -1$ is the only critical point in the interval because it is the only point where $p'(x) = h(x) = 0$.

$$p(-1) = \int_1^{-1} h(t) dt = -\int_{-1}^1 h(t) dt = -4$$

$$p(-5) = \int_1^{-5} h(t) dt = -\int_{-5}^1 h(t) dt = 0$$

$$p(6) = \int_1^6 h(t) dt = 22.5$$

Therefore, the absolute minimum value of p on $-5 \leq x \leq 6$ is $p(-1) = -4$.

5-8

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the interval $(-5, -1)$ or inequality $-5 < x < -1$.

We will allow $[-5, -1]$, $(-5, -1]$, $[-5, -1)$, or an equivalent inequality.

The second point is for the reason.

The student must earn the first point to be eligible for the second point.

The response which reasons p' is negative and increasing or p' is negative and p'' is positive will earn the second point if the connection $p' = h$ was made here or in a previous part of the problem. If there is no such connection, and the response reasons about the function p , then the second point is not earned.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

p is decreasing where $p' = h$ is negative.

p is concave up where $p' = h$ is increasing.

The graph of p is both decreasing and concave up on the open interval $(-5, -1)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the consideration of $p''(x) = h'(x)$.

We are not allowing an import of a previous connection between p'' and h' .

The explicit connection, $p'' = h'$, earns the first point.

It is not assumed that a response which only presents $p' = h$ has made that connection.

The implicit connection will be made in the explanation with a connection to h' , the differentiability of h , or the characteristic of the graph of h .

The second point is for the answer with explanation

This must include "Does not exist" and one of the following:

- limits involving the difference quotient that appears in these scoring guidelines,
- limits involving h' as the argument of the limit,

5-8

· or some correct verbal description of the characteristic of the graph of h at $x = 1$.

Note that all of the above descriptions earn the first point for the implicit consideration of $h'(x)$. Disregard notation errors.

Incorrect limit calculations will not earn the second point.

Acceptable verbal descriptions: h' does not exist because h (or "the graph") has/is (blank). Fill in the blank with a corner, sharp point, sharp turn, kink, cusp, apex of two segments, or no distinct tangent line. We will not accept: a node, doink, peak, bounce, vertex, absolute value, abrupt change in slope, not differentiable, no slope.

Note that the phrase or description must be connected to the graph of h . The phrase $p''(1)$ does not exist because there is a cusp will not earn this point. Whereas, the phrase $p''(1)$ does not exist because there is a cusp on h is acceptable and earns both points.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Considers $h'(x)$
- ☐ Answer with explanation

Solution:

$$p''(x) = h'(x)$$

Because $\lim_{x \rightarrow 1^-} \frac{h(x) - h(1)}{x - 1} = 2$, and $\lim_{x \rightarrow 1^+} \frac{h(x) - h(1)}{x - 1} = \frac{1}{5}$, $h'(1) = p''(1)$ does not exist.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is for the correct answer with supporting work.

The response must include the answer with some correct supporting work.

The continuity of p does not need to be stated.

An answer of $\frac{p(1)}{5}$ will not earn the point.

The answer of 0 with no supporting calculations will not earn the point.

The minimal answer is $\frac{0}{5}$.

Copy/calculation errors in the denominator will not earn the point. Disregard notation errors



0	1
---	---

The student response accurately includes the criteria below.

Answer

5-8

**Solution:**

$$p \text{ is continuous. } \Rightarrow \lim_{x \rightarrow 1} p(x) = p(1) \lim_{x \rightarrow 1} \frac{p(x)}{x+4} = \frac{p(1)}{1+4} = \frac{0}{5} = 0$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the correct derivative of k : $p'(x^2) \cdot 2x$.

This point is earned for any of $p'(x^2) \cdot 2x$, $h(x^2) \cdot 2x$, $h(2) \cdot 2\sqrt{2}$, or $\frac{21}{5} \cdot 2\sqrt{2}$, provided there are no previous errors shown.

An incorrect chain rule is not eligible for the third point but can still earn the second point.

The second point is for the correct Fundamental Theorem of Calculus (FTC) connection.

$$p'(x^2) = h(x^2), p'(2) = h(2), \text{ or } p'(2) = \frac{21}{5}.$$

This point would be earned with a response of $h(2) \cdot 2\sqrt{2}$, or $\frac{21}{5} \cdot 2\sqrt{2}$, even with no preceding work.

The third point is for either $\frac{21}{5} \cdot 2\sqrt{2}$ or $\frac{42}{5}\sqrt{2}$. Note, the product need not be simplified.

An answer of $\frac{42}{5}\sqrt{2}$ with no supporting work earns no points in part (f).



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$k'(x) = p'(x^2) \cdot 2x = h(x^2) \cdot 2x$$

$$k'(\sqrt{2}) = h(2) \cdot 2\sqrt{2} = \frac{21}{5} \cdot 2\sqrt{2} = \left(\frac{42}{5}\right)\sqrt{2}$$

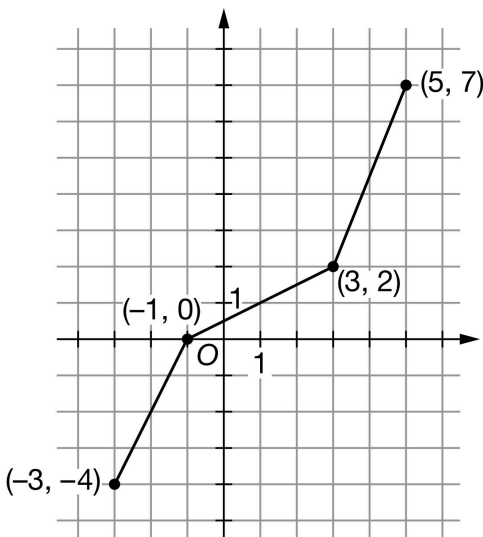
5-8

51. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g

The graph of the function g on the interval $-3 \leq x \leq 5$ consists of three line segments, as shown in the figure above. Let k be the function defined by $k(x) = \int_3^x g(t) \, dt$ for $-3 \leq x \leq 5$.

- For the function k , is $x = -1$ the location of a relative minimum, a relative maximum, or neither? Give a reason for your answer.
- Find the absolute minimum value of k on the interval $-3 \leq x \leq 5$. Justify your answer.
- On what open intervals contained in $-3 < x < 5$ is the graph of k both increasing and concave up? Give a reason for your answer.
- Find the value of $k''(-1)$, or explain why it does not exist.
- Find $\lim_{x \rightarrow 3} \frac{k(x)}{x+3}$. Show the computations that lead to your answer.
- Let h be the function defined by $h(x) = k(x^2)$. Find the value of $h'(\sqrt{3})$ or explain why it does not exist. Show the computations that lead to your answer. (Note: $\sqrt{3}$ can be keyboarded as sqrt(3))

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The response must include the wording local minimum or relative minimum. We also need to see a connection to g . The obvious is $k' = g$. A response of local minimum with the explanation: “ g (or the graph) changes from negative to positive” is an implied connection. A response of local minimum with explanation: “ k' changes from negative to positive,” without the connection of $k' = g$ is not acceptable.

5-8

Simply saying “ k changes from decreasing to increasing” is not acceptable.

In this situation, the Second Derivative Test $k''(-1) > 0$ is an incorrect statement and a response with this reason does not earn the point.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer with reason

Solution:

$x = -1$ is the location of a relative minimum of k because $k'(x) = g(x)$ changes from negative to positive there.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The only critical point method:

The first point is for the statement that $x = -1$ is the only critical point. (g is negative on $(-3, -1)$ and positive on $(-1, 5)$ is another way to state this.)

Note: Stating that $x = -1$ is the only relative minimum critical point is not enough.

The second point is for arguing that $x = -1$ then produces the absolute minimum. Note that the response does not have to repeat the justification of $x = -1$ being a relative minimum from part (a). If this is not in part (a), then it needs to be here.

The second point cannot be earned without the first point.

The third point is only earned if the first two points are earned and the student presents the correct answer.

The presentation of a coordinate point as the answer does not earn the third point.

The Candidates Test method:

The first point is for the consideration of $x = -1$. This consideration is more than just the statement $x = -1$. The response must use $x = -1$ AND an area calculation in trying to find $k(-1)$.

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

Using the y -values of the coordinate points from $g(x)$ is not an attempt to find an area and does not earn this point. These may be seen in a table or in student work as the difference of $g(x)$ values.

A common error that shows an attempt to find area is $k(-1) = 4$. This earns the first point.

The second point is for consideration of the endpoints. The words “must check the endpoints” is not enough.

Both endpoints must be used along with an attempt to find area.

The response cannot simply write the integrals with the correct upper limits. An attempt at area must be made.

5-8

Using the y -values of the coordinate points from $g(x)$ is not an attempt to find an area and does not earn this point. These may be seen in a table or in student work as the difference of $g(x)$ values.

Common errors that show an attempt to find area:

- $k(-3) = 8$ (adding the triangle areas)
- $k(5) = 5$ (finding the area of the triangle and forgetting the rectangle)
- These can be used to earn the second point.

A correct argument that eliminates $x = 5$ by stating $k(5)$ is positive will earn the second point as long as $x = -3$ is also addressed.

Note: A Candidates Test chart/table that includes $x = -3$, $x = -1$, and $x = 5$, with at least two correct areas and no supporting work will earn the first two points.

The third point is only earned if the first two points are earned and the student presents the correct answer and there are no numerical calculation mistakes in intermediate work.

A coordinate point is not an acceptable answer.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Identifies unique critical point
- ☐ Justifies critical point as location of absolute minimum
- ☐ Answer with justification
- ☐ Considers $x = -1$
- ☐ Considers both endpoints
- ☐ Answer with justification

Solution:

Because $x = -1$ is the only point on the interval $-3 \leq x \leq 5$ where $k'(x) = g(x) = 0$, $x = -1$ is the only critical point of k on the interval $-3 \leq x \leq 5$.

Because $x = -1$ is also the location of a relative minimum for k , $k(-1)$ is the absolute minimum for k on $-3 \leq x \leq 5$.

$$k(-1) = \int_3^{-1} g(t)dt = - \int_{-1}^3 g(t)dt = -4$$

Therefore, the absolute minimum value of k on $-3 \leq x \leq 5$ is $k(-1) = -4$.

Alternate solution:

The absolute minimum value will occur at an endpoint or a critical point that corresponds to a relative minimum.

$x = -1$ is the only critical point in the interval because it is the only point where $k'(x) = g(x) = 0$.

$$k(-1) = \int_3^{-1} g(t)dt = - \int_{-1}^3 g(t)dt = -4$$

5-8

$$k(-3) = \int_3^{-3} g(t) dt = - \int_{-3}^3 g(t) dt = 0$$

$$k(5) = \int_3^5 g(t) dt = 9$$

Therefore, the absolute minimum value of k on $-3 \leq x \leq 5$ is $k(-1) = -4$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the interval $(-1, 5)$ or inequality $-1 < x < 5$.

We will allow $[-1, 5]$, $(-1, 5]$, $[-1, 5)$, or an equivalent inequality.

Presenting $(-1, 3) \cup (3, 5)$ with correct reasoning earns one out of the two points. Any accepted version (open parentheses or brackets) earns the point. The “ $\cup$ ” is not necessary. Be careful, one bracket on either 3 is actually the entire interval and is eligible for both points.

The second point is for the reason.

The student must earn the first point to be eligible for the second point.

The response which reasons “ k' is positive and increasing” or “ k' is positive and k'' is positive” will earn the second point if the connection $k' = g$ was made here or in a previous part of the problem. If there is no such connection, and the response reasons about the function k , then the second point is not earned.

The interval $(-1, 7)$ with the correct justification will earn one of the two points. (The response took the y -coordinate of the right-hand point of the interval $(5, 7)$). This is considered a copy error. This is the only copy error that is allowed.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

k is increasing where $k' = g$ is positive.

k is concave up where $k' = g$ is increasing.

The graph of k is both increasing and concave up on the open interval $(-1, 5)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the consideration of $k''(x) = g'(x)$.

We are not allowing an import of a previous connection between k'' and g' .

5-8

The explicit connection, $k''(x) = g'(x)$ earns the first point.

It is not assumed that a response which only presents $k'(x) = g(x)$ has made that connection.

The implicit connection will be made in the explanation with a connection to g' , the differentiability of g , or the characteristic of the graph of g .

The second point is for the answer with explanation.

This must include “Does not exist” and one of the following:

- limits involving the difference quotient that appears in these scoring guidelines,
- limits involving g' as the argument of the limit,
- or some correct verbal description of the characteristic of the graph of g at $x = -1$.

Note that all of the above descriptions earn the first point for the implicit consideration of $g'(x)$. Disregard notation errors.

Incorrect limit calculations will not earn the second point.

Acceptable verbal descriptions: g' does not exist because g (or “the graph”) has/is (blank). Fill in the blank with a corner, sharp point, sharp turn, kink, cusp, apex of two segments, or no distinct tangent line. We will NOT accept: a node, doink, peak, bounce, vertex, absolute value, abrupt change in slope, not differentiable, no slope.

Note that the phrase or description must be connected to the graph of g . The phrase $k''(1)$ does not exist because there is a cusp will not earn this point. Whereas, the phrase $k''(1)$ does not exist because there is a cusp on g is acceptable and earns both points.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Considers $g'(x)$
- ☐ Answer with explanation

Solution:

Because $\lim_{x \rightarrow -1^-} \frac{g(x) - g(-1)}{x - (-1)} = 2$ and $\lim_{x \rightarrow -1^+} \frac{g(x) - g(-1)}{x - (-1)} = \frac{1}{2}$, $g''(-1) = k''(-1)$ does not exist.

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The point is for the correct answer with supporting work.

The response must include the answer with some correct supporting work.

The continuity of k does not need to be stated.

An answer of $\frac{k(3)}{6}$ will not earn the point.

The answer of 0 with no supporting calculations will not earn the point.

5-8

The minimal answer is $\frac{0}{6}$.

Copy/calculation errors in the denominator will not earn the point.

Disregard notation errors.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

k is continuous. $\Rightarrow \lim_{x \rightarrow 3} k(x) = k(3)$

$$\lim_{x \rightarrow 3} \frac{k(x)}{x+3} = \frac{k(3)}{3+3} = \frac{0}{6} = 0$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is for the correct derivative of $h : k'(x^2) \cdot 2x$.

This point is earned for any of $k'(x^2) \cdot 2x$, $g(x^2) \cdot 2x$, $g(3) \cdot 2\sqrt{3}$, or $2 \cdot 2\sqrt{3}$, provided there are no previous errors shown.

An incorrect chain rule is not eligible for the third point but can still earn the second point.

The second point is for the correct Fundamental Theorem of Calculus (FTC) connection.

$k'(x^2) = g(x^2)$, $k'(3) = g(3)$, or $k'(3) = 2$.

This point would be earned with a response of $g(3) \cdot 2\sqrt{3}$, or $2 \cdot 2\sqrt{3}$, even with no preceding work.

The third point is for either $2 \cdot 2\sqrt{3}$, or $4\sqrt{3}$. Note, the product need not be simplified.

An answer of $4\sqrt{3}$ with no supporting work earns no points in part (f).



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

5-8

$$h'(x) = k'(x^2) \cdot 2x = g(x^2) \cdot 2x$$

$$h'(\sqrt{3}) = g(3) \cdot 2\sqrt{3} = 2 \cdot 2\sqrt{3} = 4\sqrt{3}$$

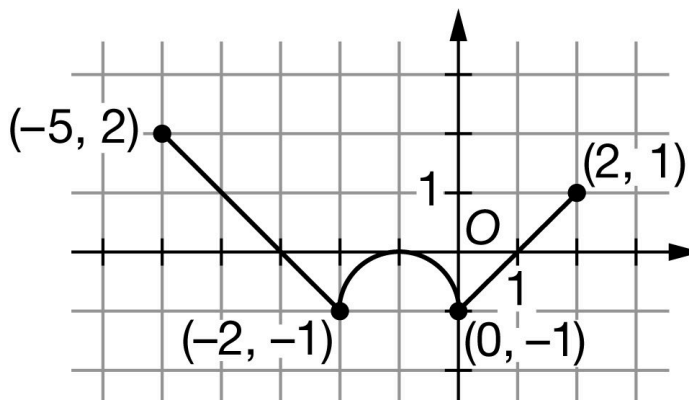
5-8

52. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g

The continuous function g has domain $-5 \leq x \leq 2$. The graph of g , consisting of two line segments and a semicircle, is shown in the figure above. The graph of g has a horizontal tangent at $x = -1$. Let h be the function defined by

$$h(x) = \int_{-2}^x g(t) \, dt \text{ for } -5 \leq x \leq 2.$$

- Find the x -coordinate of each critical point of h on the interval $-5 < x < 2$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for h . Justify each of your answers.
- For $-5 < x < 2$, on what open intervals is h increasing and concave up? Give a reason for your answer.
- Find the value of $h(0)$. Show the computations that lead to your answer.
- Find the value of $h(-5)$. Show the computations that lead to your answer.
- Find the value of $h''(-3)$, or explain why it does not exist.
- Must there exist a value of p , for $-3 < p < 1$, such that $h'(p)$ is equal to the average rate of change of h over the interval $-3 \leq x \leq 1$? Justify your answer.
- Find $\lim_{x \rightarrow -2} \frac{h'(x)}{4x}$. Show the computations that lead to your answer.
- The function f is defined by $f(x) = h(-2x)$. Find $f'(1)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = -3$, $x = -1$, and $x = 1$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-5, 2)$, or if any critical values are missing.

5-8

Disregard presentations of the endpoints or any other numbers outside the open interval $(-5, 2)$. The point is still earned provided the three critical points $x = -3$, $x = -1$ and $x = 1$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-3, 0)$, $(-1, 0)$ and $(1, 0)$, read the x -coordinates, and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$h'(x) = g(x) = 0$ at $x = -3$, $x = -1$, and $x = 1$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on h' is eligible for all three points.

If a response gives justifications based only on g , then:

- the response is eligible for all three points, if the response presents the connection between $h'(x)$ and $g(x)$, (that is, $h' = g$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of g or h' as $h' = g$, throughout all parts of this question.
- the response is eligible for at most two of the three points, if the response does not present the connection $h'(x) = g(x)$. Any additional error (for example, not justifying that at $x = -3$, h has a relative maximum) is eligible for at most one of the three points.

Read references to “the graph” as references to $g(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with h' produces a response that would earn all three points.

A response that discusses h' at a particular point (e.g., “ $h'(1)$ changes from negative to positive”) when the discussion should have been about $h'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. $h(x)$ values for $x = -3$, -1 , 1 must be included in the table or list of values. The correct values of relevant inputs include:

· $h(-5) = -\frac{3}{2}$

· $h(-3) = \frac{1}{2}$

5-8

$$\cdot h(-1) = \frac{\pi}{4} - 1$$

$$\cdot h(1) = \frac{\pi}{2} - \frac{5}{2}$$

$$\cdot h(2) = \frac{\pi}{2} - 2$$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

· At $x = -3$, h has a relative maximum because $h' = g$ is positive for $x < -3$ and $h' = g$ is negative for $x > -3$.

· At $x = -1$, h has neither because $h' = g$ is negative for $x < -1$ and $h' = g$ is negative for $x > -1$.

· At $x = 1$, h has a relative minimum because $h' = g$ is negative for $x < 1$ and $h' = g$ is positive for $x > 1$.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Relative maximum with justification
- ☐ Neither with justification
- ☐ Relative minimum with justification

Solution:

At $x = -3$, h has a relative maximum because $h'(x) = g(x)$ changes from positive to negative there.

At $x = -1$, h has neither because $h'(x) = g(x)$ does not change sign there.

At $x = 1$, h has a relative minimum because $h'(x) = g(x)$ changes from negative to positive there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, $[1, 2]$, $(1, 2]$, $[1, 2)$, and $(1, 2)$ all earn the first point.

In order to be eligible for the second point, the first point must have been earned.

A response referencing only h' is eligible to earn the second point.

A response referencing only g is not eligible to earn the second point, unless the response has established the link between $g(x)$ and $h'(x)$ previously in part (a) or part (b), in which case the response is eligible to earn the second point.

The reason “ h' is increasing” can also be expressed by either “ h'' is positive” or “the slope of h' is positive.”

A response referencing just “the function” or “it” or “the slope” is vague and does not earn the second point. Note that responses of this type are eligible for a maximum of one point in part (b) and one point in part (c); that is, a response of this type fails to earn points in both parts (b) and (c).

5-8



0	1	2
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The student response accurately includes **both** of the criteria below.

☐ Answer

☐ Reason

Solution:

h is increasing where $h' = g$ is positive.

h is concave up where $h' = g$ is increasing.

h is increasing and concave up on the open interval $(1, 2)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Correct supporting work is needed to earn this point.

The correct decimal value is $h(0) = -0.429$. If this decimal value is given, it must be correct, rounded or truncated, to three decimal places.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h(0) = -\left(2 \cdot 1 - \frac{\pi}{2} \cdot 1^2\right) = -\left(2 - \frac{\pi}{2}\right) = \frac{\pi}{2} - 2$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $\frac{1}{2} - 2$ earns both points. This is the minimal amount of work that earns both points.

As indicated by this minimal answer, the value of an integral (that is, -2 or $-\frac{1}{2}$) does not need to be explicitly linked with an integral to earn the first point.

A response of $h(-5) = -\frac{3}{2}$ with no supporting work earns the second point only.

5-8



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ $\int_{-5}^{-3} g(t) dt = 2$ – OR – $\int_{-3}^{-2} g(t) dt = -\frac{1}{2}$
- ☐ Answer

Solution:

$$\begin{aligned} h(-5) &= \int_{-2}^{-5} g(t) dt = -\int_{-5}^{-2} g(t) dt \\ &= -\int_{-5}^{-3} g(t) dt - \int_{-3}^{-2} g(t) dt = -(2) - \left(-\frac{1}{2}\right) = -\frac{3}{2} \end{aligned}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $h''(-3) = -1$ or $g'(-3) = -1$ or even simply stating a numerical value of -1 clearly marked as the answer earns the point.

No supporting work is required because the answer of -1 can easily be read from the graph.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$$h''(x) = g'(x)$$

$$h''(-3) = g'(-3) = \frac{2 - (-1)}{-5 - (-2)} = -1$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, responses must indicate that h is continuous and differentiable. An ambiguous reference to h (e.g., “the function is continuous and differentiable”) does not earn the first point. The interval on which h is continuous and differentiable does not need to be stated. If an interval is given, it must be correct with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., either by referencing the Mean Value Theorem or by restating the statement in the stem of the

5-8

problem). Examples include:

- “Yes, $h(x)$ is continuous and differentiable so MVT applies” earns both points.
- “Yes, $h(x)$ is continuous and differentiable, therefore there must be a value where $h'(p)$ equals the average rate of change of $h(x)$ ” earns both points.
- “Yes, $h'(x)$ (the derivative of $h(x)$) is continuous and differentiable, therefore there must be a value where $h'(p)$ equals the average rate of change of $h(x)$ ” does not earn the first point, but does earn the second point.
- “Yes because of MVT” does not earn the first point, but does earn the second point.
- “Yes, because $h(x)$ is continuous and differentiable” earns the first point, but not the second point.
- “Yes, because $h(x)$ is continuous” earns neither point.
- “Yes, because $h(x)$ is differentiable” earns neither point.
- “Yes” earns neither point.

A response of “No” earns neither point, regardless of any other explanation including any reference to MVT.

Special case: a response that claims “ h is not differentiable, so therefore the MVT does not apply” earns one of the two points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

$$h' = g \Rightarrow h \text{ is differentiable on } -3 \leq x \leq 1. \Rightarrow h \text{ is continuous on } -3 \leq x \leq 1.$$

Therefore, the Mean Value Theorem can be applied to h on the interval $-3 \leq x \leq 1$ to guarantee that there exists a value of p , for $-3 < p < 1$, such that $h'(p)$ equals the average rate of change of h over the interval $-3 \leq x \leq 1$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $\lim_{x \rightarrow 2} \frac{h'(x)}{4x} = \frac{1}{8}$ earns the point, as does just stating the answer of $\frac{1}{8}$.

No additional supporting work is required.

The response does not need to state that h' is continuous. If the response does state that h' is continuous, an explanation is not required.



0	1
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5-8

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h' = g \Rightarrow h' \text{ is continuous.} \Rightarrow \lim_{x \rightarrow -2} h'(x) = h'(-2)$$

$$\lim_{x \rightarrow -2} \frac{h'(x)}{4x} = \frac{h'(-2)}{4 \cdot (-2)} = \frac{g(-2)}{-8} = \frac{-1}{-8} = \frac{1}{8}$$

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x = 1$ is substituted for x , then $x = 1$ must appear unsimplified to earn the first point. Responses that earn the point include:

$$\begin{aligned} &\cdot h'(-2x) \cdot (-2) \\ &\cdot g(-2x) \cdot (-2) \\ &\cdot h'(-2 \cdot 1) \cdot (-2) \\ &\cdot g(-2 \cdot 1) \cdot (-2) \end{aligned}$$

Responses that do not earn the point include:

$$\begin{aligned} &\cdot h'(x) \cdot (-2) \\ &\cdot g(x) \cdot (-2) \\ &\cdot h'(-2) \cdot (-2) \\ &\cdot g(-2) \cdot (-2) \\ &\cdot h(-2x) \cdot (-2) \end{aligned}$$

The second point is earned for evidence of attempting to apply the Fundamental Theorem of Calculus as evidenced by linking h' and g either directly or indirectly in the presented derivative of $f(x)$ or in $f'(1)$. For example, the second point is earned by $g(-2x)$ or by $g(-2 \cdot 1)$.

The second point may be earned without earning the first point. For example, $h'(-2x) = g(-2x)$ does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because h' and g are linked.

The second point may be earned by going directly from $h'(-2)$ to -1 . The function g does not have to be stated explicitly. For example, $h'(-2 \cdot 1) \cdot (-2) = -1 \cdot (-2)$ earns the first two points. Recall, if $x = 1$ is substituted for x , then $x = 1$ must appear unsimplified to earn the first point.

The first two points may be earned by a single expression. For example, both $g(-2x) \cdot (-2)$ and $g(-2 \cdot 1) \cdot (-2)$ earn the first two points.

The third point is earned for the correct answer of 2, simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.

5-8



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$f'(x) = h'(-2x) \cdot (-2)$$

$$= g(-2x) \cdot (-2)$$

$$f'(1) = g(-2 \cdot 1) \cdot (-2) = g(-2) \cdot (-2) = (-1) \cdot (-2) = 2$$

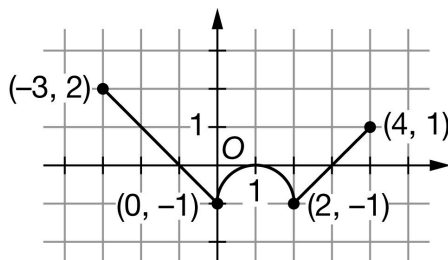
5-8

53. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f

The continuous function f has domain $-3 \leq x \leq 4$. The graph of f , consisting of two line segments and a semicircle, is shown in the figure above. The graph of f has a horizontal tangent at $x = 1$. Let g be the function defined by

$$g(x) = \int_0^x f(t) \, dt \text{ for } -3 \leq x \leq 4.$$

- Find the x -coordinate of each critical point of g on the interval $-3 < x < 4$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for g . Justify each of your answers.
- For $-3 < x < 4$, on what open intervals is g decreasing and concave up? Give a reason for your answer.
- Find the value of $g(2)$. Show the computations that lead to your answer.
- Find the value of $g(-3)$. Show the computations that lead to your answer.
- Find the value of $g''(-2)$, or explain why it does not exist.
- Must there exist a value of b , for $-3 < b < 1$, such that $g'(b)$ is equal to the average rate of change of g over the interval $-3 \leq x \leq 1$? Justify your answer.
- Find $\lim_{x \rightarrow 2} \frac{g'(x)}{4x}$. Show the computations that lead to your answer.
- The function z is defined by $z(x) = g(x^2)$. Find $z'(-2)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = -1$, $x = 1$, and $x = 3$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-3, 4)$, or if any critical values are missing.

Disregard presentations of the endpoints or any other numbers outside the open interval $(-3, 4)$. The point is still earned provided the three critical points $x = -1$, $x = 1$, and $x = 3$ are correctly listed.

5-8

If the critical points are given as ordered pairs, e.g., $(-1, 0)$, $(1, 0)$, and $(3, 0)$, read the x -coordinates, and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g'(x) = f(x) = 0 \text{ at } x = -1, x = 1, \text{ and } x = 3.$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on g' is eligible for all three points.

If a response gives justifications based only on f , then:

- the response is eligible for all three points, if the response presents the connection between $g'(x)$ and $f(x)$, (that is, $g' = f$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of f or g' as $g' = f$, throughout all parts of this question.
- the response is eligible for at most two of the three points, if the response does not present the connection $g'(x) = f(x)$. Any additional error (for example, not justifying that at $x = -1$, f has a relative maximum) is eligible for at most one of the three points.

Read references to “the graph” as references to $f(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with g' produces a response that would earn all three points.

A response that discusses g' at a particular point (e.g., “ $g'(3)$ changes from negative to positive”) when the discussion should have been about $g'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. $g(x)$ values for $x = -1, 1, 3$ must be included in the table or list of values. The correct values of relevant inputs include:

$$g(-3) = -\frac{3}{2}$$

$$g(-1) = \frac{1}{2}$$

$$g(1) = \frac{\pi}{4} - 1$$

5-8

$$g(3) = \frac{\pi}{2} - \frac{5}{2}$$

$$g(4) = \frac{\pi}{2} - 2$$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

At $x = -1$, g has a relative maximum because $g' = f$ is positive for $x < -1$ and $g' = f$ is negative for $x > -1$.

At $x = 1$, g has neither because $g' = f$ is negative for $x < 1$ and $g' = f$ is negative for $x > 1$.

At $x = 3$, g has a relative minimum because $g' = f$ is negative for $x < 3$ and $g' = f$ is positive for $x > 3$.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Relative maximum with justification
- ☐ Neither with justification
- ☐ Relative minimum with justification

Solution:

At $x = -1$, g has a relative maximum because $g'(x) = f(x)$ changes from positive to negative there.

At $x = 1$, g has neither because $g'(x) = f(x)$ does not change sign there.

At $x = 3$, g has a relative minimum because $g'(x) = f(x)$ changes from negative to positive there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, $[0, 1]$ and $(2, 3]$ or $[0, 1]$ and $(2, 3)$ both earn the first point.

In order to be eligible for the second point, the first point must have been earned.

A response referencing only g' is eligible to earn the second point.

A response referencing only f is not eligible to earn the second point, unless the response has established the link between $f(x)$ and $g'(x)$ previously in part (a) or part (b), in which case the response is eligible to earn the second point.

A response referencing just “the function” or “it” or “the slope” is vague and does not earn the second point. Note that responses of this type are eligible for a maximum of one point in part (b) and one point in part (c); that is, a response of this type fails to earn points in both parts (b) and (c).

The reason “ g' is increasing” can also be expressed by either “ g'' is positive” or “the slope of g' is positive.”

Responses giving a completely correct justification for only one of the two subintervals earn one of the two points.

5-8



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Answer

☐ Reason

Solution:

g is decreasing where $g' = f$ is negative.

g is concave up where $g' = f$ is increasing.

g is decreasing and concave up on the open intervals $(0, 1)$ and $(2, 3)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Correct supporting work is needed to earn this point.

The correct decimal value is $g(2) = -0.429$. If this decimal value is given, it must be correct, rounded or truncated, to three decimal places.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

$$g(2) = -\left(2 \cdot 1 - \frac{\pi}{2} \cdot 1^2\right) = -\left(2 - \frac{\pi}{2}\right) = \frac{\pi}{2} - 2$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $\frac{1}{2} - 2$ earns both points. This is the minimal amount of work that earns both points.

As indicated by this minimal answer, the value of an integral (that is, 2 or $-\frac{1}{2}$) does not need to be explicitly linked with an integral to earn the first point.

A response of $g(-3) = -\frac{3}{2}$ with no supporting work earns the second point only.



0	1	2
---	---	---

5-8

The student response accurately includes **both** of the criteria below.

- ☐ $\int_{-3}^{-1} f(t) dt = 2$ – OR – $\int_{-1}^0 f(t) dt = -\frac{1}{2}$
- ☐ Answer

Solution:

$$\begin{aligned} g(-3) &= \int_0^{-3} f(t) dt = -\int_{-3}^0 f(t) dt \\ &= -\int_{-3}^{-1} f(t) dt - \int_{-1}^0 f(t) dt = -2 - \left(-\frac{1}{2}\right) = -\frac{3}{2} \end{aligned}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $g''(-2) = -1$ or $f'(-2) = -1$ or even simply stating a numerical value of -1 clearly marked as the answer earns the point.

No supporting work is required because the answer of -1 can easily be read from the graph.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$$g''(x) = f'(x)$$

$$g''(-2) = f'(-2) = \frac{-1-2}{0-(-3)} = -1$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, responses must indicate that g is continuous and differentiable. An ambiguous reference to g (e.g., “the function is continuous and differentiable”) does not earn the first point. The interval on which g is continuous and differentiable does not need to be stated. If an interval is given, it must be correct with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., either by referencing the Mean Value Theorem or by restating the statement in the stem of the problem). Examples include:

- “Yes, $g(x)$ is continuous and differentiable so MVT applies” earns both points.

5-8

· “Yes, $g(x)$ is continuous and differentiable, therefore there must be a value where $g'(b)$ equals the average rate of change of $g(x)$ ” earns both points.

· “Yes, $g'(x)$ (the derivative of $g(x)$) is continuous and differentiable, therefore there must be a value where $g'(b)$ equals the average rate of change of $g(x)$ ” does not earn the first point, but does earn the second point.

· “Yes because of MVT” does not earn the first point, but does earn the second point.

· “Yes, because $g(x)$ is continuous and differentiable” earns the first point, but not the second point.

· “Yes, because $g(x)$ is continuous” earns neither point.

· “Yes, because $g(x)$ is differentiable” earns neither point.

· “Yes” earns neither point.

A response of “No” earns neither point, regardless of any other explanation including any reference to MVT.

Special case: a response that claims “ g is not differentiable, so therefore the MVT does not apply” earns one of the two points.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

$g' = f \Rightarrow g$ is differentiable on $-3 \leq x \leq 1$. $\Rightarrow g$ is continuous on $-3 \leq x \leq 1$.

Therefore, the Mean Value Theorem can be applied to g on the interval $-3 \leq x \leq 1$ to guarantee that there exists a value of b , for $-3 < b < 1$, such that $g'(b)$ equals the average rate of change of g over the interval $-3 \leq x \leq 1$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $\lim_{x \rightarrow 2} \frac{g'(x)}{4x} = \frac{-1}{8}$ earns the point, as does just stating the answer of $\frac{-1}{8}$.

No additional supporting work is required.

The response does not need to state that g' is continuous. If the response does state that g' is continuous, an explanation is not required.



0	1
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The student response accurately includes the criteria below.

Answer

5-8

**Solution:**

$$g' = f \Rightarrow g' \text{ is continuous.} \Rightarrow \lim_{x \rightarrow 2} g'(x) = g'(2)$$

$$\lim_{x \rightarrow 2} \frac{g'(x)}{4x} = \frac{g'(2)}{4 \cdot 2} = \frac{f(2)}{8} = \frac{-1}{8}$$

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x = -2$ is substituted for x , then $x = -2$ must appear unsimplified in at least one factor to earn the first point. Responses that earn the point include:

$$\cdot g'(x^2) \cdot 2x$$

$$\cdot f(x^2) \cdot 2x$$

$$\cdot g'((-2)^2) \cdot 2 \cdot (-2)$$

$$\cdot f((-2)^2) \cdot 2 \cdot (-2)$$

$$\cdot g'(4) \cdot 2 \cdot (-2)$$

$$\cdot f(4) \cdot 2 \cdot (-2)$$

$$\cdot g'((-2)^2) \cdot (-4)$$

$$\cdot f((-2)^2) \cdot (-4)$$

Responses that do not earn the point include:

$$\cdot g'(x^2) \cdot 2$$

$$\cdot f(x^2) \cdot 2$$

$$\cdot g'(x) \cdot 2x$$

$$\cdot f(x) \cdot 2x$$

$$\cdot g'(4) \cdot (-4)$$

$$\cdot f(4) \cdot (-4)$$

$$\cdot g(x^2) \cdot 2x$$

The second point is earned for evidence of attempting to apply the Fundamental Theorem of Calculus as evidenced by linking g' and f either directly or indirectly in the presented derivative of $z(x)$ or in $z'(-2)$. For example, the second point is earned by $f(x^2)$ or by $f((-2)^2)$.

The second point may be earned without earning the first point. For example, $g'(x^2) \cdot 2 = f(x^2) \cdot 2$ does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because g' and f are linked.

5-8

The second point may be earned by going directly from $g'(4)$ to 1. The function f does not have to be stated explicitly. Responses that earn the first two points include:

$$\cdot g'((-2)^2) \cdot 2 \cdot (-2) = 1 \cdot 2 \cdot (-2)$$

$$\cdot g'((-2)^2) \cdot (-4) = 1 \cdot (-4)$$

Recall, if $x = -2$ is substituted for x , then $x = -2$ must appear unsimplified in at least one factor to earn the first point.

The first two points may be earned by a single expression. For example, both $f(x^2) \cdot 2x$ and $f((-2)^2) \cdot 2 \cdot (-2)$ earn the first two points.

The third point is earned for the correct answer of -4 , simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.

Be careful of parenthesis errors in part (i). Responses of the form $g'(x)^2 \cdot 2x$ or $(g'(x))^2 \cdot 2x$ are not uncommon. These type of responses fail to earn the first point, are eligible for the second point, and fail to earn the third point.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$z'(x) = g'(x^2) \cdot 2x$$

$$= f(x^2) \cdot 2x$$

$$z'(-2) = f((-2)^2) \cdot 2 \cdot (-2) = f(4) \cdot (-4) = 1 \cdot (-4) = -4$$

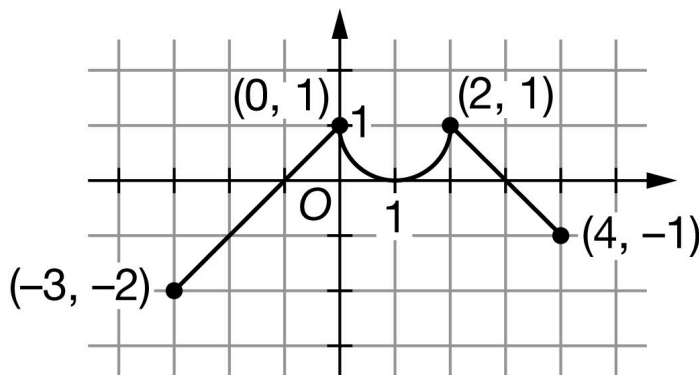
5-8

54. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of g

The continuous function g has domain $-3 \leq x \leq 4$. The graph of g , consisting of two line segments and a semicircle, is shown in the figure above. The graph of g has a horizontal tangent at $x = 1$. Let h be the function defined by

$$h(x) = \int_0^x g(t) \, dt \text{ for } -3 \leq x \leq 4.$$

- Find the x -coordinate of each critical point of h on the interval $-3 < x < 4$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for h . Justify each of your answers.
- For $-3 < x < 4$, on what open intervals is h decreasing and concave down? Give a reason for your answer.
- Find the value of $h(2)$. Show the computations that lead to your answer.
- Find the value of $h(-3)$. Show the computations that lead to your answer.
- Find the value of $h''(-1)$, or explain why it does not exist.
- Must there exist a value of r , for $-3 < r < 1$, such that $h'(r)$ is equal to the average rate of change of h over the interval $-3 \leq x \leq 1$? Justify your answer.
- Find $\lim_{x \rightarrow 2} \frac{h'(x)}{5x}$. Show the computations that lead to your answer.
- The function k is defined by $k(x) = h(x^2)$. Find $k'(-2)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = 1$, $x = 1$, and $x = 3$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-3, 4)$, or if any critical values are missing.

5-8

Disregard presentations of the endpoints or any other numbers outside the open interval $(-3, 4)$. The point is still earned provided the three critical points $x = -1$, $x = 1$, and $x = 3$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-1, 0)$, $(1, 0)$, and $(3, 0)$, read the x -coordinates, and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h'(x) = g(x) = 0 \text{ at } x = -1, x = 1, \text{ and } x = 3.$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on h' is eligible for all three points.

If a response gives justifications based only on g , then:

- the response is eligible for all three points, if the response presents the connection between $h'(x)$ and $g(x)$, (that is, $h' = g$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of g or h' as $h' = g$, throughout all parts of this question.
- the response is eligible for at most two of the three points, if the response does not present the connection $h'(x) = g(x)$. Any additional error (for example, not justifying that at $x = -1$, h has a relative minimum) is eligible for at most one of the three points.

Read references to “the graph” as references to $g(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with h' produces a response that would earn all three points.

A response that discusses h' at a particular point (e.g., “ $h'(3)$ changes from negative to positive”) when the discussion should have been about $h'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. $h(x)$ values for $x = -1$, 1 , 3 must be included in the table or list of values. The correct values of relevant inputs include:

$$\cdot h(-3) = \frac{3}{2}$$

$$\cdot h(-1) = -\frac{1}{2}$$

5-8

$$\cdot h(1) = 1 - \frac{\pi}{4}$$

$$\cdot h(3) = \frac{5}{2} - \frac{\pi}{2}$$

$$\cdot h(4) = 2 - \frac{\pi}{2}$$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

· At $x = -1$, h has a relative minimum because $h' = g$ is negative for $x < -1$ and $h' = g$ is positive for $x > -1$.

· At $x = 1$, h has neither because $h' = g$ is positive for $x < 1$ and $h' = g$ is positive for $x > 1$.

· At $x = 3$, h has a relative maximum because $h' = g$ is positive for $x < 3$ and $h' = g$ is negative for $x > 3$.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Relative minimum with justification
- ☐ Neither with justification
- ☐ Relative maximum with justification

Solution:

At $x = -1$, h has a relative minimum because $h'(x) = g(x)$ changes from negative to positive there.

At $x = 1$, h has neither because $h'(x) = g(x)$ does not change sign there.

At $x = 3$, h has a relative maximum because $h'(x) = g(x)$ changes from positive to negative there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, $[3, 4]$, $(3, 4]$, $[3, 4)$, and $(3, 4)$ all earn the first point.

In order to be eligible for the second point, the first point must have been earned.

A response referencing only h' is eligible to earn the second point.

A response referencing only g is not eligible to earn the second point, unless the response has established the link between $g(x)$ and $h'(x)$ previously in part (a) or part (b), in which case the response is eligible to earn the second point.

The reason “ h' is decreasing” can also be expressed by either “ h'' is negative” or “the slope of h' is negative.”

A response referencing just “the function” or “it” or “the slope” is vague and does not earn the second point. Note that responses of this type are eligible for a maximum of one point in part (b) and one point in part (c); that is, a response of this type fails to earn points in both parts (b) and (c).

5-8



0	1	2
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The student response accurately includes **both** of the criteria below.

☐ Answer

☐ Reason

Solution:

h is decreasing where $h' = g$ is negative.

h is concave down where $h' = g$ is decreasing.

h is decreasing and concave down on the open interval $(3, 4)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Correct supporting work is needed to earn this point.

The correct decimal value is $h(2) = 0.429$. If this decimal value is given, it must be correct, rounded or truncated, to three decimal places.



0	1
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The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h(2) = 2 \cdot 1 - \frac{\pi}{2} \cdot 1^2 = 2 - \frac{\pi}{2}$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $-\frac{1}{2} + 2$ earns both points. This is the minimal amount of work that earns both points.

As indicated by this minimal answer, the value of an integral (that is, -2 or $\frac{1}{2}$) does not need to be explicitly linked with an integral to earn the first point.

A response of $h(-3) = \frac{3}{2}$ with no supporting work earns the second point only.

5-8



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ $\int_{-3}^{-1} g(t) dt = -2$ – OR – $\int_{-1}^0 g(t) dt = \frac{1}{2}$
- ☐ Answer

Solution:

$$\begin{aligned} h(-3) &= \int_0^{-3} g(t) dt = - \int_{-3}^0 g(t) dt \\ &= - \int_{-3}^{-1} g(t) dt - \int_{-1}^0 g(t) dt = -(-2) - \frac{1}{2} = \frac{3}{2} \end{aligned}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $h'(-1) = 1$ or $g'(-1) = 1$ or even simply stating a numerical value of 1 clearly marked as the answer earns the point.

No supporting work is required because the answer of 1 can easily be read from the graph.



0	1
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The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$$h''(x) = g'(x)$$

$$h''(-1) = g'(-1) = \frac{1 - (-2)}{0 - (-3)} = 1$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, responses must indicate that h is continuous and differentiable. An ambiguous reference to h (e.g., “the function is continuous and differentiable”) does not earn the first point. The interval on which h is continuous and differentiable does not need to be stated. If an interval is given, it must be correct with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., either by referencing the Mean Value Theorem or by restating the statement in the stem of the

5-8

problem). Examples include:

- “Yes, $h(x)$ is continuous and differentiable so MVT applies” earns both points.
- “Yes, $h(x)$ is continuous and differentiable, therefore there must be a value where $h'(r)$ equals the average rate of change of $h(x)$ ” earns both points.
- “Yes, $h'(x)$ (the derivative of $h(x)$) is continuous and differentiable, therefore there must be a value where $h'(r)$ equals the average rate of change of $h(x)$ ” does not earn the first point, but does earn the second point.
- “Yes because of MVT” does not earn the first point, but does earn the second point.
- “Yes, because $h(x)$ is continuous and differentiable” earns the first point, but not the second point.
- “Yes, because $h(x)$ is continuous” earns neither point.
- “Yes, because $h(x)$ is differentiable” earns neither point.
- “Yes” earns neither point.

A response of “No” earns neither point, regardless of any other explanation including any reference to MVT.

Special case: a response that claims “ h is not differentiable, so therefore the MVT does not apply” earns one of the two points.



0	1	2
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The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

$h' = g \Rightarrow h$ is differentiable on $-3 \leq x \leq 1 \Rightarrow h$ is continuous on $-3 \leq x \leq 1$.

Therefore, the Mean Value Theorem can be applied to h on the interval $-3 \leq x \leq 1$ to guarantee that there exists a value of r , for $-3 < r < 1$, such that $h'(r)$ equals the average rate of change of h over the interval $-3 \leq x \leq 1$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $\lim_{x \rightarrow 2} \frac{h'(x)}{5x} = \frac{1}{10}$ earns the point, as does just stating the answer of $\frac{1}{10}$.

No additional supporting work is required.

The response does not need to state that h' is continuous. If the response does state that h' is continuous, an explanation is not required.



0	1
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5-8

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h' = g \Rightarrow h' \text{ is continuous.} \Rightarrow \lim_{x \rightarrow 2} h'(x) = h'(2)$$

$$\lim_{x \rightarrow 2} \frac{h'(x)}{5x} = \frac{h'(2)}{5 \cdot 2} = \frac{g(2)}{10} = \frac{1}{10}$$

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x = -2$ is substituted for x , then $x = -2$ must appear unsimplified in at least one factor in order to earn the first point. Responses that earn the point include:

$$\cdot h'(x^2) \cdot 2x$$

$$\cdot g(x^2) \cdot 2x$$

$$\cdot h'((-2)^2) \cdot 2 \cdot (-2)$$

$$\cdot g((-2)^2) \cdot 2 \cdot (-2)$$

$$\cdot h'(4) \cdot 2 \cdot (-2)$$

$$\cdot g(4) \cdot 2 \cdot (-2)$$

$$\cdot h'((-2)^2) \cdot (-4)$$

$$\cdot g((-2)^2) \cdot (-4)$$

Responses that do not earn the point include:

$$\cdot h'(x^2) \cdot 2$$

$$\cdot g(x^2) \cdot 2$$

$$\cdot h'(x) \cdot 2x$$

$$\cdot g(x) \cdot 2x$$

$$\cdot h'(4) \cdot (-4)$$

$$\cdot g(4) \cdot (-4)$$

$$\cdot h(x^2) \cdot 2x$$

The second point is earned for evidence of attempting to apply the Fundamental Theorem of Calculus as evidenced by linking h' and g either directly or indirectly in the presented derivative of $k(x)$ or in $k'(-2)$. For example, the second point is earned by $g(x^2)$ or by $g((-2)^2)$.

5-8

The second point may be earned without earning the first point. For example, $h'(x^2) \cdot 2 = g(x^2) \cdot 2$ does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because h' and Ag are linked.

The second point may be earned by going directly from $h'(4)$ to -1 . The function g does not have to be stated explicitly. Responses that earn the first two points include:

$$\cdot h'((-2)^2) \cdot 2 \cdot (-2) = -1 \cdot 2 \cdot (-2)$$

$$\cdot h'((-2)^2) \cdot (-4) = -1 \cdot (-4)$$

Recall, if $x = -2$ is substituted for x , then $x = -2$ must appear unsimplified in at least one factor to earn the first point.

The first two points may be earned by a single expression. For example, both $g(x^2) \cdot 2x$ and $g((-2)^2) \cdot 2 \cdot (-2)$ earn the first two points.

The third point is earned for the correct answer of 4 simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.

Be careful of parenthesis errors in part (i). Responses of the form $h'(x)^2 \cdot 2x$ or $(h'(x))^2 \cdot 2x$ are not uncommon. These types of responses fail to earn the first point, are eligible for the second point, and fail to earn the third point.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$k'(x) = h'(x^2) \cdot 2x$$

$$= g(x^2) \cdot 2x$$

$$k'(-2) = g((-2)^2) \cdot 2 \cdot (-2) = g(4) \cdot (-4) = -1 \cdot (-4) = 4$$

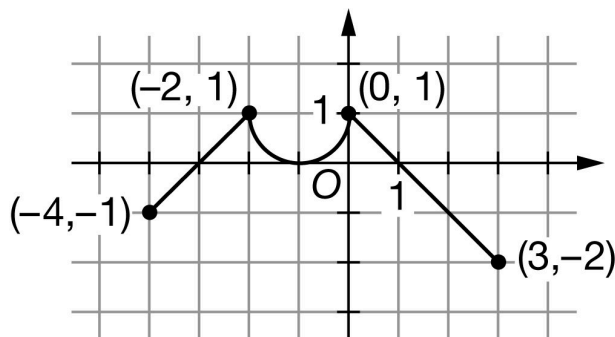
5-8

55. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of f

The continuous function f has domain $-4 \leq x \leq 3$. The graph of f , consisting of two line segments and a semicircle, is shown in the figure above. The graph of f has a horizontal tangent at $x = -1$. Let h be the function defined by

$$h(x) = \int_0^x f(t) \, dt \text{ for } -4 \leq x \leq 3.$$

- Find the x -coordinate of each critical point of h on the interval $-4 < x < 3$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for h . Justify each of your answers.
- For $-4 < x < 3$, on what open intervals is h increasing and concave down? Give a reason for your answer.
- Find the value of $h(-2)$. Show the computations that lead to your answer.
- Find the value of $h(3)$. Show the computations that lead to your answer.
- Find the value of $h''(2)$, or explain why it does not exist.
- Must there exist a value of k , for $-4 < k < 0$, such that $h'(k)$ is equal to the average rate of change of h over the interval $-4 \leq x \leq 0$? Justify your answer.
- Find $\lim_{x \rightarrow -2} \frac{h'(x)}{3x}$. Show the computations that lead to your answer.
- The function r is defined by $r(x) = h(-3x)$. Find $r'(-1)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = -3$, $x = -1$, and $x = 1$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-4, 3)$, or if any critical values are missing.

5-8

Disregard presentations of the endpoints or any other numbers outside the open interval $(-4, 3)$. The point is still earned provided the three critical points $x = -3$, $x = -1$, and $x = 1$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(-3, 0)$, $(-1, 0)$, and $(1, 0)$, read the x -coordinates, and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h'(x) = f(x) = 0 \text{ at } x = -3, x = -1, \text{ and } x = 1.$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on h' is eligible for all three points.

If a response gives justifications based only on f , then:

- the response is eligible for all three points, if the response presents the connection between $h'(x)$ and $f(x)$, (that is, $h' = f$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of f or h' as $h' = f$, throughout all parts of this question.
- the response is eligible for at most two of the three points, if the response does not present the connection $h'(x) = f(x)$. Any additional error (for example, not justifying that at $x = -3$, h has a relative minimum) is eligible for at most one of the three points.

Read references to “the graph” as references to $f(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with h' produces a response that would earn all three points.

A response that discusses h' at a particular point (e.g., “ $h'(-3)$ changes from negative to positive”) when the discussion should have been about $h'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. $h(x)$ values for $x = -3$, -1 , 1 must be included in the table or list of values. The correct values of relevant inputs include:

$$\cdot h(-4) = \frac{\pi}{2} - 2$$

$$\cdot h(-3) = \frac{\pi}{2} - \frac{5}{2}$$

5-8

$$\cdot h(-1) = \frac{\pi}{4} - 1$$

$$\cdot h(1) = \frac{1}{2}$$

$$\cdot h(3) = -\frac{3}{2}$$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

· At $x = -3$, h has a relative minimum because $h' = f$ is negative for $x < -3$ and $h' = f$ is positive for $x > -3$.

· At $x = -1$, h has neither because $h' = f$ is positive for $x < -1$ and $h' = f$ is positive for $x > -1$.

· At $x = 1$, h has a relative maximum because $h' = f$ is positive for $x < 1$ and $h' = f$ is negative for $x > 1$.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ Relative minimum with justification
- ☐ Neither with justification
- ☐ Relative maximum with justification

Solution:

At $x = -3$, h has a relative minimum because $h'(x) = f(x)$ changes from negative to positive there.

At $x = -1$, h has neither because $h'(x) = f(x)$ does not change sign there.

At $x = 1$, h has a relative maximum because $h'(x) = f(x)$ changes from positive to negative there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, $[-2, -1]$ and $(0, 1]$ or $[-2, -1)$ and $(0, 1)$ both earn the first point.

In order to be eligible for the second point, the first point must have been earned.

A response referencing only h' is eligible to earn the second point.

A response referencing only f is not eligible to earn the second point, unless the response has established the link between $f(x)$ and $h'(x)$ previously in part (a) or part (b), in which case the response is eligible to earn the second point.

The reason “ h' is decreasing” can also be expressed by either “ h'' is negative” or “the slope of h' is negative.”

A response referencing just “the function” or “it” or “the slope” is vague and does not earn the second point. Note that responses of this type are eligible for a maximum of one point in part (b) and one point in part (c); that is, a response of this type fails to earn points in both parts (b) and (c).

5-8

Responses giving a completely correct justification for only one of the two subintervals earn one of the two points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Answer
- ☐ Reason

Solution:

h is increasing where $h' = f$ is positive.

h is concave down where $h' = f$ is decreasing.

h is increasing and concave down on the open intervals $(-2, -1)$ and $(0, 1)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Correct supporting work is needed to earn this point.

The correct decimal value is $h(-2) = -0.429$. If this decimal value is given, it must be correct, rounded or truncated, to three decimal places.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$$h(-2) = -\left(2 \cdot 1 - \frac{\pi}{2} \cdot 1^2\right) = -\left(2 - \frac{\pi}{2}\right) = \frac{\pi}{2} - 2$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $\frac{1}{2} - 2$ earns both points. This is the minimal amount of work that earns both points.

As indicated by this minimal answer, the value of an integral (that is, -2 or $\frac{1}{2}$) does not need to be explicitly linked with an integral to earn the first point.

A response of $h(3) = -\frac{3}{2}$ with no supporting work earns the second point only.

5-8



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ $\int_0^1 f(t) dt = \frac{1}{2}$ – OR – $\int_1^3 f(t) dt = -2$
- ☐ Answer

Solution:

$$\begin{aligned} h(3) &= \int_0^3 f(t) dt \\ &= \int_0^1 f(t) dt + \int_1^3 f(t) dt = \frac{1}{2} + (-2) = -\frac{3}{2} \end{aligned}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $h''(2) = -1$ or $f'(2) = -1$ or even simply stating a numerical value of -1 clearly marked as the answer earns the point.

No supporting work is required because the answer of -1 can easily be read from the graph.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$$\begin{aligned} h''(x) &= f'(x) \\ h''(2) &= f'(2) = \frac{1 - (-2)}{0 - 3} = -1 \end{aligned}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, responses must indicate that h is continuous and differentiable. An ambiguous reference to h (e.g., “the function is continuous and differentiable”) does not earn the first point. The interval on which h is continuous and differentiable does not need to be stated. If an interval is given, it must be correct with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., either by referencing the Mean Value Theorem or by restating the statement in the stem of the

5-8

problem). Examples include:

- “Yes, $h(x)$ is continuous and differentiable so MVT applies” earns both points.
- “Yes, $h(x)$ is continuous and differentiable, therefore there must be a value where $h'(k)$ equals the average rate of change of $h(x)$ ” earns both points.
- “Yes, $h'(x)$ (the derivative of $h(x)$) is continuous and differentiable, therefore there must be a value where $h'(k)$ equals the average rate of change of $h(x)$ ” does not earn the first point, but does earn the second point.
- “Yes because of MVT” does not earn the first point, but does earn the second point.
- “Yes, because $h(x)$ is continuous and differentiable” earns the first point, but not the second point.
- “Yes, because $h(x)$ is continuous” earns neither point.
- “Yes, because $h(x)$ is differentiable” earns neither point.
- “Yes” earns neither point.

A response of “No” earns neither point, regardless of any other explanation including any reference to MVT.

Special case: a response that claims “ h is not differentiable, so therefore the MVT does not apply” earns one of the two points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

$$h' = f \Rightarrow h \text{ is differentiable on } -4 \leq x \leq 0. \Rightarrow h \text{ is continuous on } -4 \leq x \leq 0.$$

Therefore, the Mean Value Theorem can be applied to h on the interval $-4 \leq x \leq 0$ to guarantee that there exists a value of k , for $-4 < k < 0$, such that $h'(k)$ equals the average rate of change of h over the interval $-4 \leq x \leq 0$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $\lim_{x \rightarrow -2} \frac{h'(x)}{3x} = \frac{-1}{6}$ earns the point, as does just stating the answer of $\frac{-1}{6}$.

No additional supporting work is required.

The response does not need to state that h' is continuous. If the response does state that h' is continuous, an explanation is not required.



0	1
---	---

5-8

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$h' = f \Rightarrow h' \text{ is continuous. } \Rightarrow \lim_{x \rightarrow -2} h'(x) = h'(-2)$$

$$\lim_{x \rightarrow -2} \frac{h'(x)}{3x} = \frac{h'(-2)}{3 \cdot (-2)} = \frac{f(-2)}{-6} = \frac{1}{-6}$$

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x = -1$ is substituted for x , then $x = -1$ must appear unsimplified to earn the first point. Responses that earn the point include:

$$\begin{aligned} &\cdot h'(-3x) \cdot (-3) \\ &\cdot f(-3x) \cdot (-3) \\ &\cdot h'(-3 \cdot -1) \cdot (-3) \\ &\cdot f(-3 \cdot -1) \cdot (-3) \end{aligned}$$

Responses that do not earn the point include:

$$\begin{aligned} &\cdot h'(x) \cdot (-3) \\ &\cdot f(x) \cdot (-3) \\ &\cdot h'(3) \cdot (-3) \\ &\cdot f(3) \cdot (-3) \\ &\cdot h(-3x) \cdot (-3) \end{aligned}$$

The second point is earned for evidence of attempting to apply the Fundamental Theorem of Calculus as evidenced by linking h' and f either directly or indirectly in the presented derivative of $r(x)$ or in $r'(-1)$. For example, the second point is earned by $f(-3x)$ or by $f(-3 \cdot -1)$.

The second point may be earned without earning the first point. For example, $h'(-3x) = f(-3x)$ does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because h' and f are linked.

The second point may be earned by going directly from $h'(3)$ to -2 . The function f does not have to be stated explicitly. For example, $h'(-3 \cdot -1) \cdot (-3) = -2 \cdot (-3)$ earns the first two points. Recall, if $x = -1$ is substituted for x , then $x = -1$ must appear unsimplified to earn the first point.

The first two points may be earned by a single expression. For example, both $f(-3x) \cdot (-3)$ and $f(-3 \cdot -1) \cdot (-3)$ earn the first two points.

The third point is earned for the correct answer of 6, simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.

5-8



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

$$r'(x) = h'(-3x) \cdot (-3)$$

$$= f(-3x) \cdot (-3)$$

$$r'(-1) = f((-3) \cdot (-1)) \cdot (-3) = f(3) \cdot (-3) = -2 \cdot (-3) = 6$$

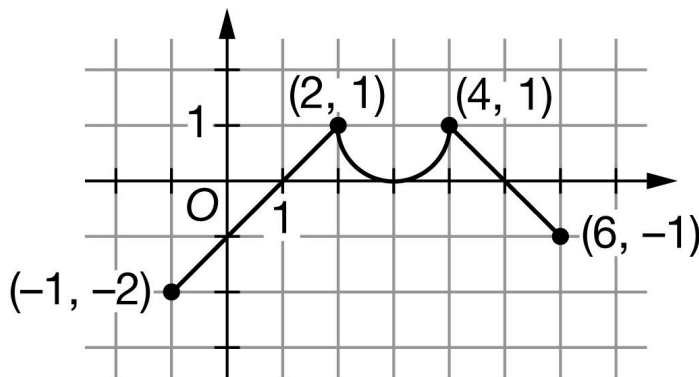
5-8

56. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Graph of h

The continuous function h has domain $-1 \leq x \leq 6$. The graph of h , consisting of two line segments and a semicircle, is shown in the figure above. The graph of h has a horizontal tangent at $x = 3$. Let g be the function defined by

$$g(x) = \int_2^x h(t) \, dt \text{ for } -1 \leq x \leq 6.$$

- Find the x -coordinate of each critical point of g on the interval $-1 \leq x \leq 6$.
- Classify each critical point from part (a) as the location of a relative minimum, a relative maximum, or neither for g . Justify each of your answers.
- For $-1 \leq x \leq 6$, on what open intervals is g decreasing and concave up? Give a reason for your answer.
- Find the value of $g(4)$. Show the computations that lead to your answer.
- Find the value of $g(-1)$. Show the computations that lead to your answer.
- Find the value of $g''(0)$, or explain why it does not exist.
- Must there exist a value of d , for $-1 < d < 3$, such that $g'(d)$ is equal to the average rate of change of g over the interval $-1 \leq x \leq 3$? Justify your answer.
- Find $\lim_{x \rightarrow 4} \frac{g'(x)}{3x}$. Show the computations that lead to your answer.
- The function a is defined by $a(x) = g(2x)$. Find $a'(0)$. Show the computations that lead to your answer.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn this point, the response must include all three critical values: $x = 1$, $x = 3$, and $x = 5$.

No supporting work is required, although any presented work must be correct.

The point is not earned if the response presents any extra numbers in the open interval $(-1, 6)$, or if any critical values are missing.

5-8

Disregard presentations of the endpoints or any other numbers outside the open interval $(-1, 6)$. The point is still earned provided the three critical points $x = 1$, $x = 3$, and $x = 5$ are correctly listed.

If the critical points are given as ordered pairs, e.g., $(1, 0)$, $(3, 0)$, and $(5, 0)$, read the x -coordinates, and disregard the y -coordinates.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$g'(x) = h(x) = 0$ at $x = 1$, $x = 3$, and $x = 5$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Each critical point from part (a) must be accompanied by a correct justification to earn the point.

A response giving justifications based on g' is eligible for all three points.

If a response gives justifications based only on h , then:

- the response is eligible for all three points, if the response presents the connection between $g'(x)$ and $h(x)$, (that is, $g' = h$). If you do not see this in part (b), look for it in part (a). If you find this connection in either part, read all subsequent occurrences of h or g' as $g' = h$, throughout all parts of this question.

- the response is eligible for at most two of the three points, if the response does not present the connection $g'(x) = h(x)$. Any additional error (for example, not justifying that at $x = 1$, g has a relative minimum) is eligible for at most one of the three points.

Read references to “the graph” as references to $h(x)$ as indicated by the stem.

A response referencing “the function” or “it” or “the slope” is vague. A response of this type can earn a maximum of one point in this part. This type of response earns that one point if replacing every occurrence of “the function” or “it” or “the slope” with g' produces a response that would earn all three points.

A response that discusses g' at a particular point (e.g., “ $g'(1)$ changes from negative to positive”) when the discussion should have been about $g'(x)$ can earn at most two of the three points because of poor communication. A response of this type that otherwise correctly classifies and justifies all three critical points earns two points, otherwise correctly classifying and justifying two critical points earns one point, and otherwise correctly classifying and justifying one critical point earns no points.

A sign chart by itself is not sufficient to earn any points. A sign chart must be accompanied by sufficient written explanation of what it means.

Responses using a Candidates Test are eligible for all three points. $g(x)$ value for $x = 1, 3, 5$ must be included in the table or list of values. The correct values of relevant inputs include:

- $g(-1) = \frac{3}{2}$

- $g(1) = -\frac{1}{2}$

5-8

$$g(3) = 1 - \frac{\pi}{4}$$

$$g(5) = \frac{5}{2} - \frac{\pi}{4}$$

$$g(6) = 2 - \frac{\pi}{4}$$

Special Case: The response presents a communication error in describing the change in sign at the critical points that results in a “global” rather than a “local” argument. These earn two of three points. For example, a response that presents all three of the following statements in part (b) earn two points:

- At $x = 1$, g has a relative minimum because $g' = h$ is negative for $x < 1$ and $g' = h$ is positive for $x > 1$.
- At $x = 3$, g has neither because $g' = h$ is positive for $x < 3$ and $g' = h$ is positive for $x > 3$.
- At $x = 5$, g has a relative maximum because $g' = h$ is positive for $x < 5$ and $g' = h$ is negative for $x > 5$.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ Relative minimum with justification
- ☐ Neither with justification
- ☐ Relative maximum with justification

Solution:

At $x = 1$, g has a relative minimum because $g'(x) = h(x)$ changes from negative to positive there.

At $x = 3$, g has neither because $g'(x) = h(x)$ does not change sign there.

At $x = 5$, g has a relative maximum because $g'(x) = h(x)$ changes from positive to negative there.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned by the correct answer. In addition, responses may give closed, half-open, and open intervals; that is, $[-1, 1]$, $(-1, 1]$, $[-1, 1)$, and $(-1, 1)$ all earn the first point.

In order to be eligible for the second point, the first point must have been earned.

A response referencing only g' is eligible to earn the second point.

A response referencing only h is not eligible to earn the second point, unless the response has established the link between $h(x)$ and $g'(x)$ previously in part (a) or part (b), in which case the response is eligible to earn the second point.

The reason “ g' is increasing” can also be expressed by either “ g'' is positive” or “the slope of g' is positive.”

A response referencing just “the function” or “it” or “the slope” is vague and does not earn the second point. Note that responses of this type are eligible for a maximum of one point in part (b) and one point in part (c); that is, a response of this type fails to earn points in both parts (b) and (c).

5-8



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

☐ Answer

☐ Reason

Solution:

g is decreasing where $g' = h$ is negative.

g is concave up where $g' = h$ is increasing.

g is decreasing and concave up on the open interval $(-1, 1)$.

Part D

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

Correct supporting work is needed to earn this point.

The correct decimal value is $g(4) = 0.429$. If this decimal value is given, it must be correct, rounded or truncated, to three decimal places.



0	1
---	---

The student response accurately includes the criteria below.

☐ Answer

Solution:

$$g(4) = 2 \cdot 1 - \frac{\pi}{2} \cdot 1^2 = 2 - \frac{\pi}{2}$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $-\frac{1}{2} + 2$ earns both points. This is the minimal amount of work that earns both points.

As indicated by this minimal answer, the value of an integral (that is, -2 or $\frac{1}{2}$) does not need to be explicitly linked with an integral to earn the first point.

A response of $g(-1) = \frac{3}{2}$ with no supporting work earns the second point only.

5-8



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ $\int_{-1}^1 h(t) \, dt = -2$ – OR – $\int_1^2 h(t) \, dt = \frac{1}{2}$
- ☐ Answer

Solution:

$$\begin{aligned} g(-1) &= \int_2^{-1} h(t) \, dt = - \int_{-1}^2 h(t) \, dt \\ &= - \int_{-1}^1 h(t) \, dt - \int_1^2 h(t) \, dt = -(-2) - \frac{1}{2} = \frac{3}{2} \end{aligned}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $g''(0) = 1$ or $h'(0) = 1$ or even simply stating a numerical value of 1 clearly marked as the answer earns the point.

No supporting work is required because the answer of 1 can easily be read from the graph.



0	1
---	---

The student response accurately includes the criteria below.

- ☐ Answer

Solution:

$$\begin{aligned} g''(x) &= h'(x) \\ g''(0) &= h'(0) = \frac{1}{2} = \frac{1 - (-2)}{2 - (-1)} \end{aligned}$$

Part G

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

To earn the first point, responses must indicate that g is continuous and differentiable. An ambiguous reference to g (e.g., “the function is continuous and differentiable”) does not earn the first point. The interval on which g is continuous and differentiable does not need to be stated. If an interval is given, it must be correct with the possible exception of inclusion/exclusion of the endpoints.

To earn the second point, responses must provide both an affirmative response to the question (can be implicit) and indicate that the Mean Value Theorem applies (e.g., either by referencing the Mean Value Theorem or by restating the statement in the stem of the problem). Examples include:

5-8

- “Yes, $g(x)$ is continuous and differentiable so MVT applies” earns both points.
- “Yes, $g(x)$ is continuous and differentiable, therefore there must be a value where $g'(d)$ equals the average rate of change of $g(x)$ ” earns both points.
- “Yes, $g'(x)$ (the derivative of $g(x)$) is continuous and differentiable, therefore there must be a value where $g'(d)$ equals the average rate of change of $g(x)$ ” does not earn the first point, but does earn the second point.
- “Yes because of MVT” does not earn the first point, but does earn the second point.
- “Yes, because $g(x)$ is continuous and differentiable” earns the first point, but not the second point.
- “Yes, because $g(x)$ is continuous” earns neither point.
- “Yes, because $g(x)$ is differentiable” earns neither point.
- “Yes” earns neither point.

A response of “No” earns neither point, regardless of any other explanation including any reference to MVT.

Special case: a response that claims “ g is not differentiable, so therefore the MVT does not apply” earns one of the two points.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ Conditions
- ☐ Conclusion using Mean Value Theorem

Solution:

$g' = h \Rightarrow g$ is differentiable on $-1 \leq x \leq 3$. $\Rightarrow g$ is continuous on $-1 \leq x \leq 3$.

Therefore, the Mean Value Theorem can be applied to g on the interval $-1 \leq x \leq 3$ to guarantee that there exists a value of d , for $-1 < d < 3$, such that $g'(d)$ equals the average rate of change of g over the interval $-1 \leq x \leq 3$.

Part H

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

A response of $\lim_{x \rightarrow 4} \frac{g'(x)}{3x} = \frac{1}{12}$ earns the point, as does just stating the answer of $\frac{1}{12}$.

No additional supporting work is required.

The response does not need to state that g' is continuous. If the response does state that g' is continuous, an explanation is not required.



0	1
---	---

The student response accurately includes the criteria below.

5-8

☐ Answer
Solution:

$$g' = h \Rightarrow g' \text{ is continuous. } \Rightarrow \lim_{x \rightarrow 4} g'(x) = g'(4)$$

$$\lim_{x \rightarrow 4} \frac{g'(x)}{3x} = \frac{g'(4)}{3 \cdot 4} = \frac{h(4)}{12} = \frac{1}{12}$$

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned only for the correct application of the chain rule. If $x = 0$ is substituted for x , then $x = 0$ must appear unsimplified to earn the first point. Responses that earn the point include:

$$\cdot g'(2x) \cdot 2$$

$$\cdot h(2x) \cdot 2$$

$$\cdot g'(2 \cdot 0) \cdot 2$$

$$\cdot h(2 \cdot 0) \cdot 2$$

Responses that do not earn the point include:

$$\cdot g'(x) \cdot 2$$

$$\cdot h(x) \cdot 2$$

$$\cdot g'(0) \cdot 2$$

$$\cdot h(0) \cdot 2$$

$$\cdot g(2x) \cdot 2$$

The second point is earned for evidence of attempting to apply the Fundamental Theorem of Calculus as evidenced by linking g' and h either directly or indirectly in the presented derivative of $a(x)$ or in $a'(0)$. For example, the second point is earned by $h(2x)$ or by $h(2 \cdot 0)$.

The second point may be earned without earning the first point. For example, $g'(2x) = h(2x)$ does not earn the first point because the chain rule is applied incorrectly, but does earn the second point because g' and h are linked.

The second point may be earned by going directly from $g'(0)$ to -1 . The function h does not have to be stated explicitly. For example, $g'(2 \cdot 0) \cdot 2 = -1 \cdot 2$ earns the first two points. Recall, if $x = 0$ is substituted for x , then $x = 0$ must appear unsimplified to earn the first point.

The first two points may be earned by a single expression. For example, both $h(2x) \cdot 2$ and $h(2 \cdot 0) \cdot 2$ earn the first two points.

The third point is earned for the correct answer of -2 , simplified or unsimplified. Responses that fail to earn both of the first two points are not eligible to earn the third point.



0	1	2	3
---	---	---	---

5-8

The student response accurately includes **all three** of the criteria below.

- ☐ Chain rule
- ☐ Fundamental Theorem of Calculus
- ☐ Answer

Solution:

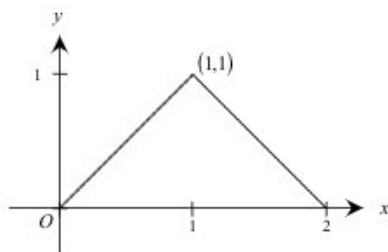
$$a'(x) = g'(2x) \cdot 2$$

$$= h(2x) \cdot 2$$

$$a'(0) = g'(2 \cdot 0) \cdot 2 = g'(0) \cdot 2 = h(0) \cdot 2 = -1 \cdot 2 = -2$$

5-8

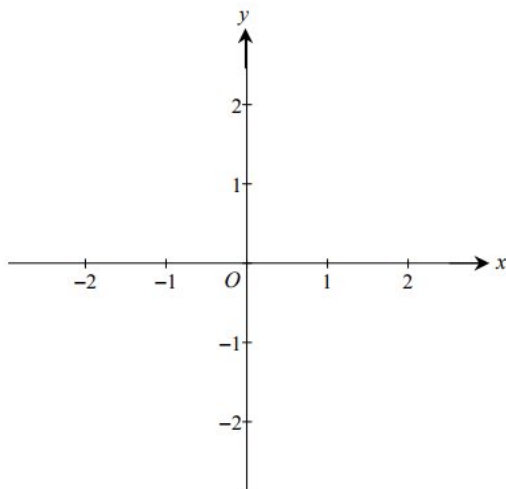
57.



Note: This is the graph of the derivative of f , not the graph of f .

The figure above shows graph of f' , the derivative of f . The domain of f is the set of all x such that $0 < x < 2$.

- Write an expression for $f'(x)$ in terms of x .
- Given that $f(1) = 0$, write an expression for $f(x)$ in terms of x .
- In the xy -plane provided below, sketch the graph of $y = f(x)$.

**Part A**

1 point is earned for x if $x \leq 1$

1 point is earned for $2 - x$ if $x \geq 1$

$$f'(x) = \begin{cases} x & \text{if } 0 < x \leq 1 \\ 2 - x & \text{if } 1 \leq x < 2 \end{cases}$$

or

$$f'(x) = 1 - |x - 1| \text{ if } 0 < x < 2$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for x if $x \leq 1$

5-8

1 point is earned for $2 - x$ if $x \geq 1$

$$f'(x) = \begin{cases} x & \text{if } 0 < x \leq 1 \\ 2 - x & \text{if } 1 \leq x < 2 \end{cases}$$

or

$$f'(x) = 1 - |x - 1| \text{ if } 0 < x < 2$$

Part B

1 point is earned for anti derivative for $x \leq 1$

1 point is earned for anti derivative for $x \geq 1$

2 points are earned for constants

$$f(x) = \begin{cases} \frac{x^2}{2} + C_1 & \text{if } 0 < x \leq 1 \\ 2x - \frac{x^2}{2} + C_2 & \text{if } 1 \leq x < 2 \end{cases}$$

$$0 = f(1) = \frac{1}{2} + C_1 \Rightarrow C_1 = -\frac{1}{2}$$

$$0 = f(1) = 2 - \frac{1}{2} + C_2 \Rightarrow C_2 = -\frac{3}{2}$$

$$f(x) = \begin{cases} \frac{x^2}{2} - \frac{1}{2} & \text{if } 0 < x \leq 1 \\ 2x - \frac{x^2}{2} - \frac{3}{2} & \text{if } 1 \leq x < 2 \end{cases}$$

or

$$f(x) = x - \frac{1}{2}(x - 1)|x - 1| - 1$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for anti derivative for $x \leq 1$

1 point is earned for anti derivative for $x \geq 1$

2 points are earned for constants

5-8

$$f(x) = \begin{cases} \frac{x^2}{2} + C_1 & \text{if } 0 < x \leq 1 \\ 2x - \frac{x^2}{2} + C_2 & \text{if } 1 \leq x < 2 \end{cases}$$

$$0 = f(1) = \frac{1}{2} + C_1 \Rightarrow C_1 = -\frac{1}{2}$$

$$0 = f(1) = 2 - \frac{1}{2} + C_2 \Rightarrow C_2 = -\frac{3}{2}$$

$$f(x) = \begin{cases} \frac{x^2}{2} - \frac{1}{2} & \text{if } 0 < x \leq 1 \\ 2x - \frac{x^2}{2} - \frac{3}{2} & \text{if } 1 \leq x < 2 \end{cases}$$

or

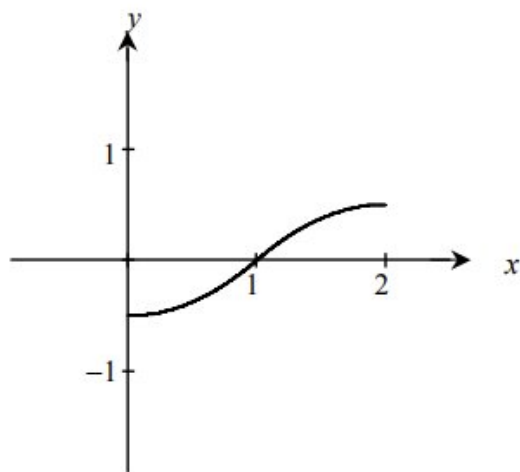
$$f(x) = x - \frac{1}{2}(x-1)|x-1| - 1$$

Part C

1 point is earned for increasing and continuous on $(0, 2)$

1 point is earned for concave up on $(0, 1)$ and concave down on $(1, 2)$

1 point is earned for $f(1) = 0$



0	1	2	3
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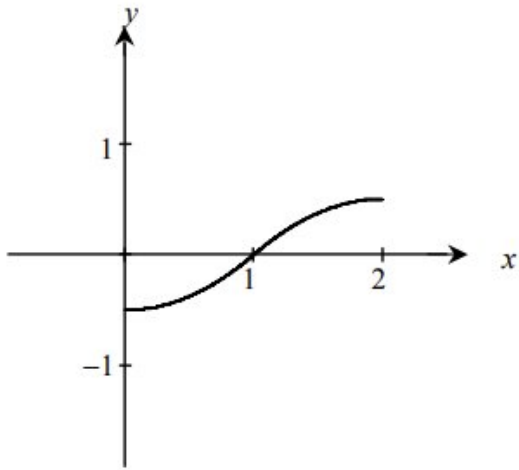
The student response earns all of the following points:

1 point is earned for increasing and continuous on $(0, 2)$

1 point is earned for concave up on $(0, 1)$ and concave down on $(1, 2)$

1 point is earned for $f(1) = 0$

5-8

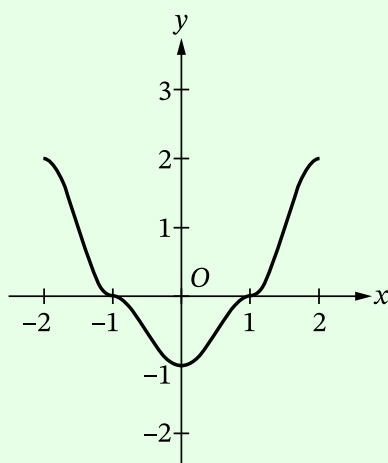


5-8

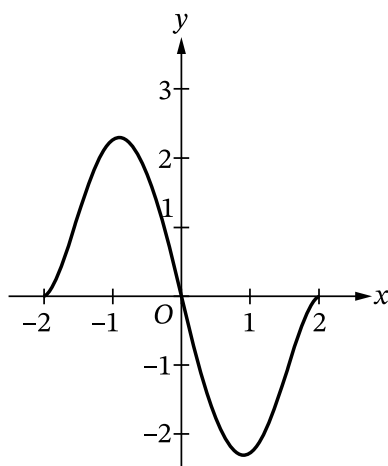
58. Let f be a differentiable function that is decreasing on the interval $(-1, 1)$ and increasing on the intervals $(-2, -1)$ and $(1, 2)$. Which of the following could be the graph of f' , the derivative of f , on the interval $(-2, 2)$?

5-8

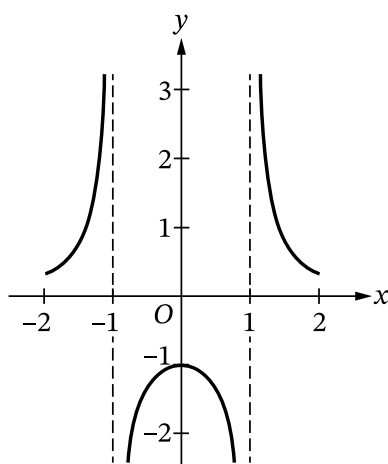
(A)



(B)

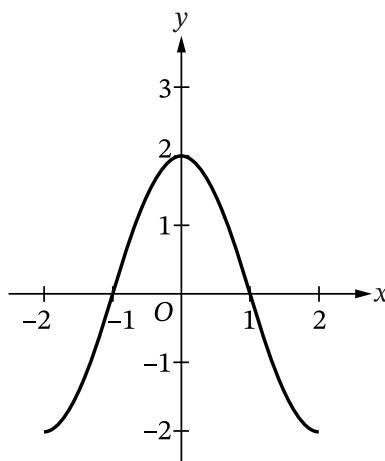


(C)



5-8

(D)

**Answer A**

Correct. If f is decreasing on the interval $(-1, 1)$, then $f'(x) \leq 0$ for x in $(-1, 1)$. If f is increasing on the intervals $(-2, -1)$ and $(1, 2)$, then $f'(x) \geq 0$ for x in the intervals $(-2, -1)$ and $(1, 2)$. This graph satisfies all those properties and therefore could be the graph of f' on the interval $(-2, 2)$.

59. The function f is defined by $f(x) = 1 + 9x + 3x^2 - x^3$. On what open interval is f increasing at a decreasing rate?

- (A) $(-\infty, -1)$
- (B) $(-1, 1)$ only
- (C) $(-1, 3)$
- (D) $(1, 3)$ only

**Answer D**

Correct. The function f is increasing at a decreasing rate on open intervals, where $f'(x) > 0$ and $f''(x) < 0$.

$$f'(x) = 9 + 6x - 3x^2 = 3(3 + 2x - x^2) = 3(3 - x)(1 + x)$$

This is a parabola opening downward with zeros at $x = -1$ and $x = 3$, so $f'(x) > 0$ on the interval $(-1, 3)$.

$$f''(x) = 6 - 6x = 6(1 - x)$$

This is a line with negative slope and zero at $x = 1$, so $f''(x) < 0$ on $(1, \infty)$. The intersection of the two intervals is $(1, 3)$.

5-8

60.  Let f be the function with derivative defined by $f'(x) = 2 + (2x - 8) \sin(x + 3)$. How many points of inflection does the graph of f have on the interval $0 < x < 9$?

- (A) One
(B) Two
(C) Three
(D) Four

**Answer D**

Number of extrema of f' on the interval

61.  The first derivative of the function h is given by $h'(x) = \sin x + \cos(x^2) + x$, and the second derivative of h is given by $h''(x) = \cos x - 2x \sin(x^2) + 1$. On what open intervals contained in $-3 < x < 2$ is the graph of h both increasing and concave down?

- (A) $(0.969, 1.697)$ only
(B) $(-2.499, -1.829)$ and $(0.969, 1.697)$
(C) $(-0.495, 2)$
(D) $(-1.311, -0.166)$

**Answer A**

Correct. The calculator is used to graph h' and h'' on the interval $-3 < x < 2$. The graph of h will be both increasing and concave down on intervals where the graph of h' is above the x -axis (graph of h increasing) and where the graph of h'' is below the x -axis (graph of h concave down). There is one such interval. The calculator is used to find the zeros of h' and h'' .

62. Let f be the function defined by $f(x) = \frac{1}{3}x^3 - 3x^2 - 16x$. On which of the following intervals is the graph of f both decreasing and concave down?

- (A) $(-\infty, 3)$
(B) $(-2, 3)$ only
(C) $(3, 8)$
(D) $(8, \infty)$

**Answer B**

Correct. The graph of f will be both decreasing and concave down where $f'(x) < 0$ and $f''(x) < 0$.

$$f'(x) = x^2 - 6x - 16 = (x + 2)(x - 8)$$

$$f''(x) = 2x - 6 = 2(x - 3)$$

Therefore, $f'(x) < 0$ for $-2 < x < 8$ and $f''(x) < 0$ for $x < 3$. The intersection of these two intervals is $-2 < x < 3$.

5-8

63.

x	$0 < x < 3$	$x = 3$	$3 < x < 9$	$x = 9$	$9 < x < 11$	$x = 11$	$11 < x < 16$
$f'(x)$	Positive	Undefined	Negative	-3	Negative	0	Positive
$f''(x)$	Positive	Undefined	Negative	0	Positive	0	Positive

The function f is continuous on the interval $(0, 16)$, and f is twice differentiable except at $x = 3$, where the derivatives are undefined. Information about the first and second derivatives of f for values of x in the interval $(0, 16)$ is given in the table above. At what values of x in the interval $(0, 16)$ does the graph of f have a point of inflection?

(A) $x = 9$ only

(B) $x = 3$ and $x = 9$

(C) $x = 3$ and $x = 11$

(D) $x = 9$ and $x = 11$

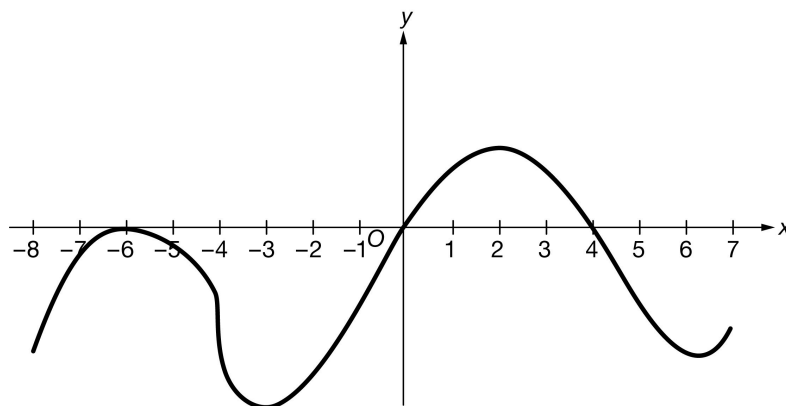


Answer B

Correct. The sign of f'' changes from positive to negative at $x = 3$, indicating that the graph of f changes from concave up to concave down at $x = 3$. Also, the sign of f'' changes from negative to positive at $x = 9$, indicating that the graph of f changes from concave down to concave up at $x = 9$. Therefore, the graph of f has points of inflection at both $x = 3$ and $x = 9$. There is not a point of inflection at $x = 11$ even though $f''(11) = 0$ because f'' does not change sign at $x = 11$.

5-8

64.

Graph of f'

The graph of f' , the derivative of the continuous function f , is shown above on the interval $-8 < x < 7$. The graph of f' has horizontal tangent lines at $x = -6$, $x = -3$, $x = 2$, and $x = 6.3$, and a vertical tangent line at $x = -4$. On which of the following intervals is the graph of f both decreasing and concave up?

- (A) $(-8, 0)$ and $(4, 7)$
 (B) $(-6, -3)$ and $(4, 6.3)$ only
 (C) $(-4, -3)$ and $(4, 6.3)$ only
 (D) $(-8, -6)$, $(-3, 0)$, and $(6.3, 7)$ only

**Answer D**

Correct. The graph of f will be both decreasing and concave up where $f'(x) < 0$ and f' is increasing. That happens on each of these three open intervals.

65.

x	$0 < x < 2$	$x = 2$	$2 < x < 4$	$x = 4$	$4 < x < 6$	$x = 6$	$6 < x < 8$	$x = 8$	$8 < x < 9$
$f'(x)$	Unknown	0	Negative	0	Negative	DNE	Positive	0	Unknown
$f''(x)$	Negative	-1	Negative	0	Positive	DNE	Negative	0	Unknown

The function f is continuous on the interval $(0, 9)$ and is twice differentiable except at $x = 6$, where the derivatives do not exist (DNE). Information about the first and second derivatives of f for some values of x in the interval $(0, 9)$ is given in the table above. Which of the following statements could be false?

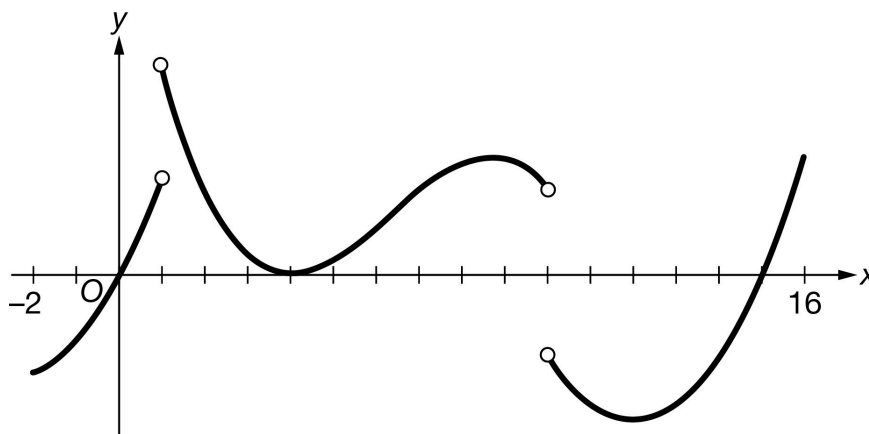
- (A) The function f has a relative maximum at $x = 2$.
 (B) The graph of f has a point of inflection at $x = 4$.
 (C) The function f has a relative minimum at $x = 6$.
 (D) The graph of f has a point of inflection at $x = 8$.

**Answer D**

Correct. While $f''(8) = 0$, it cannot be known if the second derivative actually changes sign at $x = 8$. For example, the function f may behave like $-(x - 8)^4$ near $x = 8$, in which case there is no point of inflection at $x = 8$ and the statement would be false.

5-8

66.

Graph of f'

The graph of f' , the derivative of the continuous function f , is shown above on the interval $-2 < x < 16$. Which of the following statements is true about f on the interval $-2 < x < 16$?

- (A) f has two relative extrema, and the graph of f has three points of inflection.
- (B) f has three relative extrema, and the graph of f has three points of inflection.
- (C) f has three relative extrema, and the graph of f has four points of inflection. ✓
- (D) f has five relative extrema, and the graph of f has four points of inflection.

Answer C

Correct. The function f has three critical points at $x = 0$, $x = 4$, and $x = 15$ where $f'(x) = 0$, and two critical points at $x = 1$ and $x = 10$ where f' is undefined. There is a relative minimum at $x = 0$ because the sign of f' changes from negative to positive there. The function f has neither a relative minimum nor a relative maximum at $x = 1$ because the sign of f' is positive on both sides of $x = 1$. The function f also has neither a relative minimum nor a relative maximum at $x = 4$ because the sign of f' is positive on both sides of $x = 4$. There is a relative maximum at $x = 10$ where the sign of f' changes from positive to negative. There is a relative minimum at $x = 15$ where the sign of f' changes from negative to positive. Therefore, f has three relative extrema on the interval $-2 < x < 16$.

There is a point of inflection at $x = 1$ where the graph of f' changes from increasing to decreasing. There is a point of inflection at $x = 4$ where the graph of f' changes from decreasing to increasing. There is a point of inflection at approximately $x = 8.5$ where the graph of f' changes from increasing to decreasing. The final point of inflection is at $x = 12$ where the graph of f' changes from decreasing to increasing. Therefore, the graph of f has four points of inflection on the interval $-2 < x < 16$.

5-8

67.  The first derivative of the function h is given by $h'(x) = 3 \ln(2 + \cos(2x)) - x$, and the second derivative of h is given by $h''(x) = \frac{-6 \sin(2x)}{2 + \cos(2x)} - 1$. On what open intervals contained in $-2 < x < 2$ is the graph of h both increasing and concave down?

(A) $(-2, -1.486)$ and $(-0.250, 1.085)$ ✓

(B) $(-2, -1.486)$ and $(-0.250, 1.656)$

(C) $(-2, 1.085)$

(D) $(-1.047, -0.250)$

Answer A

Correct. The calculator is used to graph h' and h'' on the interval $-2 < x < 2$. The graph of h will be both increasing and concave down on intervals where the graph of h' is above the x -axis (graph of h increasing) and where the graph of h'' is below the x -axis (graph of h concave down). There are two such intervals. The calculator is used to find the zeros of h' and h'' .

68. Let f be the function defined by $f(x) = \frac{1}{3}x^3 - 4x^2 - 9x + 5$. On which of the following intervals is the graph of f both decreasing and concave down?

(A) $(-\infty, 4)$

(B) $(-1, 4)$ ✓

(C) $(4, 9)$

(D) $(9, \infty)$

Answer B

Correct. The graph of f will be both decreasing and concave down where $f'(x) < 0$ and $f''(x) < 0$.

$$f'(x) = x^2 - 8x - 9 = (x + 1)(x - 9)$$

$$f''(x) = 2x - 8 = 2(x - 4)$$

Therefore, $f'(x) < 0$ for $-1 < x < 9$ and $f''(x) < 0$ for $x < 4$. The intersection of these two intervals is $-1 < x < 4$.

5-8

69.

x	$0 < x < 5$	$x = 5$	$5 < x < 8$	$x = 8$	$8 < x < 12$	$x = 12$	$12 < x < 16$
$f'(x)$	Positive	Undefined	Negative	-2	Negative	0	Positive
$f''(x)$	Positive	Undefined	Negative	0	Positive	0	Positive

The function f is continuous on the interval $(0, 16)$, and f is twice differentiable except at $x = 5$ where the derivatives are undefined. Information about the first and second derivatives of f for values of x in the interval $(0, 16)$ is given in the table above. At what values of x in the interval $(0, 16)$ does the graph of f have a point of inflection?

(A) $x = 8$ only

(B) $x = 5$ and $x = 8$

(C) $x = 5$ and $x = 12$

(D) $x = 8$ and $x = 12$

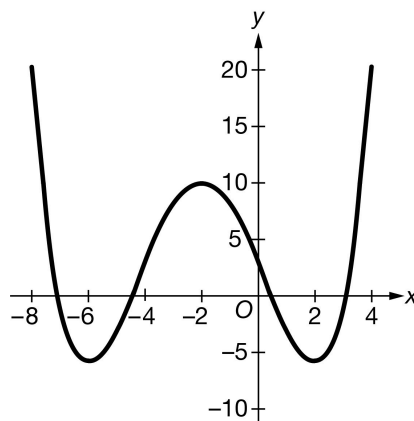


Answer B

Correct. The sign of f'' changes from positive to negative at $x = 5$, indicating that the graph of f changes from concave up to concave down at $x = 5$. Also, the sign of f'' changes from negative to positive at $x = 8$, indicating that the graph of f changes from concave down to concave up at $x = 8$. Therefore, the graph of f has a point of inflection at both $x = 5$ and $x = 8$. There is not a point of inflection at $x = 12$, even though $f''(12) = 0$, because f'' does not change sign at $x = 12$.

5-8

70.

Graph of f'

The graph of f' , the derivative of the continuous function f , is shown above on the interval $[-8, 4]$. The graph of f' has horizontal tangent lines at $x = -6$, $x = -2$, and $x = 2$. On which of the following intervals is the graph of f both decreasing and concave up?

(A) $(-8, -6)$ (B) $(-6, -4.5)$ (C) $(-2, 0)$ (D) $(1, 2)$ **Answer B**

Correct. The graph of f will be both decreasing and concave up where $f'(x) < 0$ and f' is increasing. That happens on this interval.

5-8

71.

x	$0 < x < 1$	$x = 1$	$1 < x < 2$	$x = 2$	$2 < x < 3$	$x = 3$	$3 < x < 4$	$x = 4$	$4 < x < 5$
$f'(x)$	Positive	0	Negative	DNE	Positive	0	Positive	0	Unknown
$f''(x)$	Unknown	-5	Negative	DNE	Negative	0	Positive	0	Unknown

The function f is continuous on the interval $(0, 5)$ and is twice differentiable except at $x = 2$, where the derivatives do not exist (DNE). Information about the first and second derivatives of f for some values of x in the interval $(0, 5)$ is given in the table above. Which of the following statements could be false?

- (A) The function f has a relative maximum at $x = 1$.
 (B) The function f has a relative minimum at $x = 2$.
 (C) The graph of f has a point of inflection at $x = 3$.
 (D) The graph of f has a point of inflection at $x = 4$.

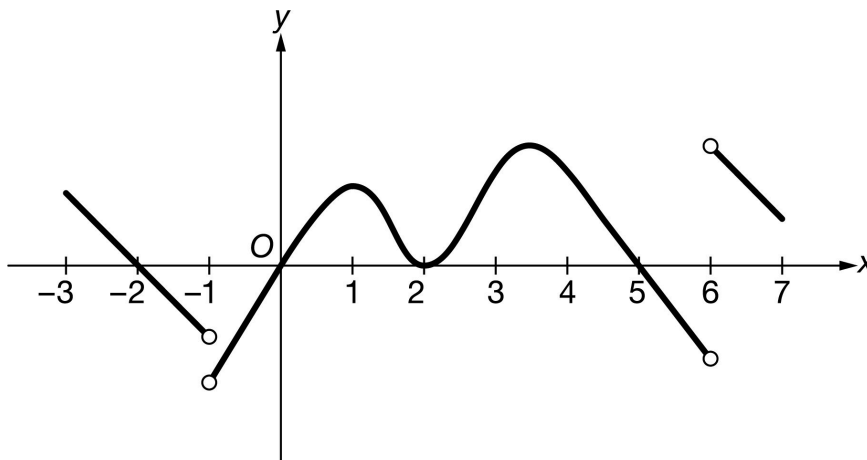


Answer D

Correct. While $f''(4) = 0$, it cannot be known if the second derivative actually changes sign at $x = 4$. For example, the function may behave like $g(x) = -(x - 4)^6$ near $x = 4$, in which the value of second derivative is 0 at $x = 4$, but the point is a relative maximum rather than a point of inflection.

5-8

72.

Graph of f'

The graph of f' , the derivative of the continuous function f , is shown above on the interval $-3 < x < 7$. Which of the following statements is true about f on the interval $-3 < x < 7$?

- (A) f has three relative extrema, and the graph of f has one point of inflection.
- (B) f has three relative extrema, and the graph of f has four points of inflection.
- (C) f has four relative extrema, and the graph of f has two points of inflection.
- (D) f has four relative extrema, and the graph of f has four points of inflection. ✓

Answer D

Correct. The function f has four critical points at $x = -2$, $x = 0$, $x = 2$, and $x = 5$, where $f'(x) = 0$, and two critical points, $x = -1$ and $x = 6$, where f' is undefined.

There are relative maxima at $x = -2$ and $x = 5$ because the sign of f' changes from positive to negative there.

The function f has neither a relative minimum nor a relative maximum at $x = -1$ because the sign of f' is negative on both sides of $x = -1$.

There are relative minima at $x = 0$ and $x = 6$ because the sign of f' changes from negative to positive there.

The function f has neither a relative minimum nor a relative maximum at $x = 2$ because the sign of f' is positive on both sides of $x = 2$.

Therefore, f has four relative extrema on the interval $-3 < x < 7$.

There are points of inflection at $x = -1$, $x = 1$, and $x = 2$, where the graph of f' changes from decreasing to increasing.

There is a point of inflection at approximately $x = 3.5$, where the graph of f' changes from increasing to decreasing.

Therefore, there are four points of inflection on the interval $-3 < x < 7$.

5-8

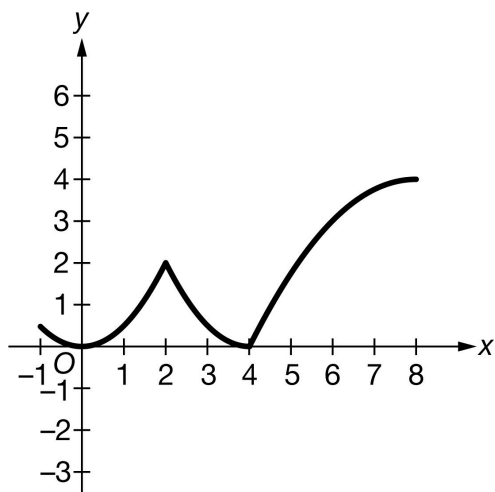
73.

x	0	$0 < x < 2$	2	$2 < x < 4$	4	$4 < x < 8$
$f'(x)$	0	Negative	Does not exist	Positive	Does not exist	Negative
$f''(x)$	Negative	Negative	Does not exist	Negative	Does not exist	Positive

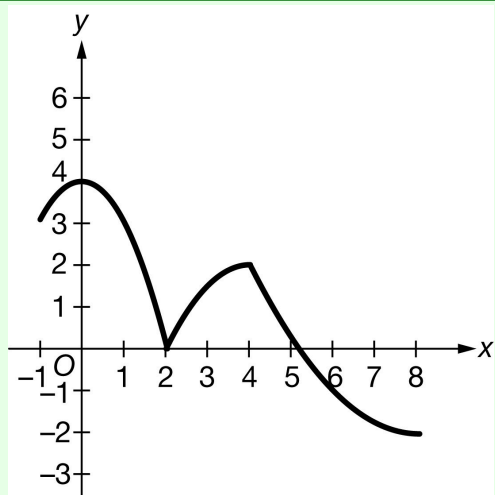
The first and second derivatives of a function f have the properties given in the table above. Which of the following could be the graph of f ?

5-8

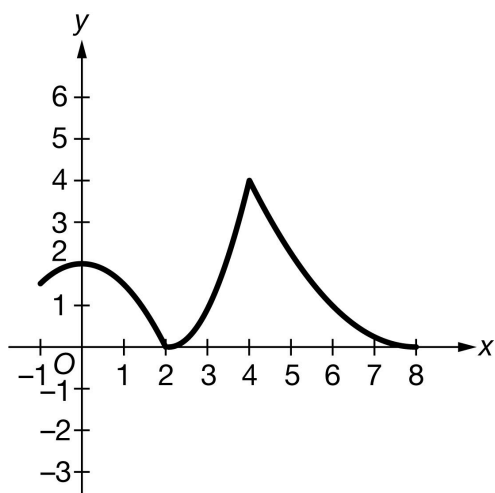
(A)



(B)

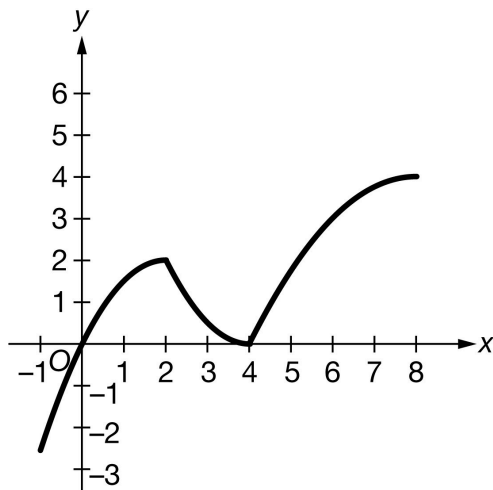


(C)



5-8

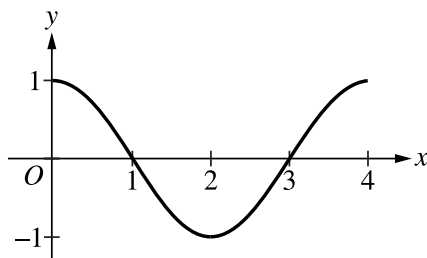
(D)

**Answer B**

Correct. The graph of f has a local maximum at $x = 0$, because $f'(0) = 0$ and $f''(0)$ is negative. The graph of f is decreasing and concave down on the interval $0 < x < 2$, because both $f'(x)$ and $f''(x)$ are negative on this interval. The graph of f is increasing and concave down on the interval $2 < x < 4$, because $f'(x)$ is positive and $f''(x)$ is negative on this interval. The graph of f is decreasing and concave up on the interval $4 < x < 8$, because $f'(x)$ is negative and $f''(x)$ is positive on this interval. The function f is not differentiable at $x = 2$ and $x = 4$. This graph exhibits all those behaviors and thus could be the graph of f .

5-8

74.

Graph of f

The graph of a twice-differentiable function f is shown for $0 < x < 4$. The graph of f has a horizontal tangent at $x = 2$ and points of inflection at $x = 1$ and $x = 3$. On what intervals do $f'(x)$ and $f''(x)$ have the same sign?

- (A) $0 < x < 1$ only
(B) $2 < x < 4$
(C) $0 < x < 1$ and $2 < x < 3$ ✓
(D) $1 < x < 2$ and $3 < x < 4$

Answer C

Correct. $f'(x)$ and $f''(x)$ are both negative where the graph of f is decreasing and concave down. This occurs on the interval $0 < x < 1$. $f'(x)$ and $f''(x)$ are both positive where the graph of f is increasing and concave up. This occurs on the interval $2 < x < 3$.

5-8

75. The advertising budget of a company is modeled by the differentiable function D , where $D(t)$ is measured in dollars and t is the number of years since the company was founded. For $0 \leq t \leq 10$, the company's advertising budget increased each year, but the rate of change of the advertising budget decreased each year. Which of the following could be a table of values of D ?

(A)

t	1	5	7
$D(t)$	50,000	55,000	53,000

(B)

t	1	5	7
$D(t)$	50,000	54,000	60,000

(C)

t	1	5	7
$D(t)$	50,000	53,000	55,000

(D)

t	1	5	7
$D(t)$	50,000	58,000	60,000



Answer D

Correct. The function D is continuous and differentiable. By the Mean Value Theorem, there is a c_1 in the interval $[1, 5]$ and a c_2 in the interval $[5, 7]$ such that $D'(c_1) = \frac{58,000 - 50,000}{5 - 1} = 2000$ and $D'(c_2) = \frac{60,000 - 58,000}{7 - 5} = 1000$.

Since $c_1 < c_2$ and $D'(c_1) > D'(c_2)$, this is consistent with the statement that the rate of change of the budget decreased each year, so this response could be a table of values of D .

5-8

76. Let f be a continuous function defined on the interval $0 \leq x \leq 7$. The function f and its derivatives have the properties indicated in the table shown, where DNE indicates the x -values at which the derivatives of f do not exist.

x	$x = 0$	$0 < x < 2$	$x = 2$	$2 < x < 4$	$x = 4$	$4 < x < 7$	$x = 7$
$f(x)$	-1	Negative	-4	Negative	-2	Negative	0
$f'(x)$	DNE	Negative	DNE	Positive	0	Positive	DNE
$f''(x)$	DNE	Positive	DNE	Negative	0	Positive	DNE

Part A

On what open intervals contained in $0 < x < 7$, if any, is the graph of f both increasing and concave down? Give a reason for your answer.

Part B

Find the average rate of change of f over the interval $0 \leq x \leq 4$. Can the Mean Value Theorem be applied to f on the interval $0 \leq x \leq 4$ to guarantee a value of c , for $0 < c < 4$, such that $f'(c)$ is equal to this average rate of change? Give a reason for your answer.

Part C

Use a left Riemann sum with the three subintervals indicated by the data in the table to approximate the value of $\int_0^7 f(x) dx$.

Part D

Let g be the function defined by $g(x) = x \cdot f(x)$. Is g increasing or decreasing at $x = 1$? Give a reason for your answer.

Part A Point 1

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P1** is earned only by an answer of $2 < x < 4$ or this interval including one or both endpoints.
- Special case: A response of “ f is increasing on $2 < x < 4$ and $4 < x < 7$ because $f'(x) > 0$ there” earns **P1**.
- Special case: A response of “ f is concave down on $2 < x < 4$ because $f''(x) < 0$ there” earns **P1**.

Model Solution:

f is increasing and concave down on the interval $2 < x < 4$.

5-8



0	1
---	---

The response accurately includes the criteria below.

- Answer

Part A Point 2

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response must have earned **P1** to be eligible for **P2**.
- To earn **P2**, a response must correctly cite both $f'(x) > 0$ and $f''(x) < 0$.
- Special case: A response of “ f is increasing on $2 < x < 4$ and $4 < x < 7$ because $f'(x) > 0$ there” does not earn **P2**.
- Special case: A response of “ f is concave down on $2 < x < 4$ because $f''(x) < 0$ there” does not earn **P2**.

Model Solution:

The graph of function f is increasing on an interval when $f'(x) > 0$ and concave down on an interval when $f''(x) < 0$.



0	1
---	---

The response accurately includes the criteria below.

- Reason

Part B Point 3

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- To earn **P3**, a response must include a difference and a quotient. Any of the following is sufficient to earn **P3**:

$$\frac{f(4)-f(0)}{4-0} = -\frac{1}{4}, \frac{-2-(-1)}{4}, \text{ or } \frac{-1}{4-0}.$$

Model Solution:

$$\frac{f(4)-f(0)}{4-0} = \frac{-2-(-1)}{4} = -\frac{1}{4}$$



0	1
---	---

The response accurately includes the criteria below.

- Average rate of change

5-8

Part B Point 4

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response does not need to have earned **P3** to be eligible for **P4**.
- To earn **P4**, a response must note that f is not differentiable and conclude “no.”
- To earn **P4**, a response need not explicitly name the Mean Value Theorem, but if a theorem is named, it must be correct.

Model Solution:

No, the Mean Value Theorem cannot be applied to f on the interval $0 \leq x \leq 4$ because f is not differentiable at $x = 2$.



0	1
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The response accurately includes the criteria below.

- Answer with reason

Part C Point 5

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- Read “=” as “ $\approx$ ” for **P5**.
- The form of a left Riemann sum includes three terms, each of which includes a product of two factors. To earn **P5**, at least five of the six factors must be correct.
- A response of $(f(0))(2) + (f(2))(2) + (f(4))(3)$ earns **P5**.
- A response of $(-1)(2) + (-4)(2) + (-2)(3)$ earns **P5**.
- An unsupported answer of -16 does not earn **P5**.
- A completely correct right Riemann sum with accompanying work (e.g., $(-4)(2) + (-2)(2) + (0)(3) = -12$) earns **P5**. A response with any errors or missing factors in the right Riemann sum does not earn **P5**.

Model Solution:

$$\int_0^7 f(x)dx \approx f(0)(2-0) + f(2)(4-2) + f(4)(7-4)$$



0	1
---	---

The response accurately includes the criteria below.

- Form of left Riemann sum

Part C Point 6

5-8

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response of $(f(0))(2) + (f(2))(2) + (f(4))(3)$ is eligible to earn **P6** with correct values for $f(0)$, $f(2)$, and $f(4)$ imported from the table. If any of the six factors is incorrect, the response does not earn **P6**.
- A response of $(-1)(2) + (-4)(2) + (-2)(3)$ earns **P6**.
- A response that includes $(-1)(2) + (-4)(2) + (-2)(3)$ or equivalent banks **P6** (i.e., subsequent errors in simplification will not be considered in scoring for **P6**).
- An unsupported answer of -16 does not earn **P6**.
- A right Riemann sum (e.g., $(-4)(2) + (-2)(2) + (0)(3) = -12$) does not earn **P6**.

Model Solution:

$$(-1)(2) + (-4)(2) + (-2)(3) = -16$$



0	1
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The response accurately includes the criteria below.

- Answer with supporting work

Part D Point 7

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P7** is earned for considering either $g'(x)$ or $g'(1)$.

Model Solution:

$$g'(x) = 1 \cdot f(x) + x \cdot f'(x)$$

$$g'(1) = f(1) + 1 \cdot f'(1)$$



0	1
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The response accurately includes the criteria below.

- Considers $g'(x)$

Part D Point 8

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- **P8** is earned with either $g'(x) = f(x) + x \cdot f'(x)$ or $g'(1) = f(1) + 1 \cdot f'(1)$.

5-8

Model Solution:

$$g'(x) = 1 \cdot f(x) + x \cdot f'(x)$$

$$g'(1) = f(1) + 1 \cdot f'(1)$$



0	1
---	---

The response accurately includes the criteria below.

- $g'(x)$ or $g'(1)$.

Part D Point 9

Select a point value to view scoring criteria, solutions, and examples or to score the response.

Scoring Notes:

- A response must have earned **P7** and **P8** to be eligible for **P9**.
- An eligible response that states either “decreasing because $g'(1) < 0$ ” or “decreasing because $g'(x) < 0$ for $0 < x < 2$ ” earns **P9**.
- A response that states “ $g(1)$ is decreasing” does not earn **P9**.
- A response that estimates negative values for $f(1)$ and/or $f'(1)$ using the table and reaches a correct conclusion earns **P9**.

Model Solution:

Because $g'(1) < 0$ and g' is continuous on $0 < x < 2$, g is decreasing on an interval containing $x = 1$.



0	1
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The response accurately includes the criteria below.

- Answer with reason

5-8

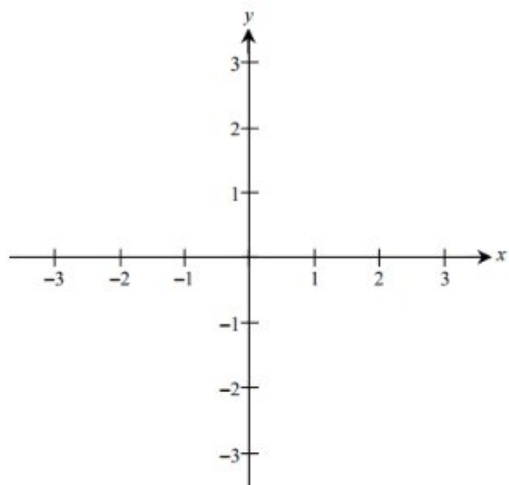
77. Let f be the function defined by $f(x) = xe^{1-x}$ for all real numbers x .

(a) Find each interval on which f is increasing.

(b) Find the range of f .

(c) Find the x -coordinate of each point of inflection of the graph of f .

(d) Using the results found in parts (a), (b), and (c), sketch the graph of f in the xy -plane provided below. (Indicate all intercepts.)



Part A

1 point is earned for derivative

1 point is earned for answer

$$f'(x) = xe^{1-x}(-1) + e^{1-x} = (1-x)e^{1-x}$$

f increases on $(-\infty, 1]$



0	1	2
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The student response earns all of the following points:

1 point is earned for derivative

1 point is earned for answer

$$f'(x) = xe^{1-x}(-1) + e^{1-x} = (1-x)e^{1-x}$$

f increases on $(-\infty, 1]$

Part B

1 point is earned for left hand endpoint

5-8

1 point is earned for right hand endpoint

Note: 0/2 for All reals without supporting work.

$$f(1) = 1; \lim_{x \rightarrow -\infty} f(x) = -\infty$$

Range: $(-\infty, 1]$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for left hand endpoint

1 point is earned for right hand endpoint

Note: 0/2 for All reals without supporting work.

$$f(1) = 1; \lim_{x \rightarrow -\infty} f(x) = -\infty$$

Range: $(-\infty, 1]$

Part C

1 point is earned for $f''(x)$ for student's $f'(x)$

1 point is earned for answer

$$\begin{aligned} f''(x) &= e^{1-x}(-1) + (1-x)e^{1-x}(-1) \\ &= (x-2)e^{1-x} \end{aligned}$$

Point of inflection at $x = 2$



0	1	2
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The student response earns all of the following points:

1 point is earned for $f''(x)$ for student's $f'(x)$

1 point is earned for answer

$$\begin{aligned} f''(x) &= e^{1-x}(-1) + (1-x)e^{1-x}(-1) \\ &= (x-2)e^{1-x} \end{aligned}$$

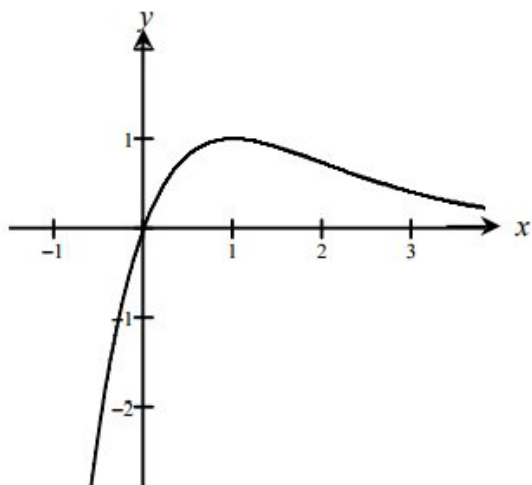
Point of inflection at $x = 2$

5-8

Part D

3 points are earned for answer without mistake

1 point is **deducted** for each error: intercept, domain, increasing, decreasing, range, inflection points.



✓

0	1	2	3
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The student response earns all of the following points:

3 points are earned for answer without mistake

1 point is **deducted** for each error: intercept, domain, increasing, decreasing, range, inflection points.

